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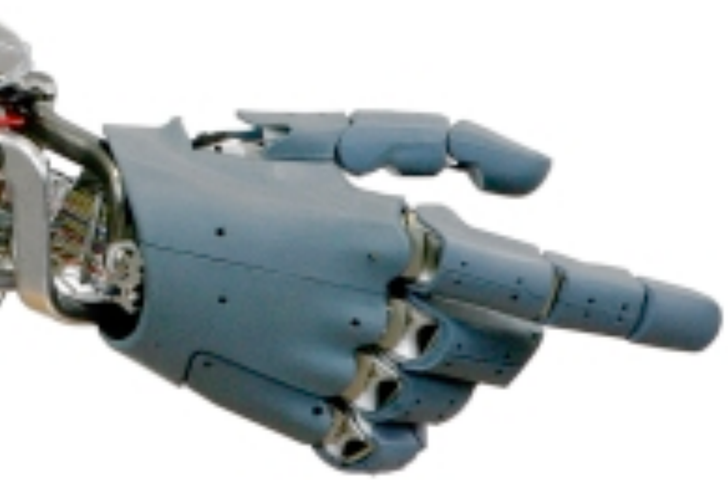
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Second Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamical Model

Robot Control

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Direct Kinematics of Robot Manipulators

For Rigid Objects in 3D Space

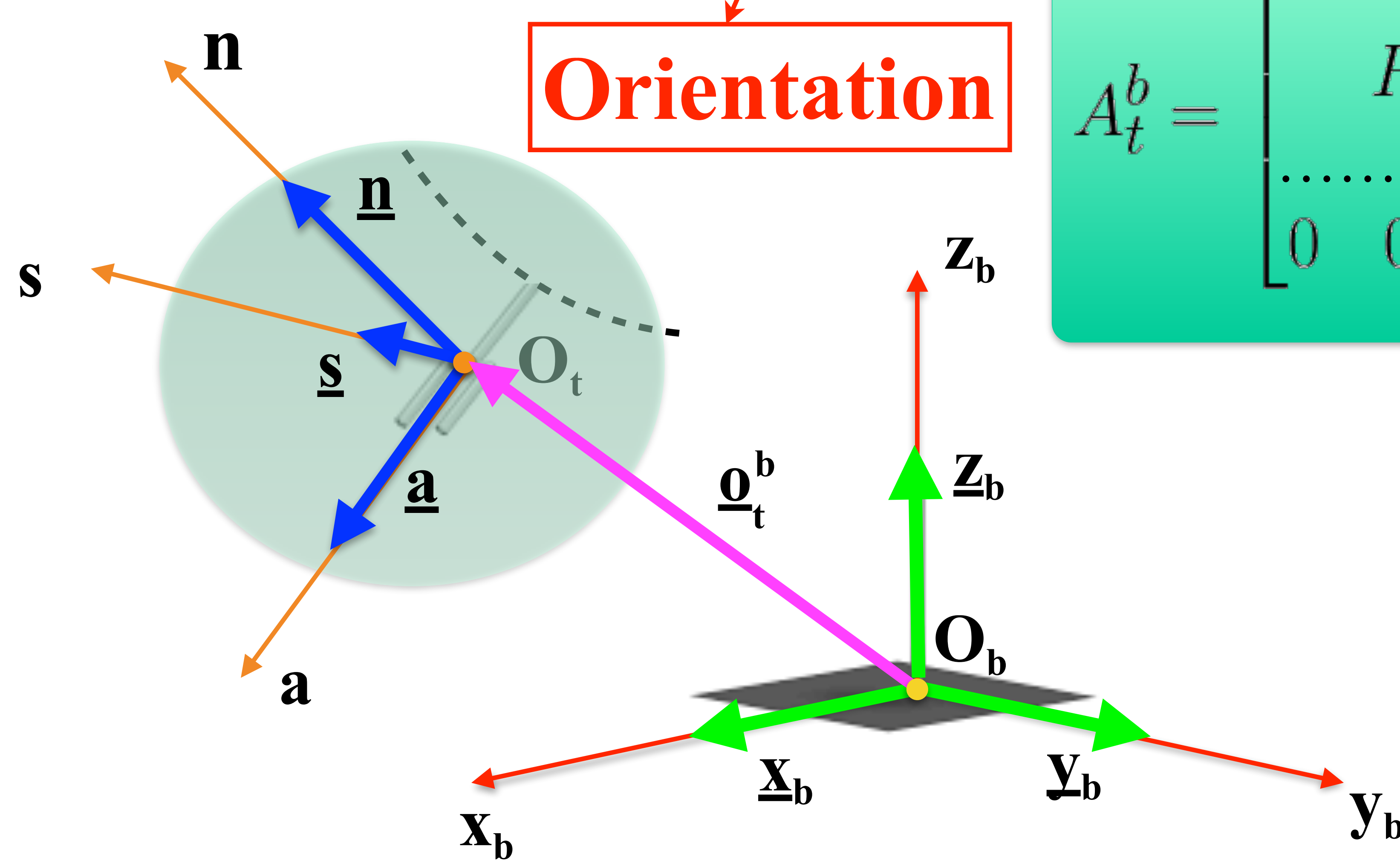
Description of the Pose of a Rigid Object with respect to the Base Reference Frame:

Homogeneous Transformation Matrix

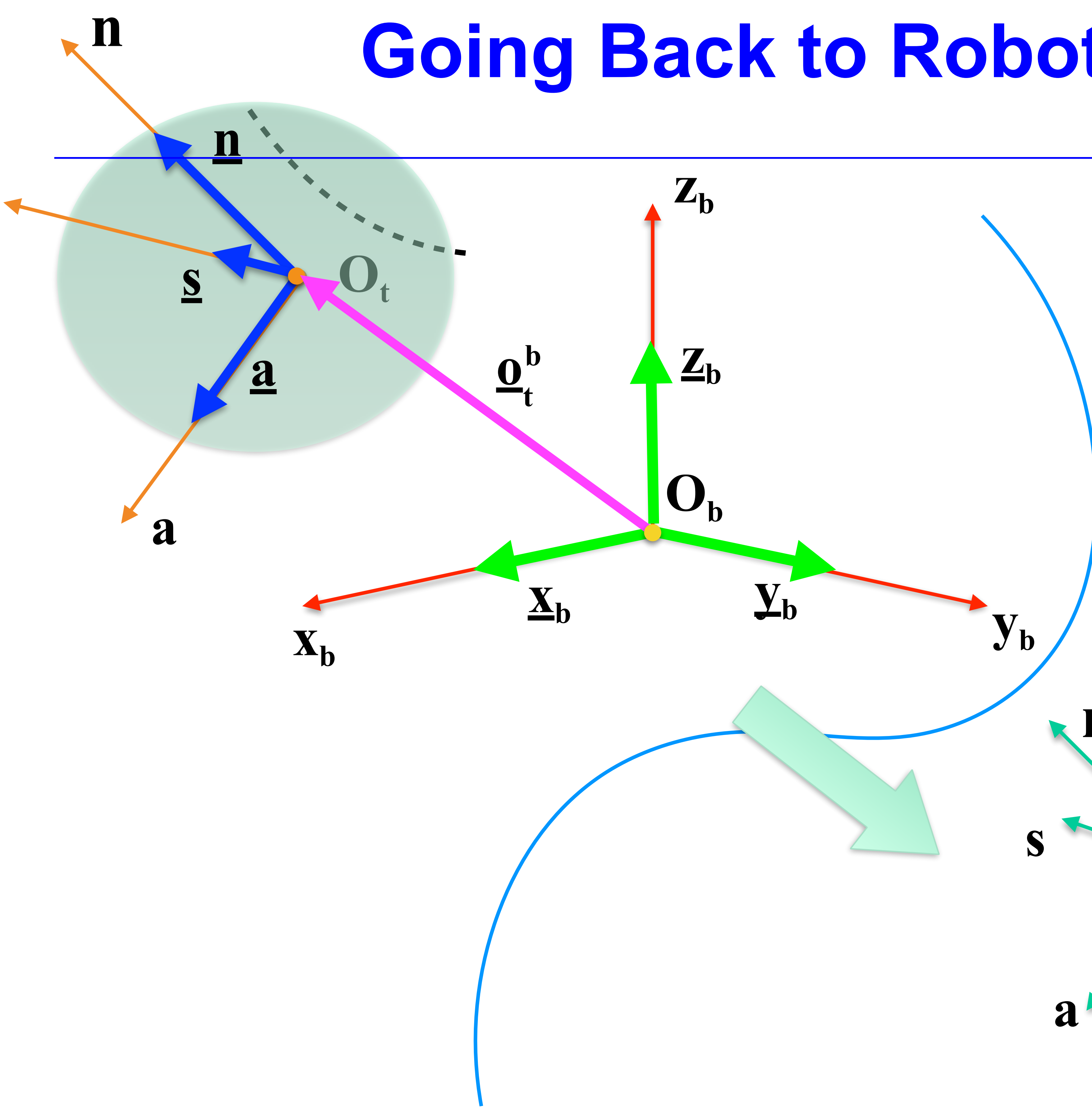
Position

Orientation

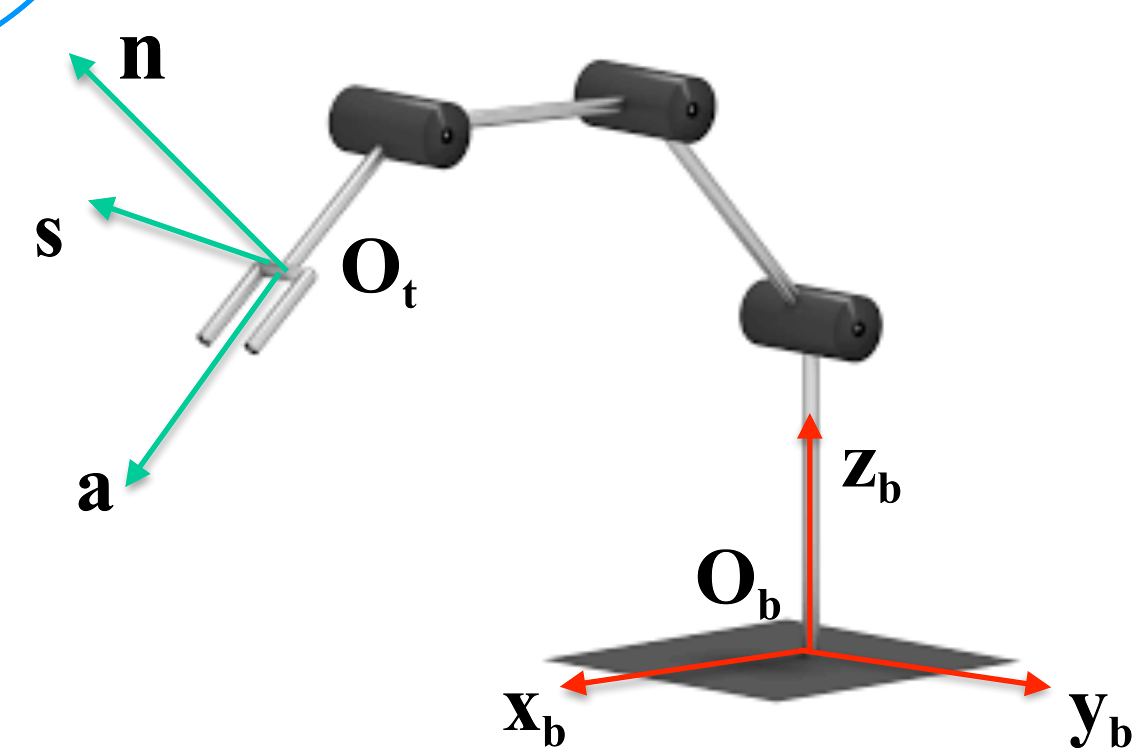
$$A_t^b = \left[\begin{array}{ccc:c} & R_t^b & & \underline{o}_t^b \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right] = \left[\begin{array}{c:c:c:c} \underline{n}^b & \underline{s}^b & \underline{a}^b & \underline{o}_t^b \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$



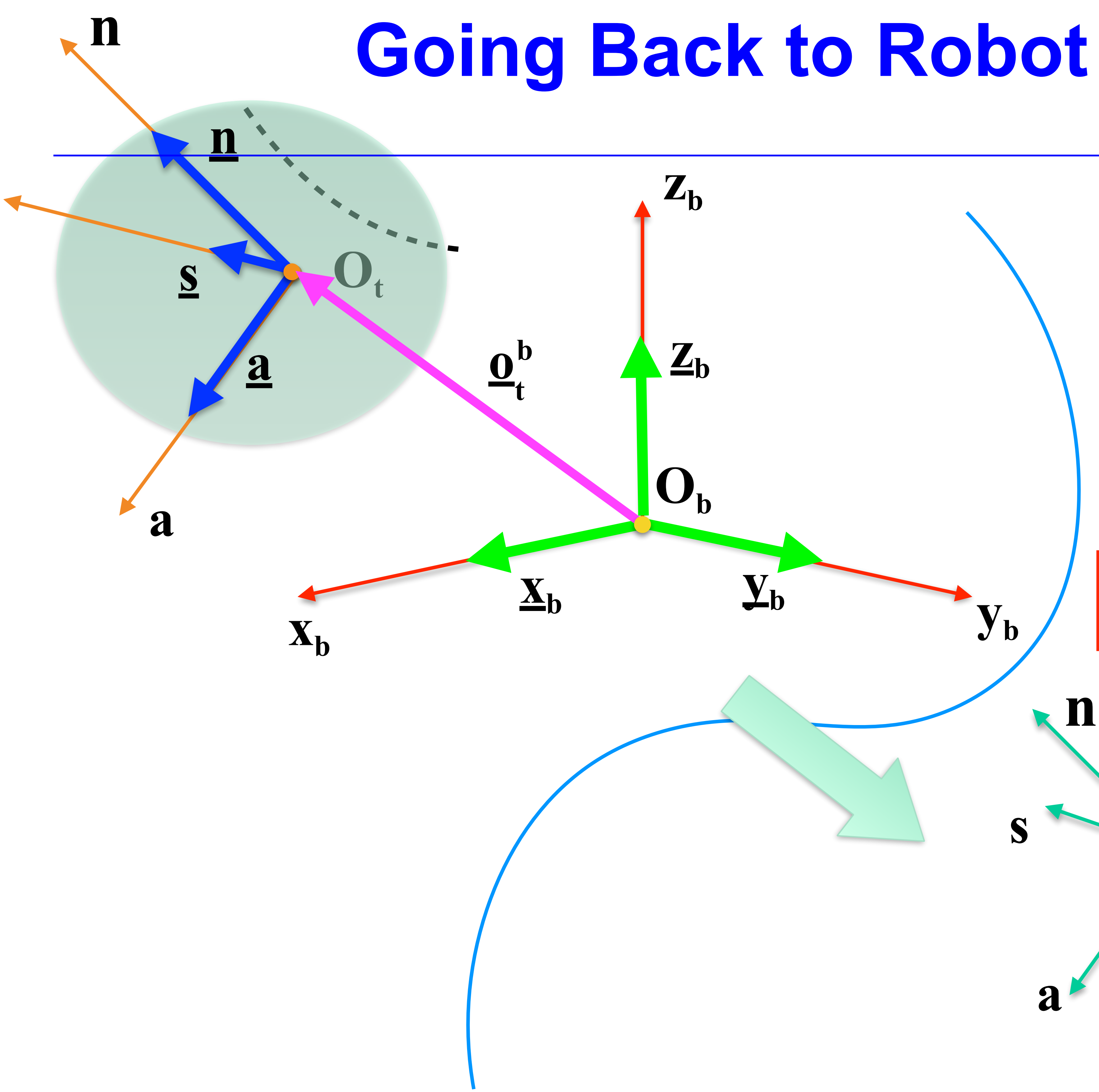
Going Back to Robot Manipulators



Description of the Pose of the End-Effector w.r.t. the Base Reference Frame



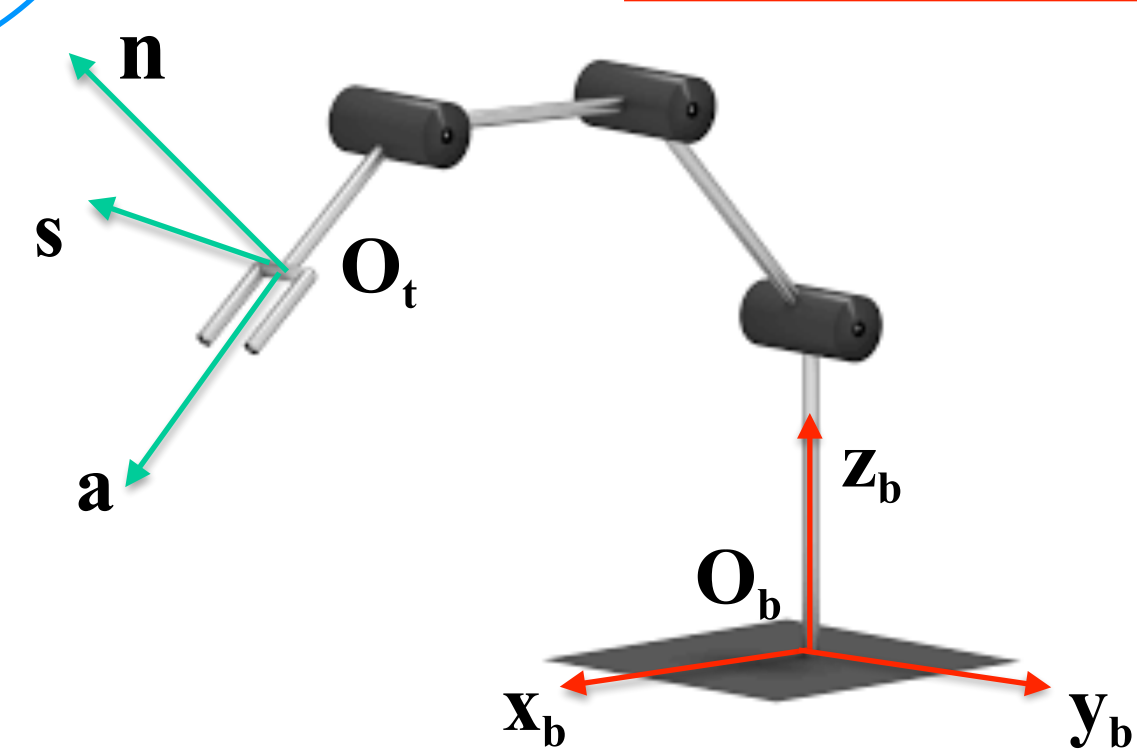
Going Back to Robot Manipulators



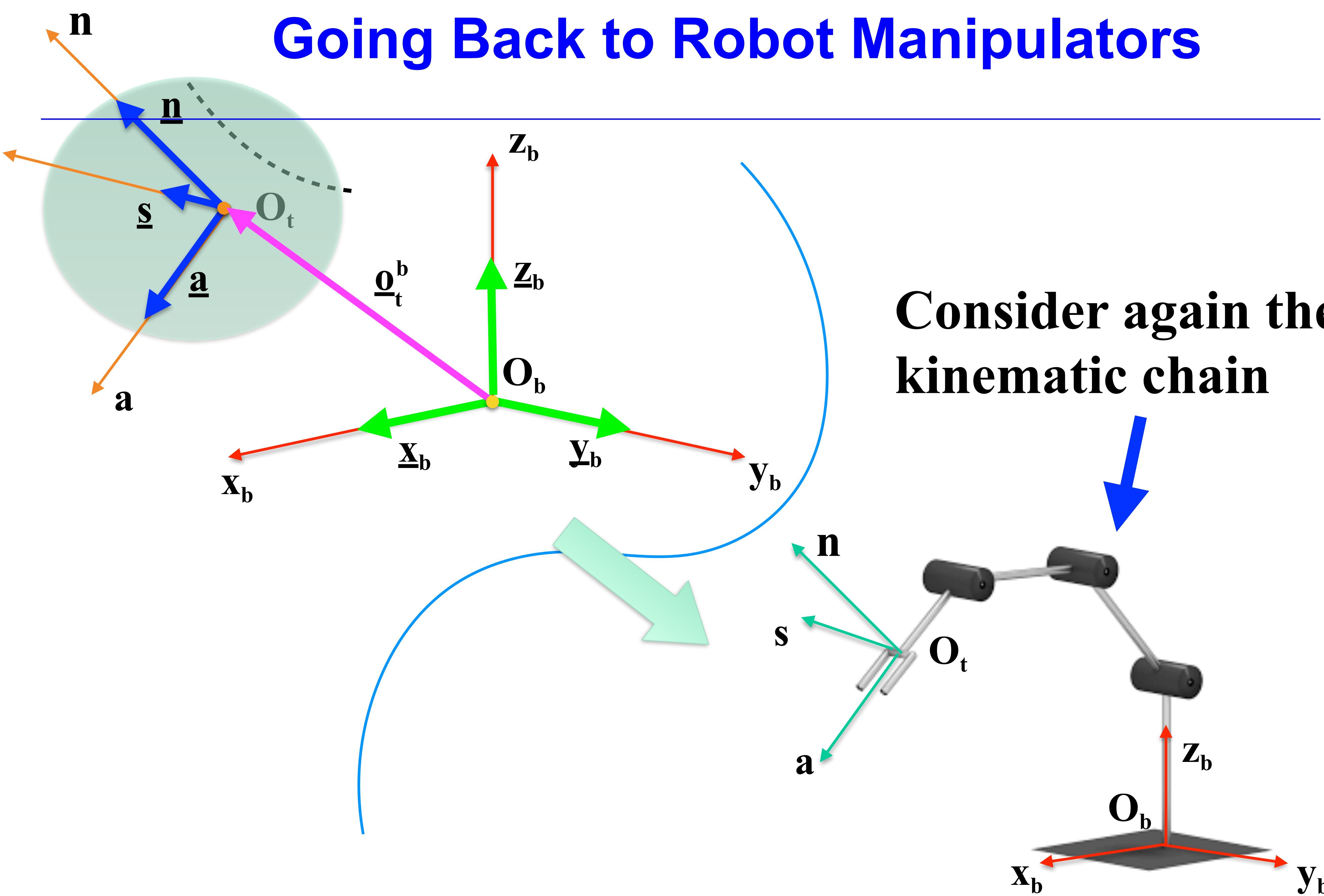
Description of the <u>Pose</u> <u>of the End-Effector</u> w.r.t. the Base Reference Frame

Position

Orientation



Going Back to Robot Manipulators

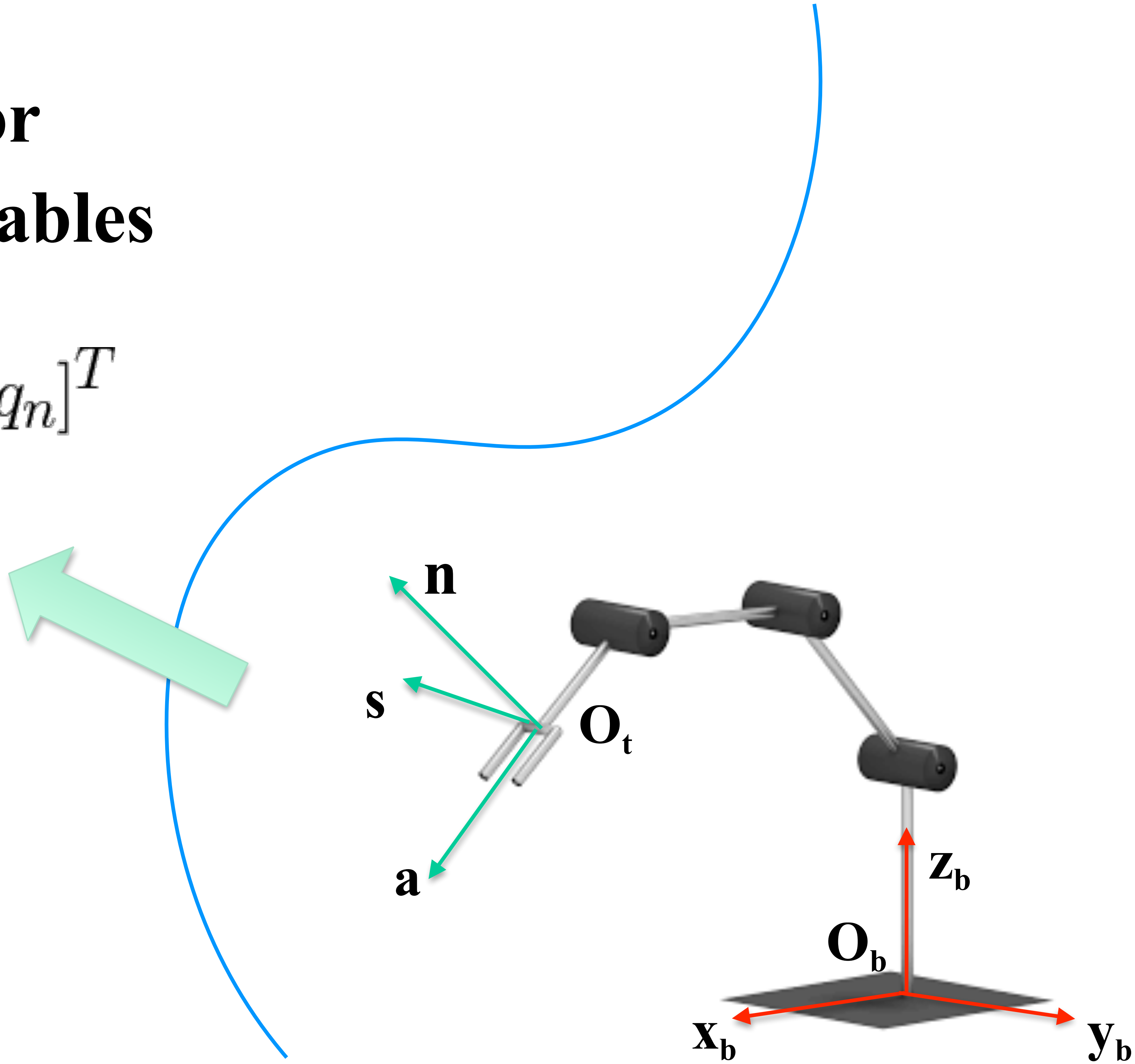


Consider again the kinematic chain

Going Back to Robot Manipulators

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2,\, \ldots,\, q_n]^T$$

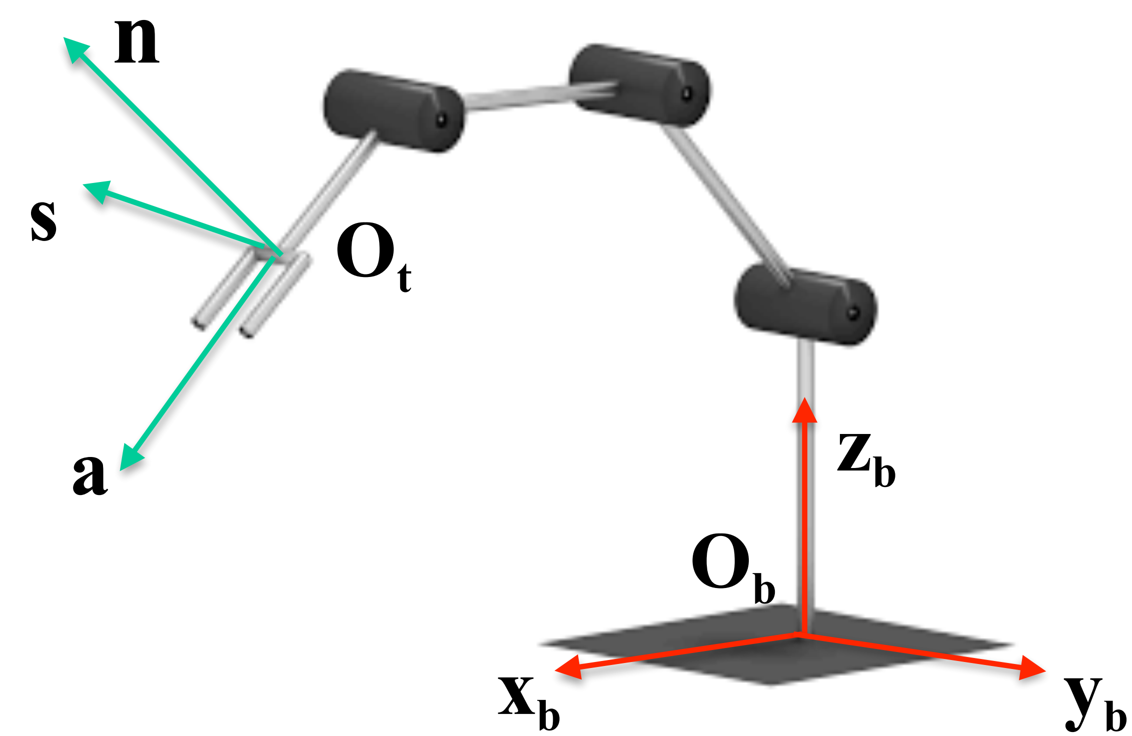


Going Back to Robot Manipulators

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2,\, \ldots,\, q_n]^T$$

$$\underline{q} = [q_1,\, q_2,\, q_3]^T = [\theta_1,\, \theta_2,\, \theta_3]^T$$

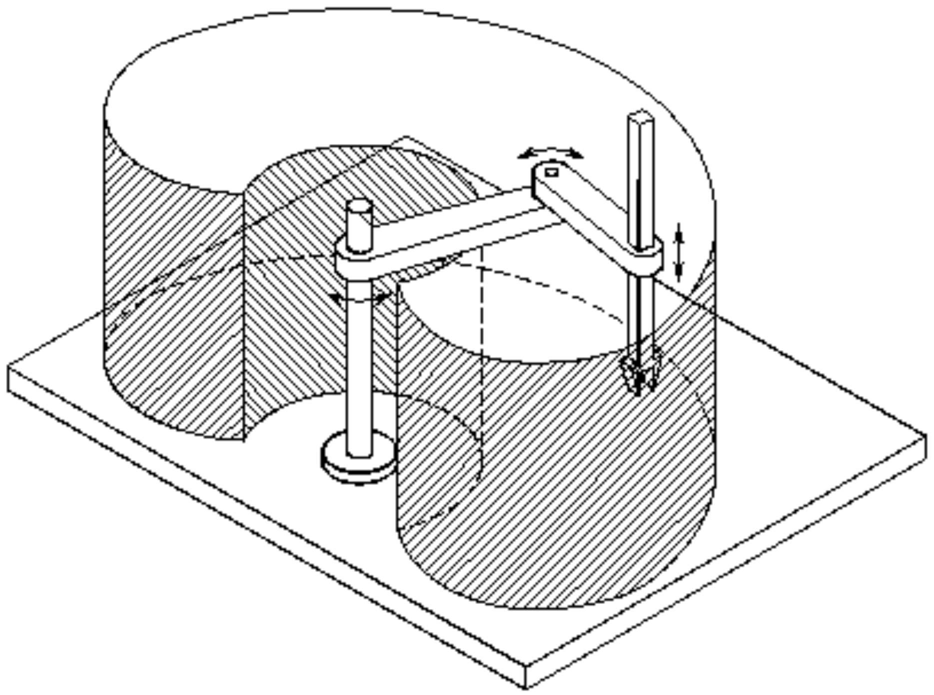


Going Back to Robot Manipulators

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2,\, \ldots,\, q_n]^T$$

$$\underline{q} = [q_1,\, q_2,\, q_3]^T = [\theta_1,\, \theta_2,\, x_3]^T$$

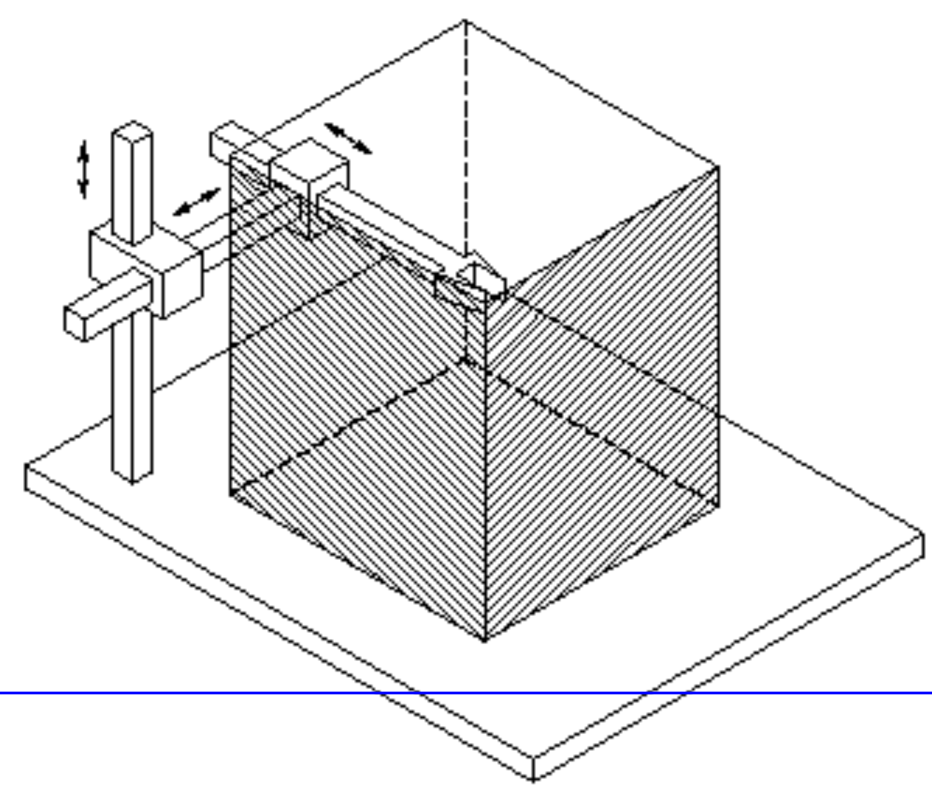


Going Back to Robot Manipulators

Define the vector of the joint variables

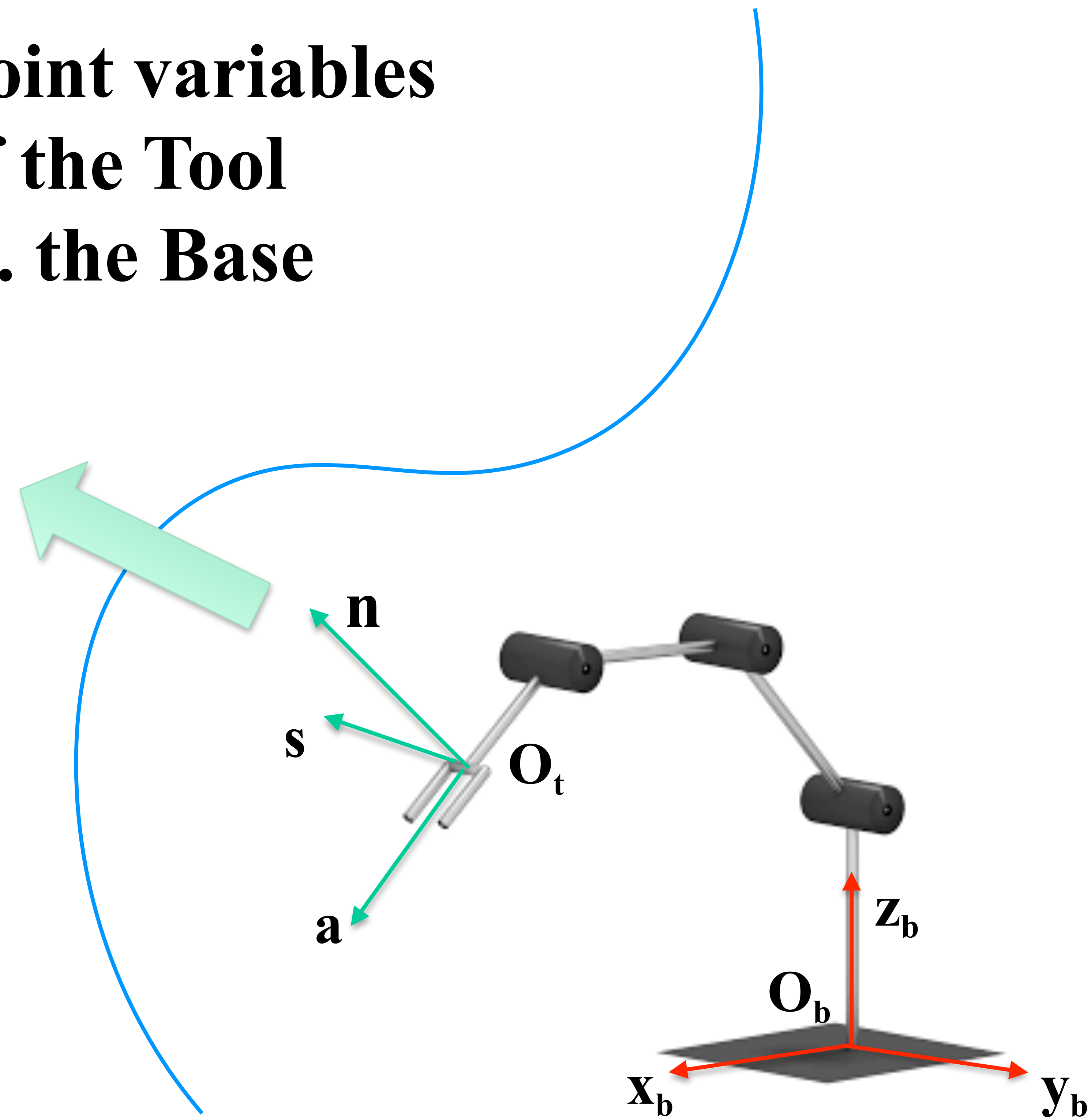
$$\underline{q} = [q_1,\, q_2,\, \ldots,\, q_n]^T$$

$$\underline{q} = [q_1,\, q_2,\, q_3]^T = [x_1,\, x_2,\, x_3]^T$$



Going Back to Robot Manipulators

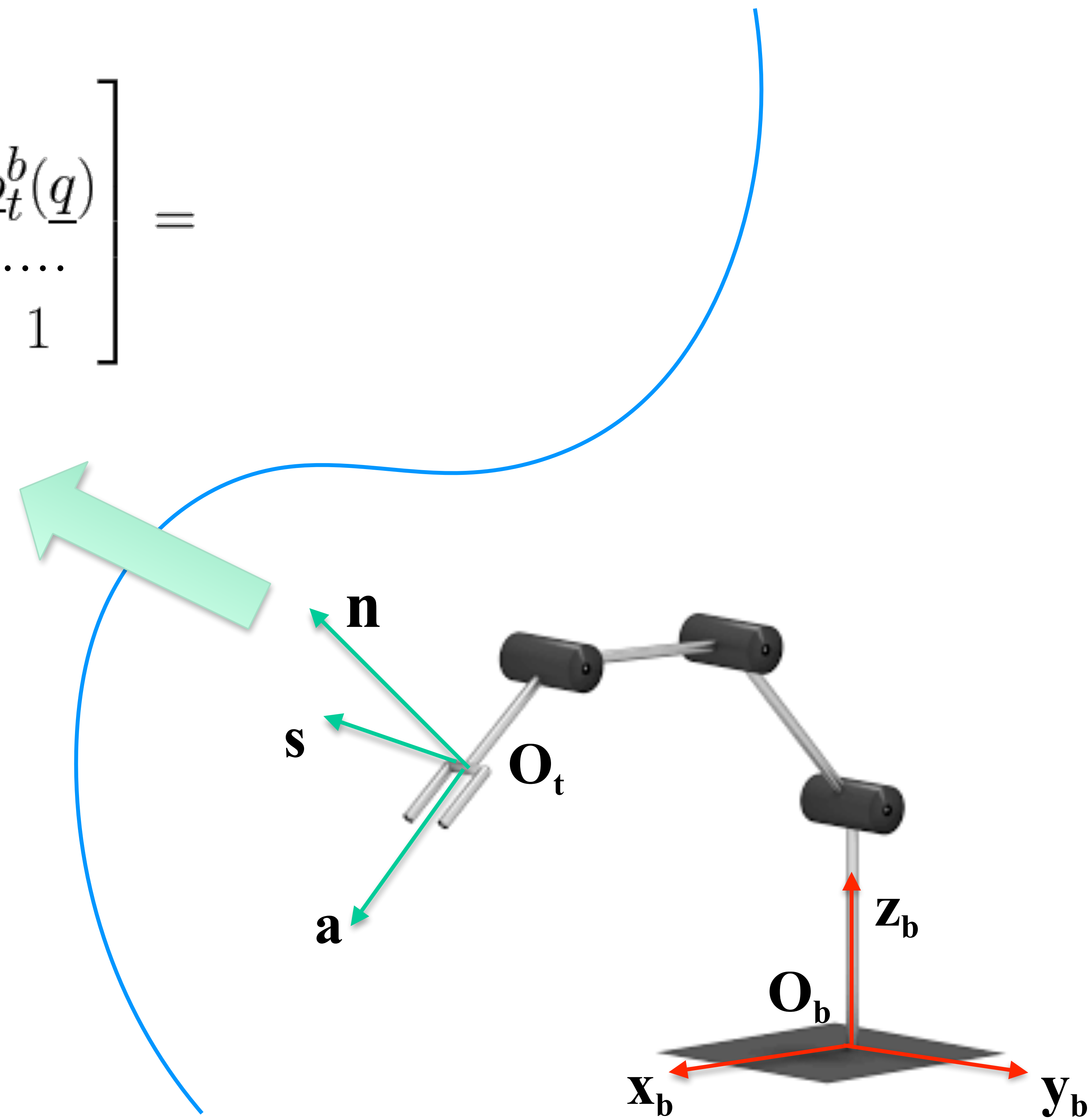
Use the vector of the joint variables to describe the Pose of the Tool Reference Frame w.r.t. the Base Reference Frame



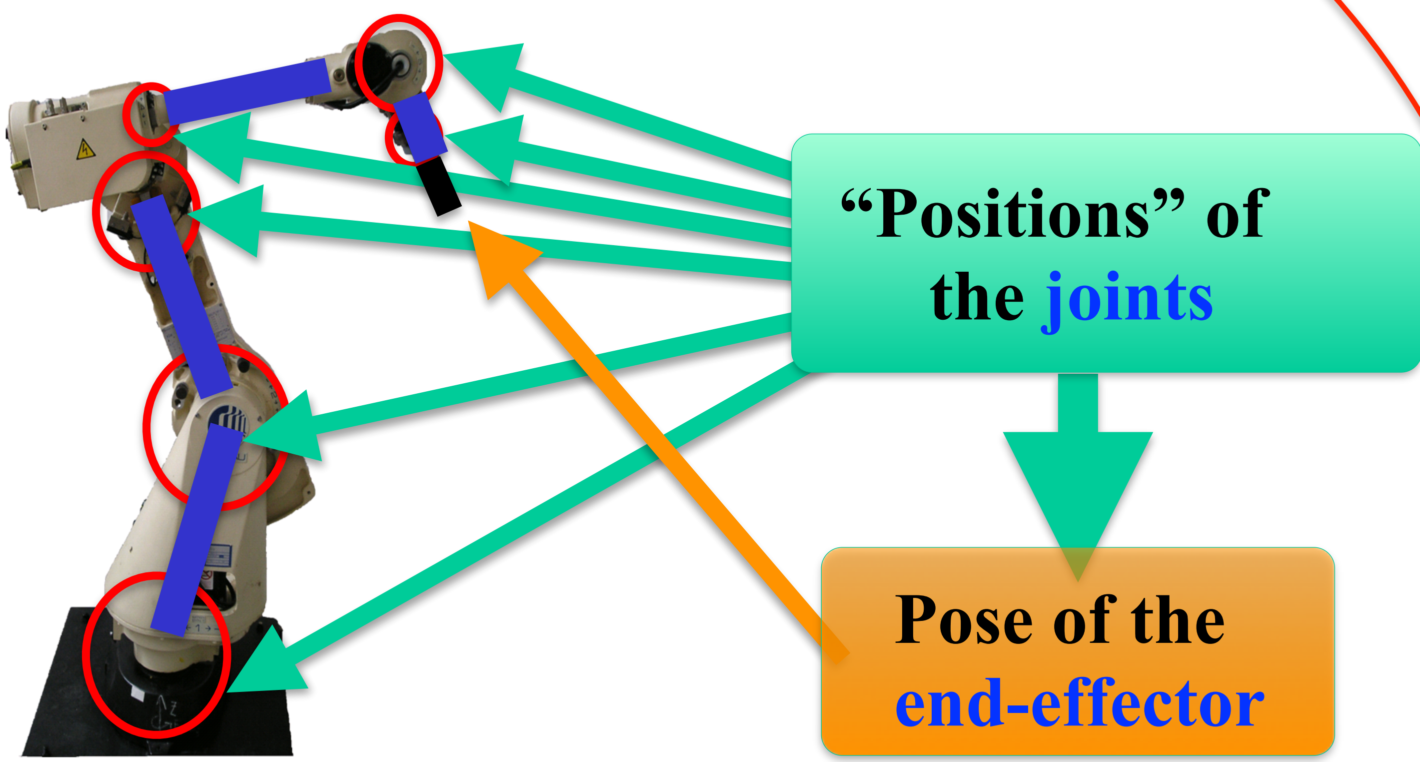
Going Back to Robot Manipulators

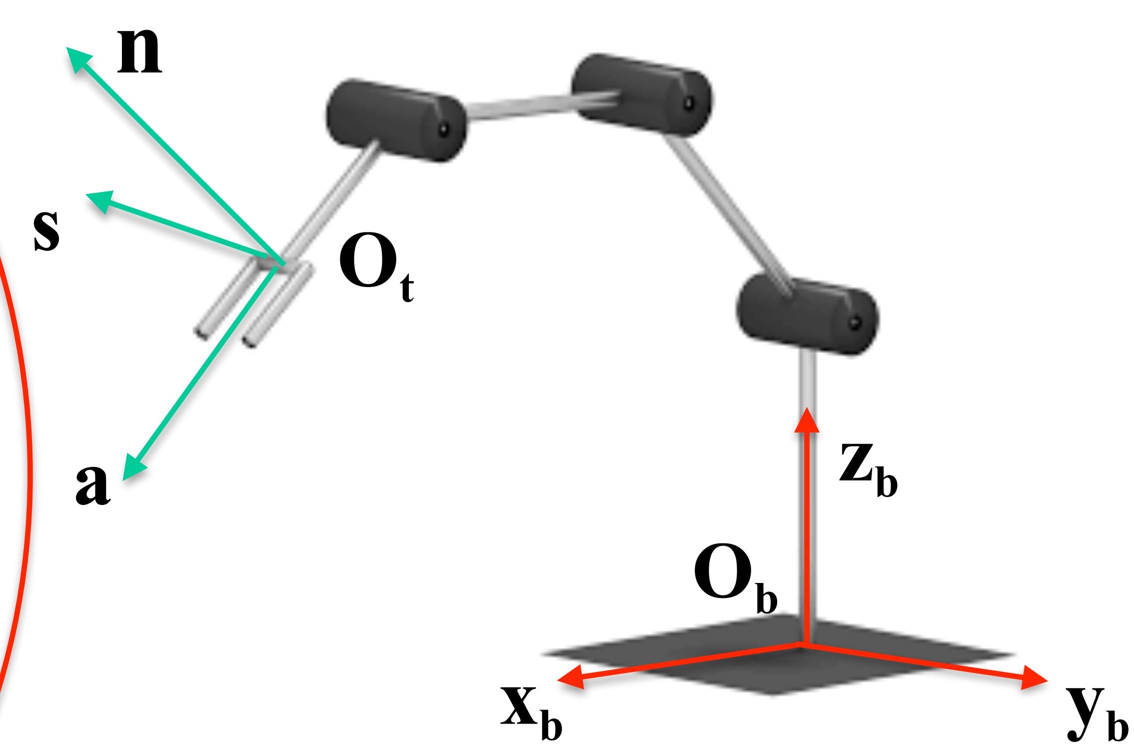
$$A_t^b = T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} & R_t^b(\underline{q}) & & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right] =$$

$$\left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$



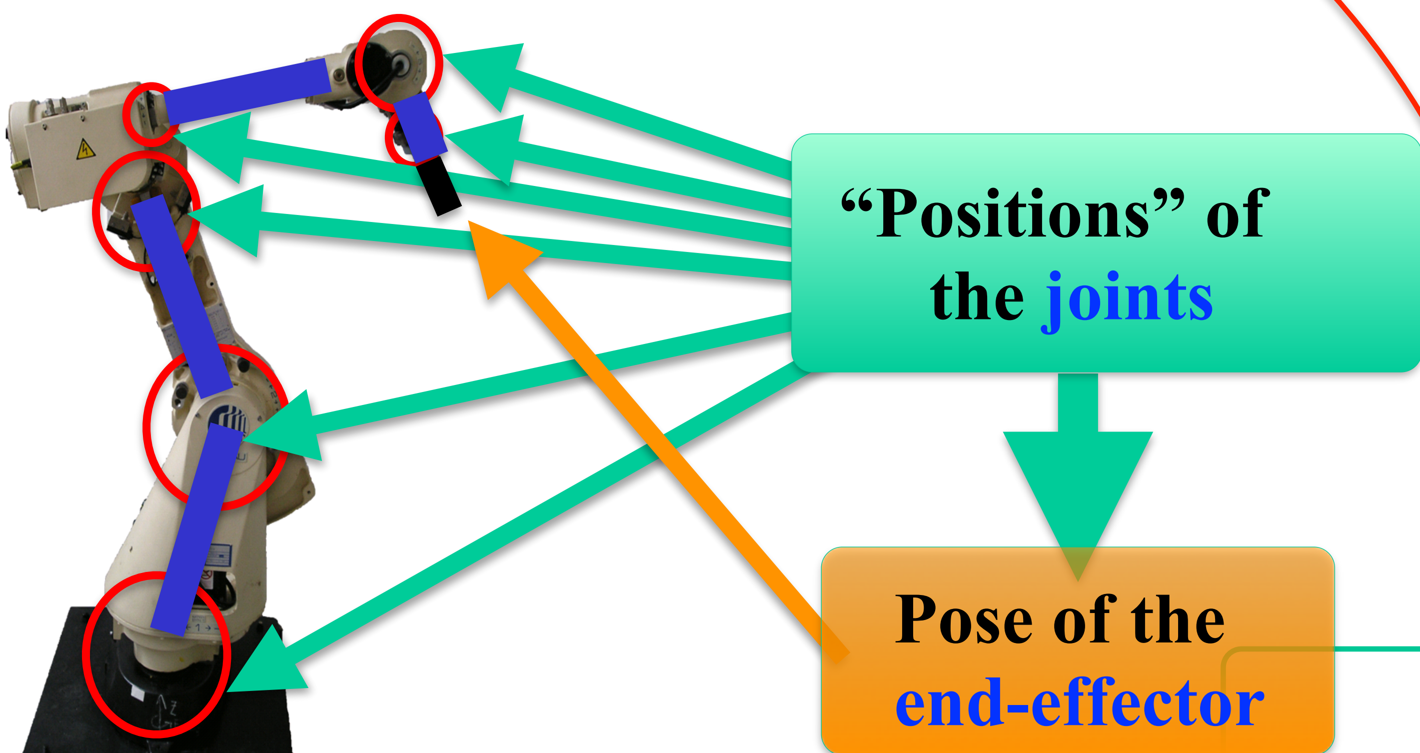
Direct Kinematics Description *(from Lecture 2)*





Find the pose of the Tool Reference Frame w.r.t. the Base Reference Frame

How to Describe the Direct Kinematics



Steps to take:

Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2,\ \ldots,\ q_n]^T$$

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \begin{bmatrix} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Direct Kinematics - Geometric Approach: Summary

How to Describe the Direct Kinematics: Complete Procedure

Preliminary Step:

1. **Select the Base Reference Frame and attach the Tool reference Frame to the End-Effector**

Other Steps to take:

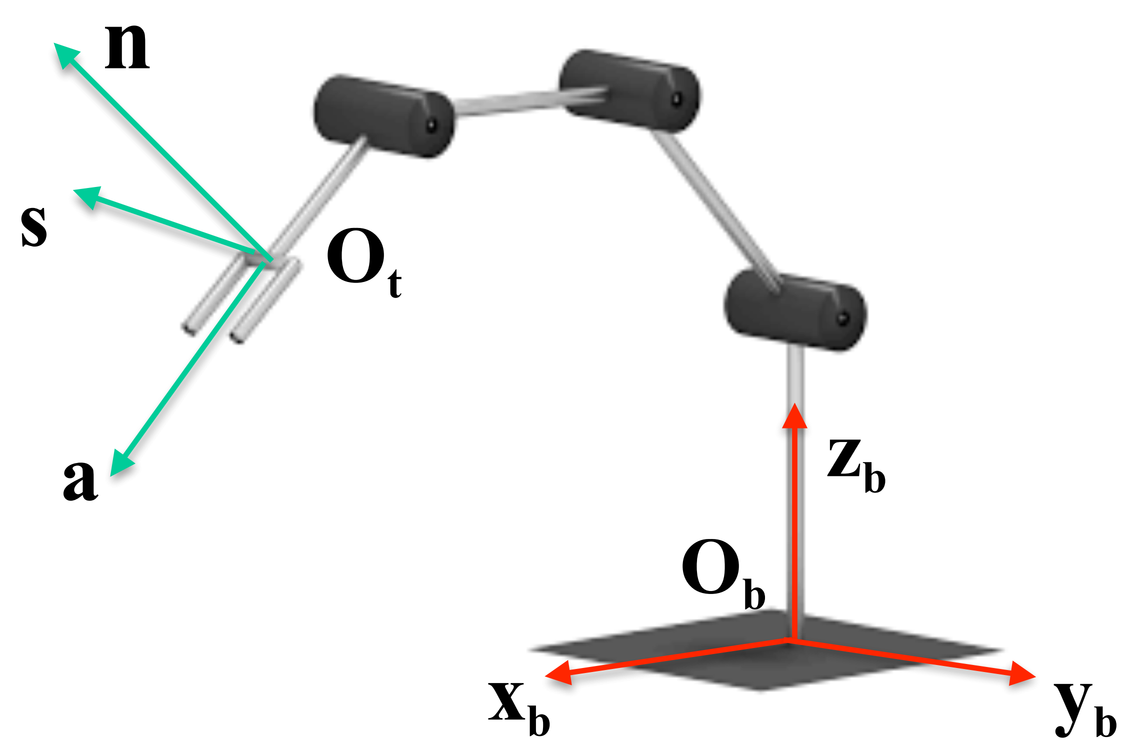
2. **Define the vector of the joint variables**

$$\underline{q} = [q_1,\, q_2,\, \ldots,\, q_n]^T$$

3. **Determine the Homogeneous Transformation Matrix**

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

How to Describe the Direct Kinematics: Complete Procedure



Preliminary Step:

1. Select the Base Reference Frame and attach the Tool reference Frame to the End-Effector

Other Steps to take:

2. Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2,\ \dots,\ q_n]^T$$

3. Determine the Homogeneous Transformation Matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

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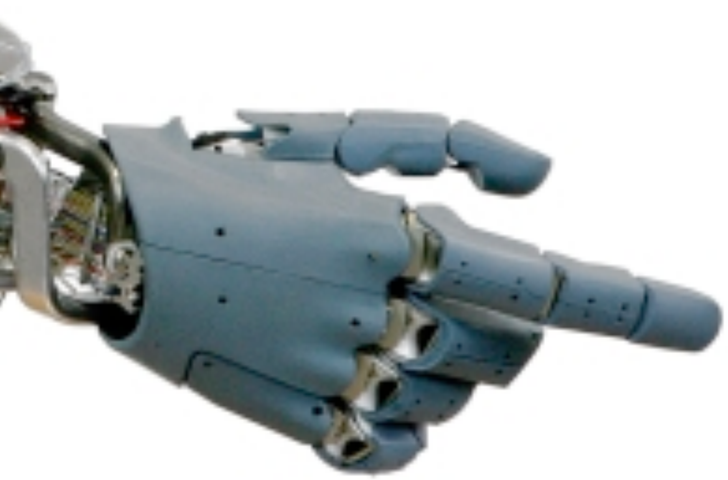


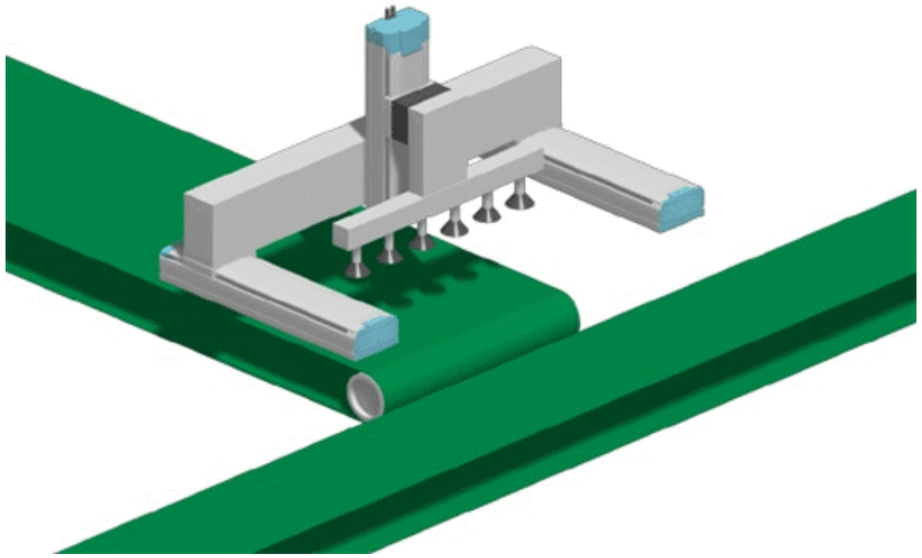
Direct Kinematics - Cartesian Robots

Cartesian Robot Manipulators

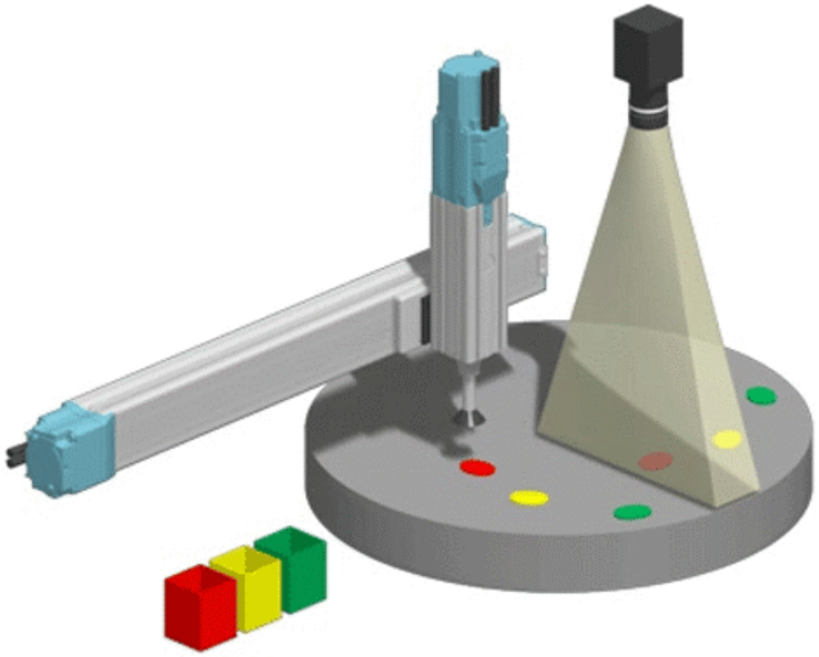
Source: HTE Automation, IAI 3-axes robot

Source: https://flt-us.com, AGR area gantry robot

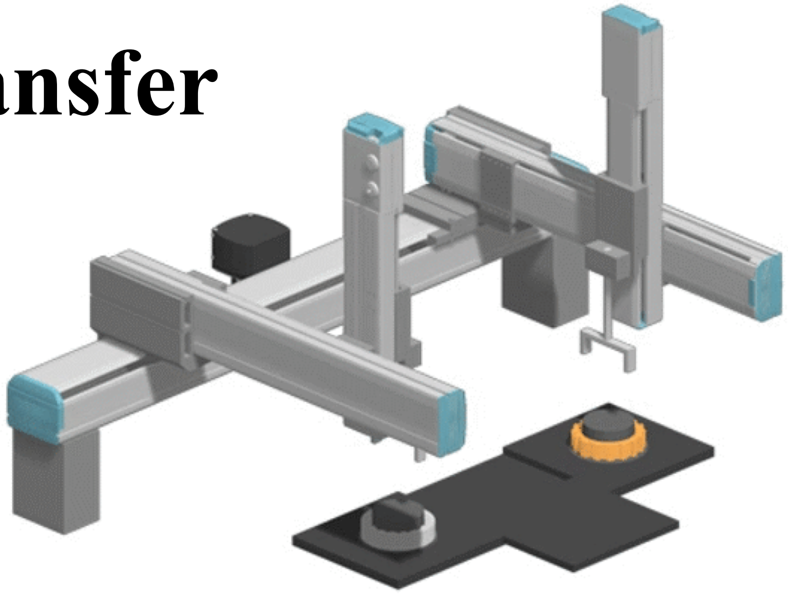
Cartesian Robot Manipulators: Applications



Process-to-process transfer



Pick and place



Part assembly

Source: https://global.yamaha-motor.com/business/robot/lineup/application/xyx/

Cartesian Robot Manipulators: Applications

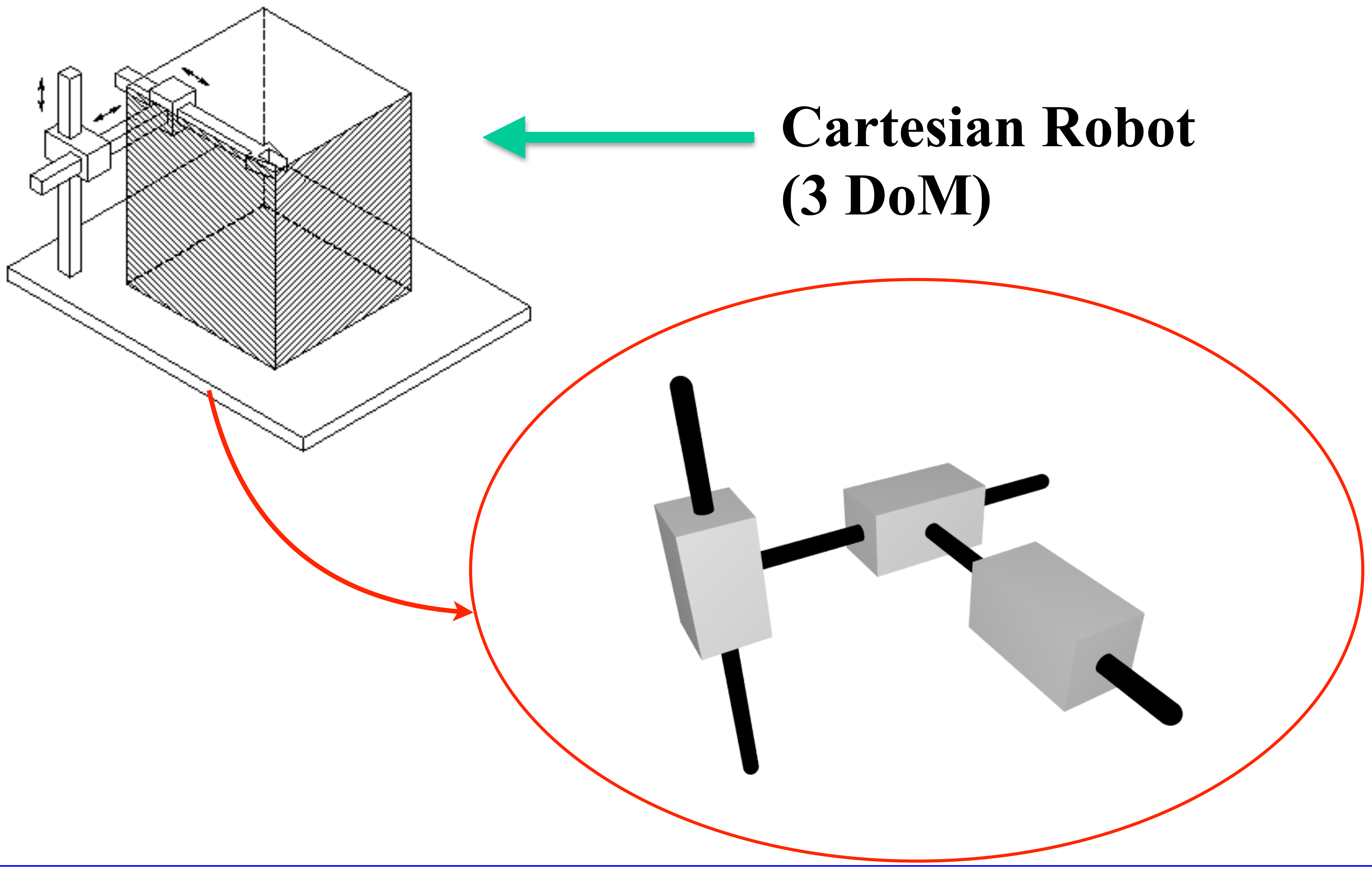
Example: Dispensing Task (Screw pump dispensing)

https://www.youtube.com/watch?v=rnreEviX0ak

Example: Gantry Robot Logistics Application

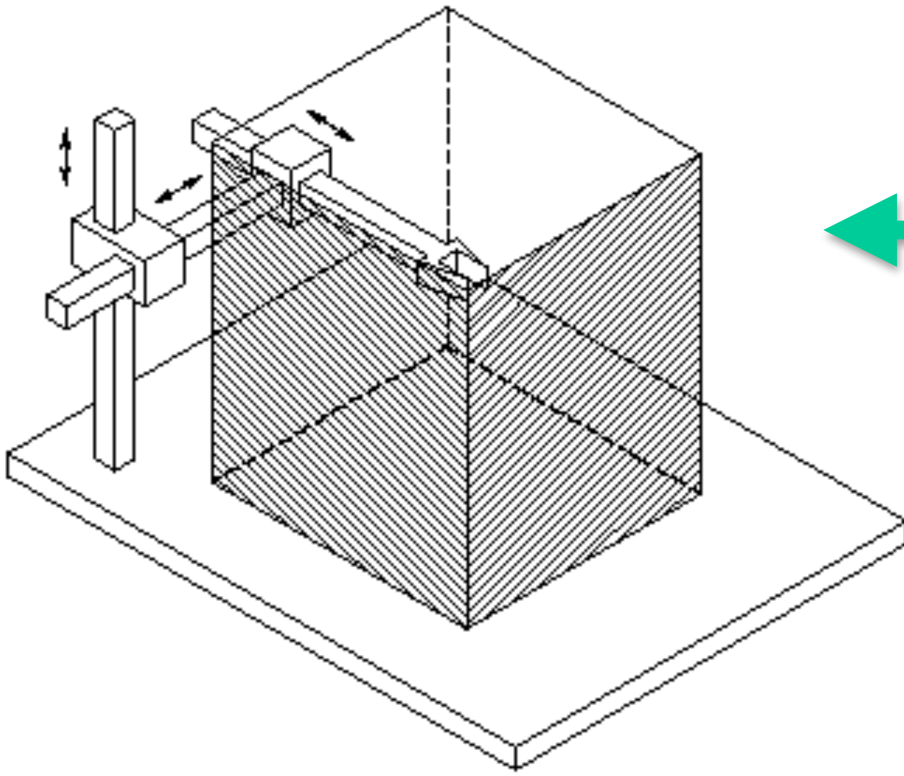
https://www.youtube.com/watch?v=f17BAu9xqV0

Description of the Direct Kinematics of Cartesian Robot Manipulators

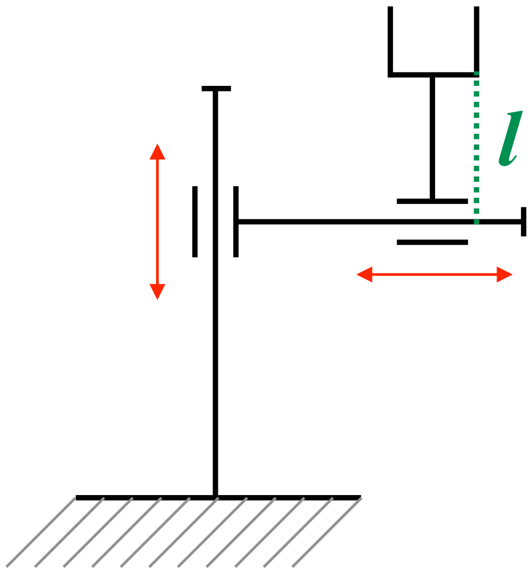


Cartesian Robot
(3 DoM)

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1

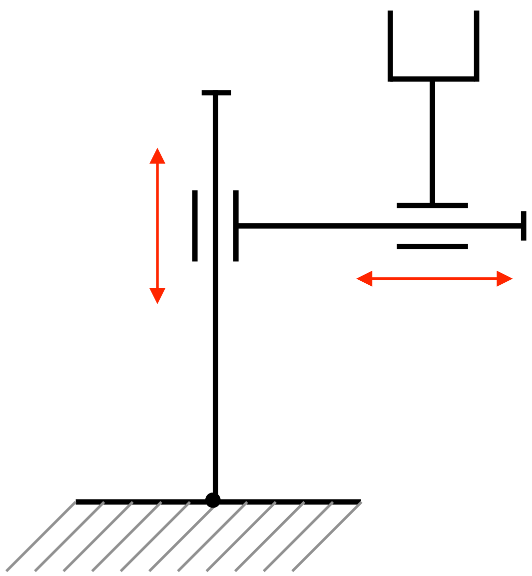


Cartesian Robot (3 DoM)



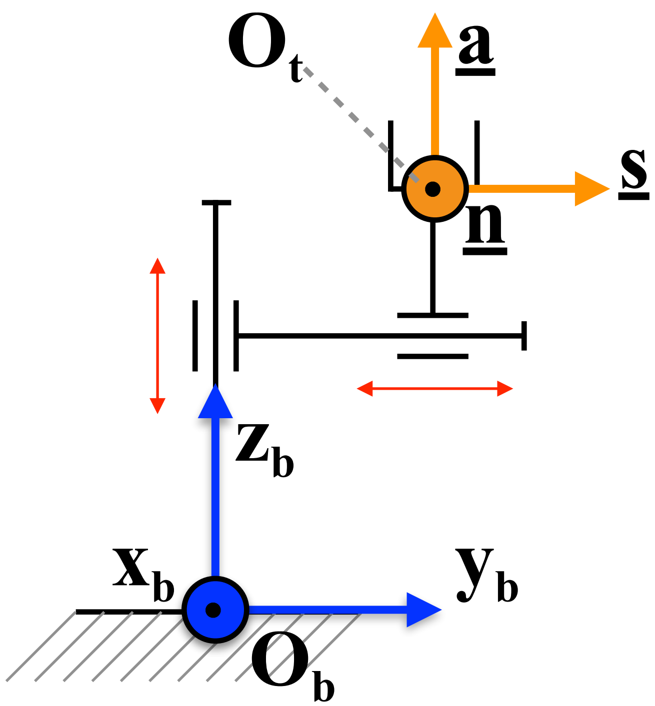
Planar Cartesian Robot (2 DoM)

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1



Assign the Base Reference Frame and the Tool Reference

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1



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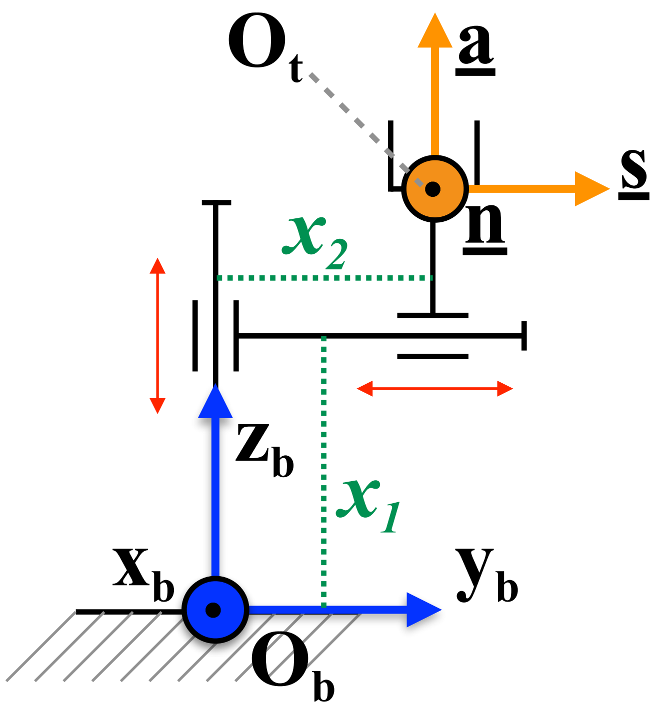
Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1

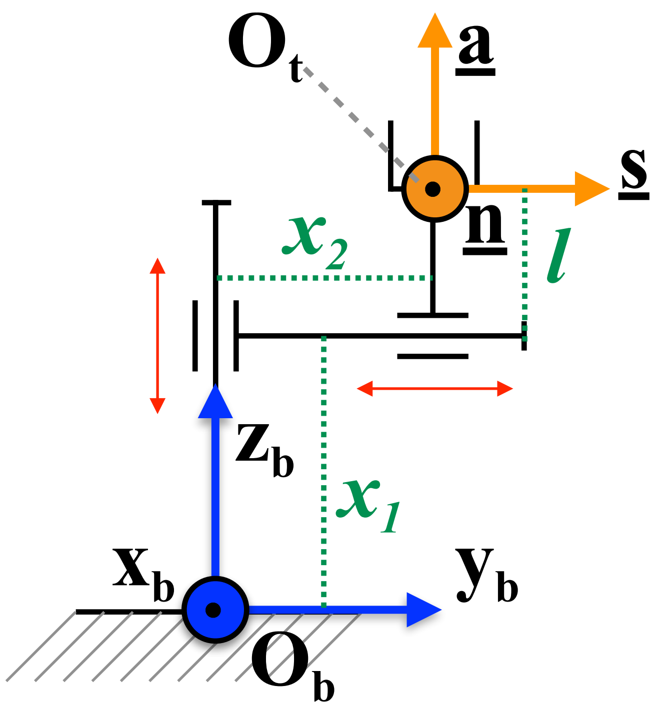


Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2]^T = [x_1,\, x_2]^T$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1



Assign the Base Reference Frame and the Tool Reference

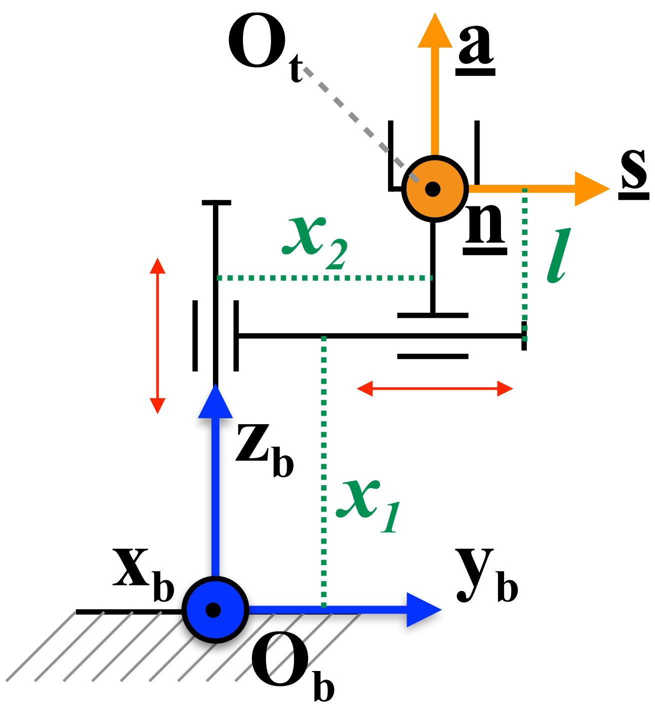
Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2]^T = [x_1,\ x_2]^T$$

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1



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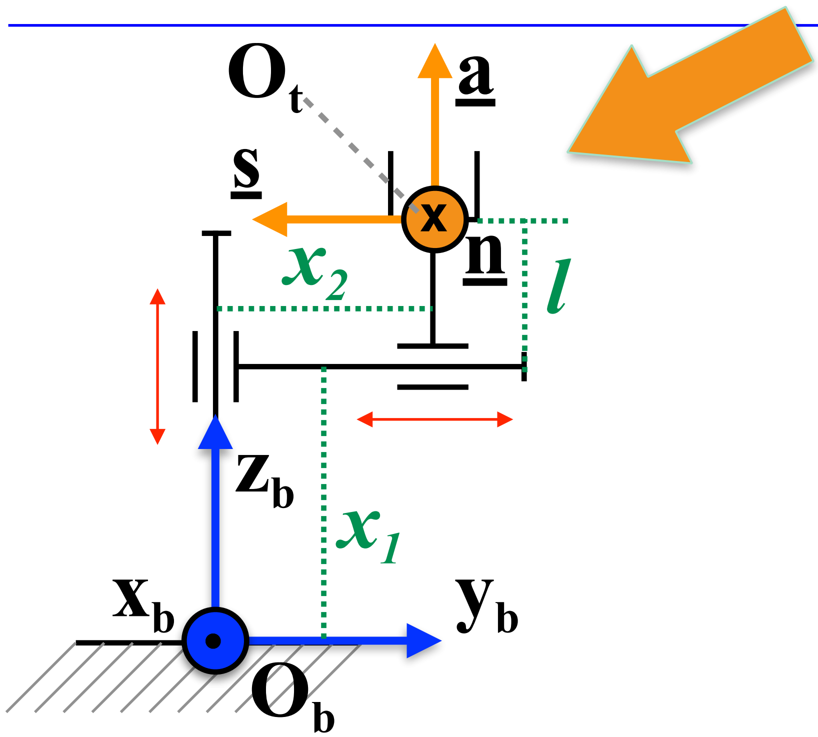
$$T_t^b(\underline{q}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & x_2 \\ 0 & 0 & 1 & x_1 + l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 1



Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

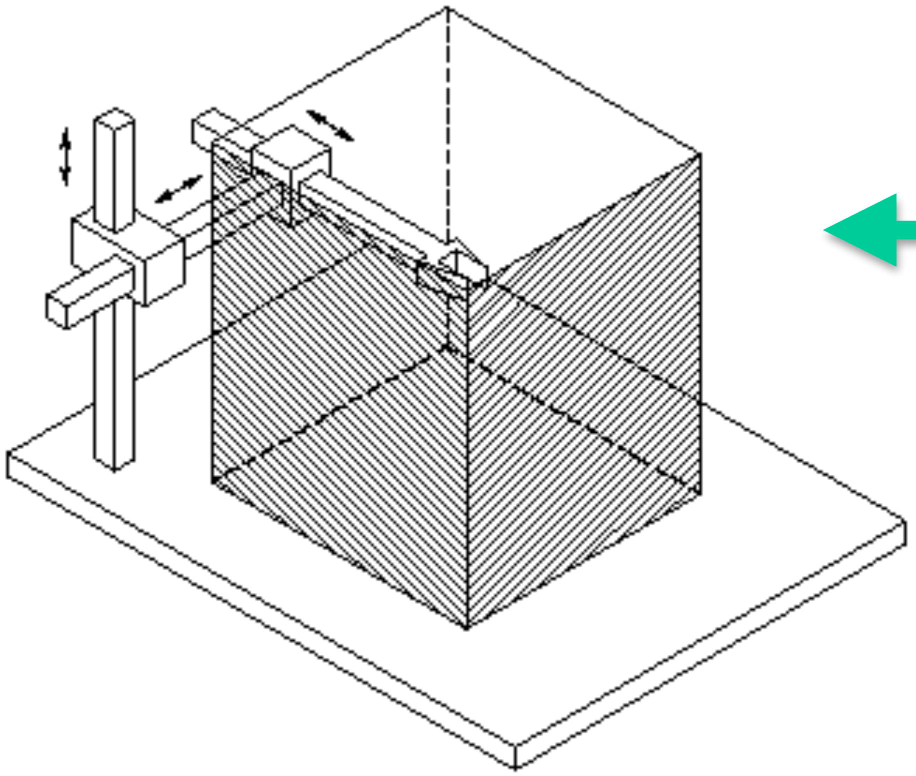
$$\underline{q} = [q_1,\ q_2]^T = [x_1,\ x_2]^T$$

Determine the homogeneous transformation matrix

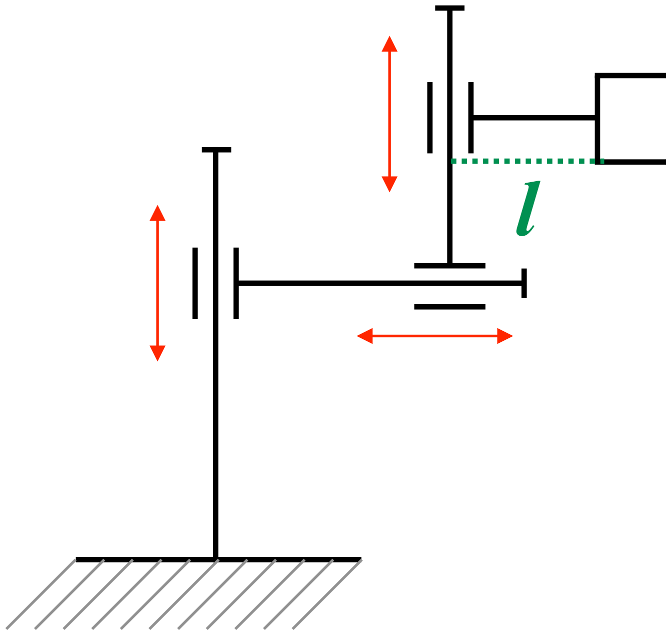
$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

$$T_t^b(\underline{q}) = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & x_2 \\ 0 & 0 & 1 & x_1 + l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 2

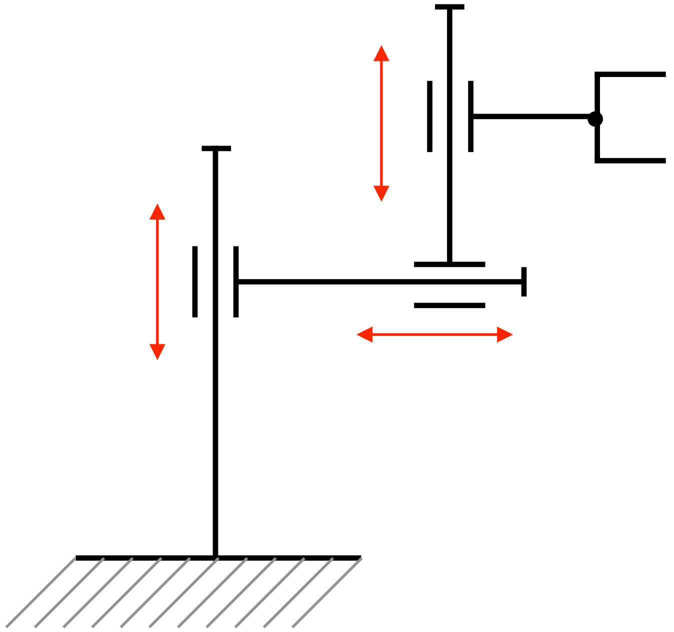


Cartesian Robot (3 DoM)



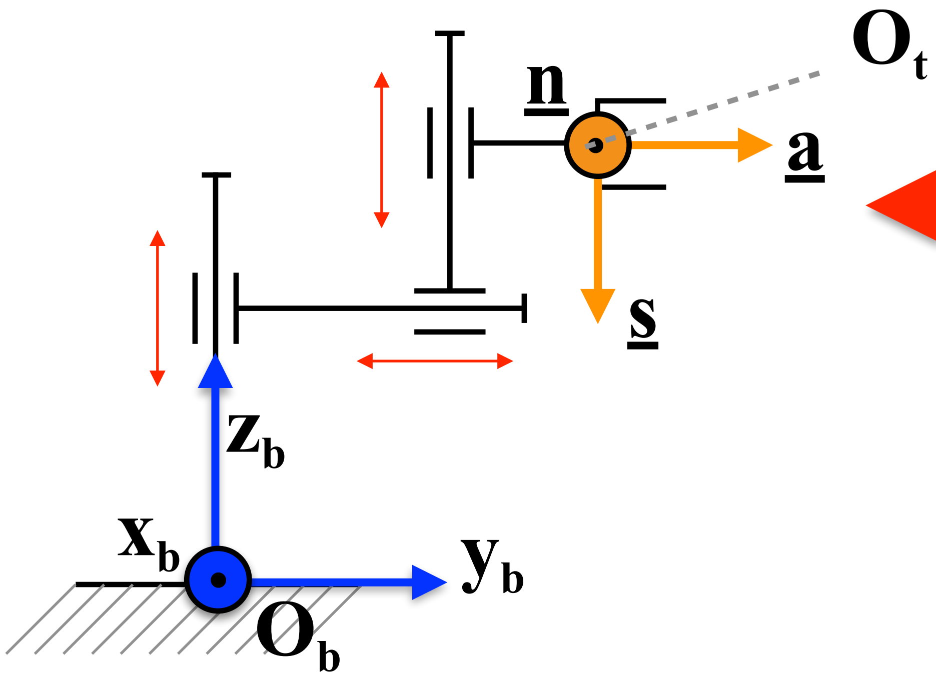
Planar Cartesian Robot (3 DoM)

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 2



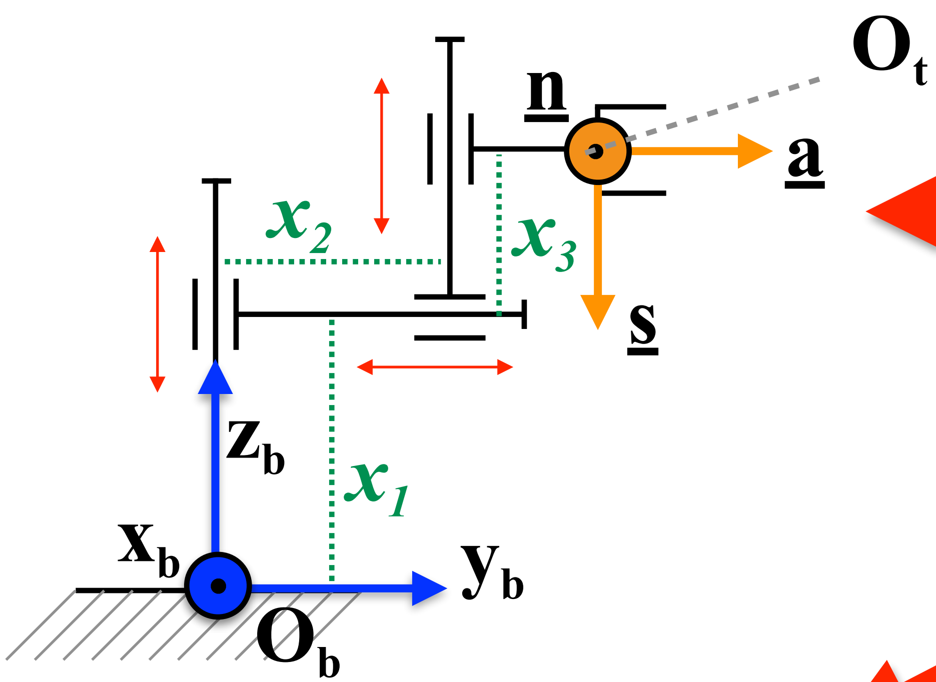
Assign the Base Reference Frame and the Tool Reference

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 2



Assign the Base Reference Frame and the Tool Reference

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 2

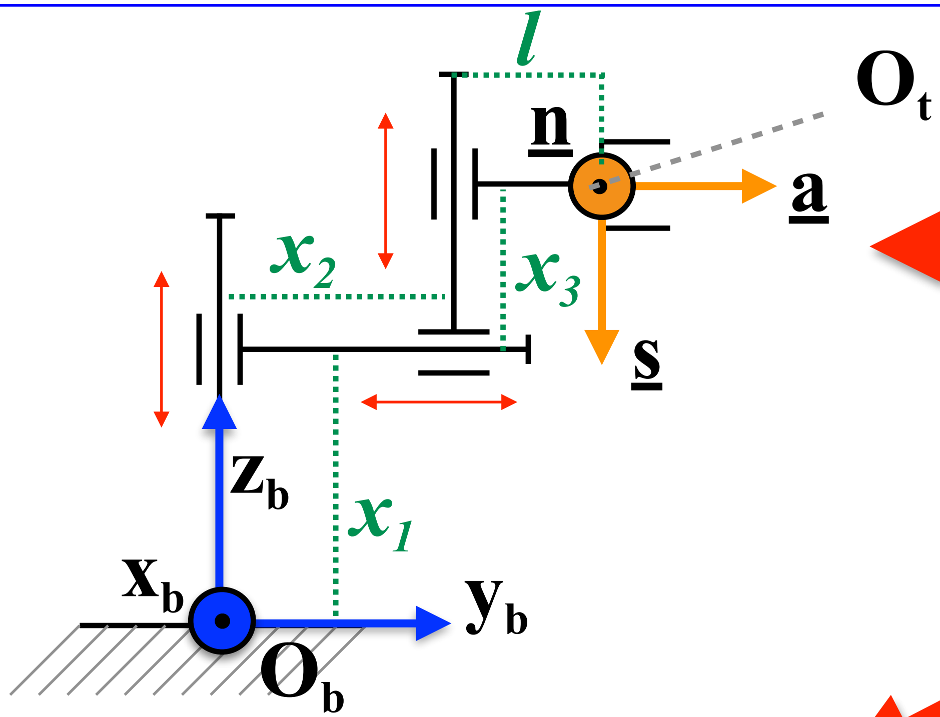


Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2,\ q_3]^T = [x_1,\ x_2,\ x_3]^T$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 2



Assign the Base Reference Frame and the Tool Reference

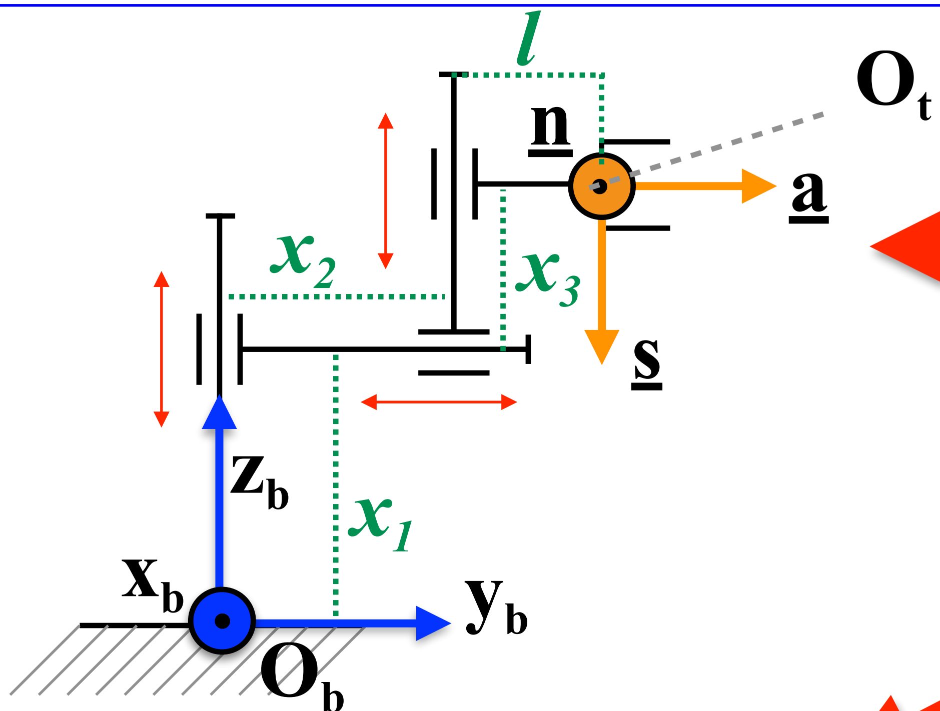
Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2,\ q_3]^T = [x_1,\ x_2,\ x_3]^T$$

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 2



Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

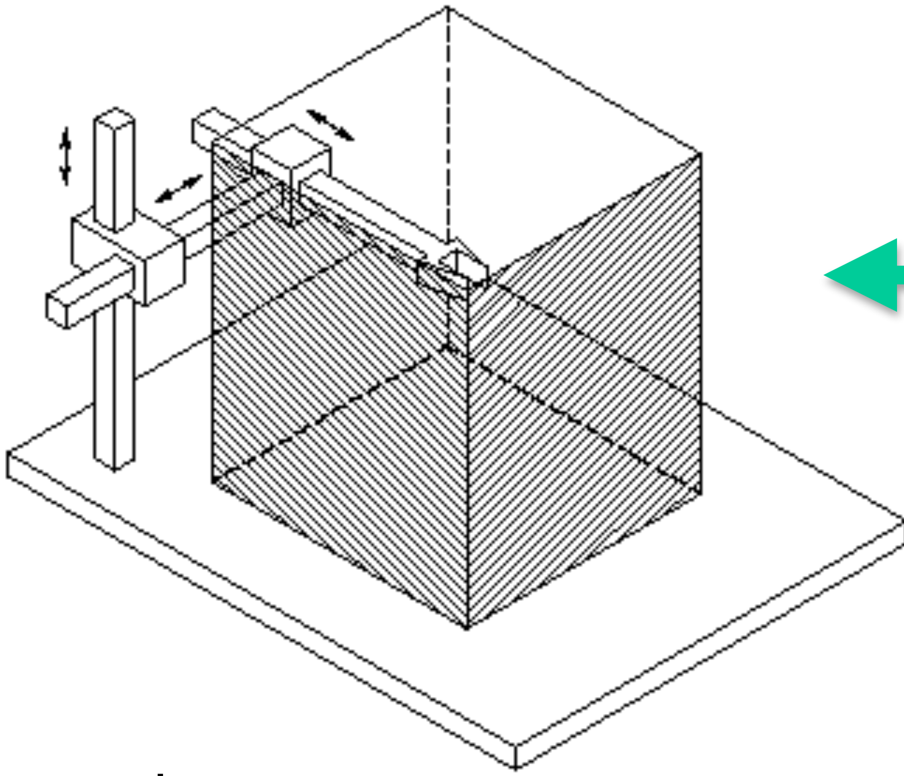
Determine the homogeneous transformation matrix

$$\underline{q} = [q_1,\ q_2,\ q_3]^T = [x_1,\ x_2,\ x_3]^T$$

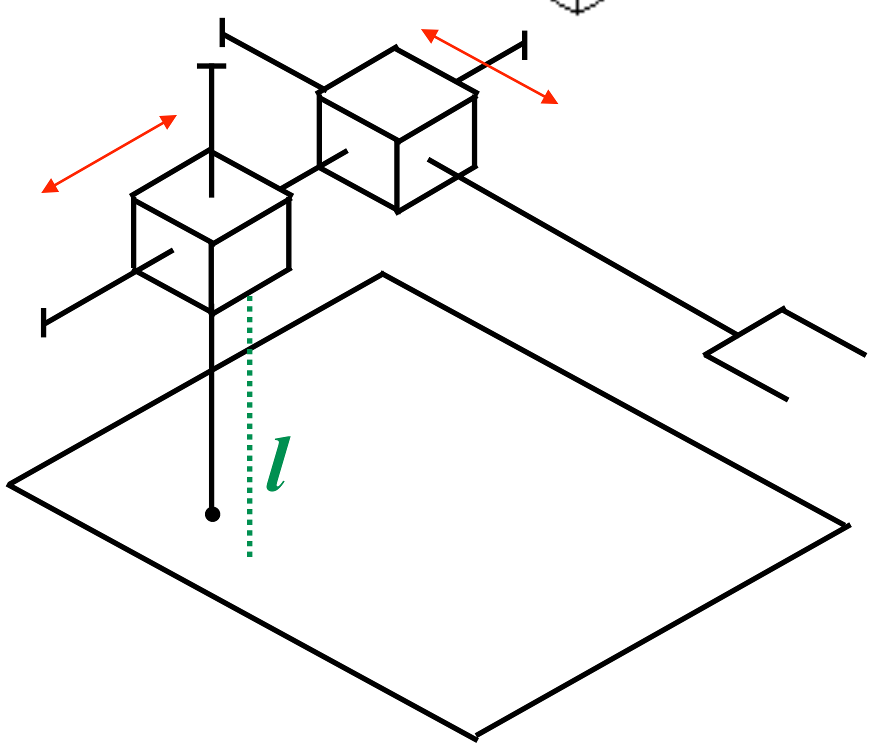
$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

$$T_t^b(\underline{q}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & x_2 + l \\ 0 & -1 & 0 & x_1 + x_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3



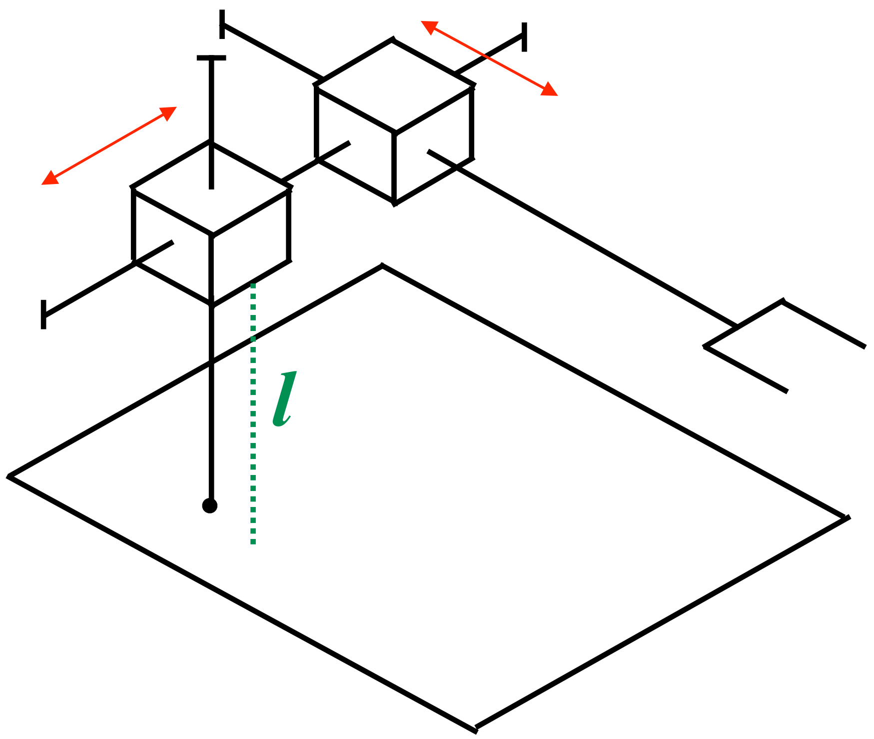
Cartesian Robot (3 DoM)



Cartesian Robot (2 DoM)

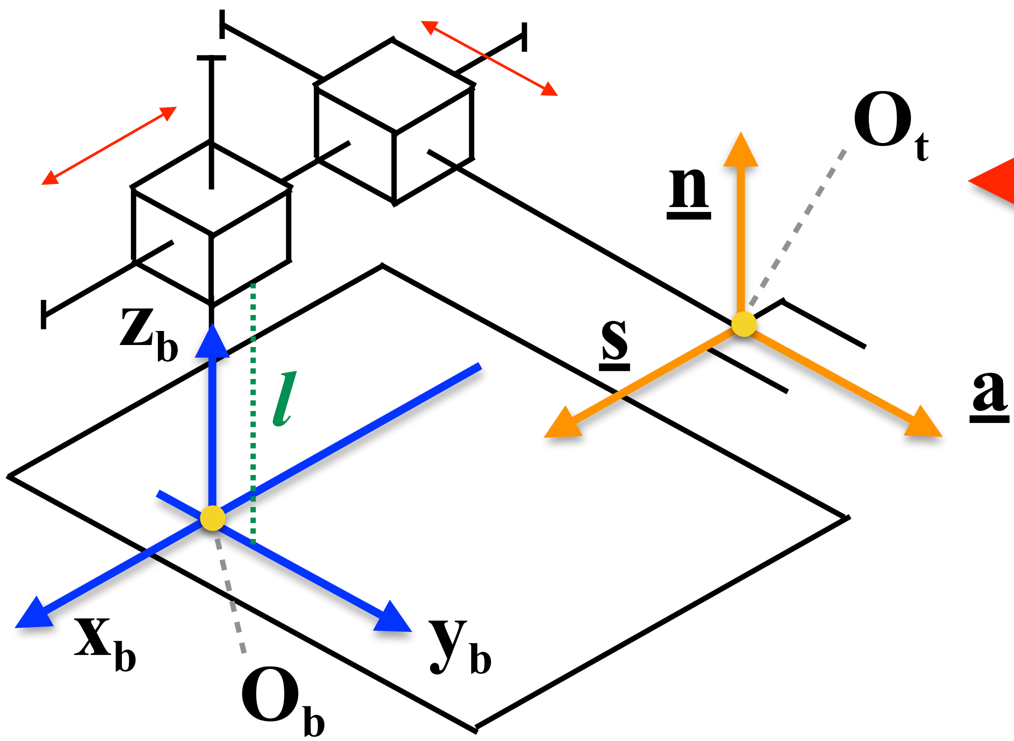
Note: neglect the prismatic joint height, length and depth

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3



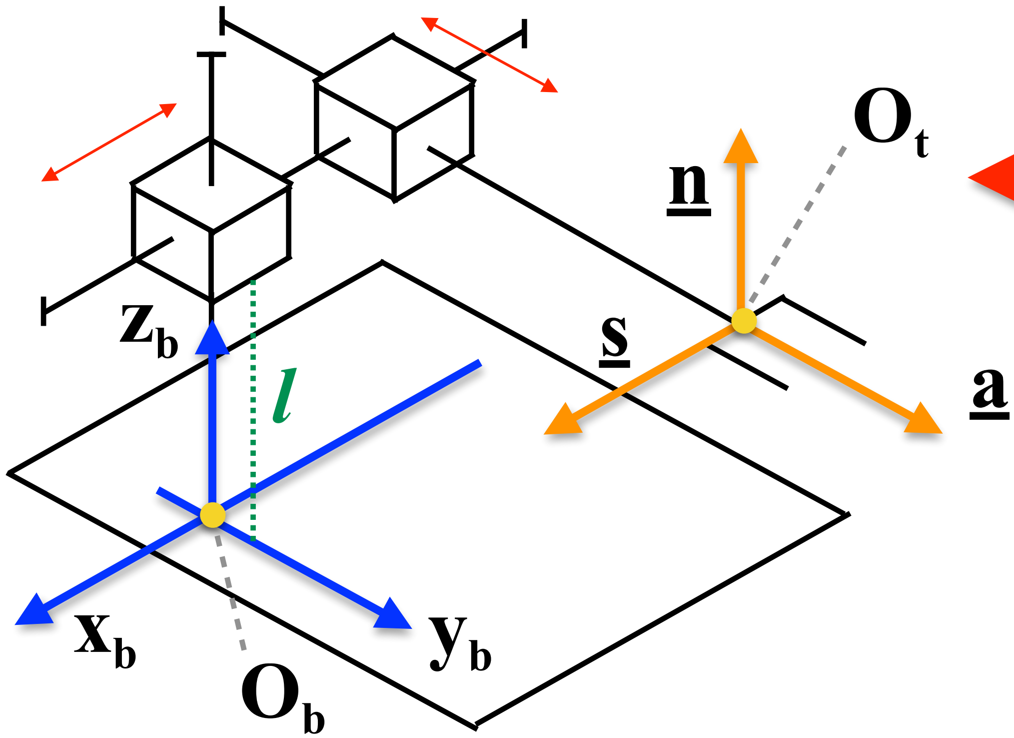
Assign the Base Reference Frame and the Tool Reference

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3



Assign the Base Reference Frame and the Tool Reference

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3

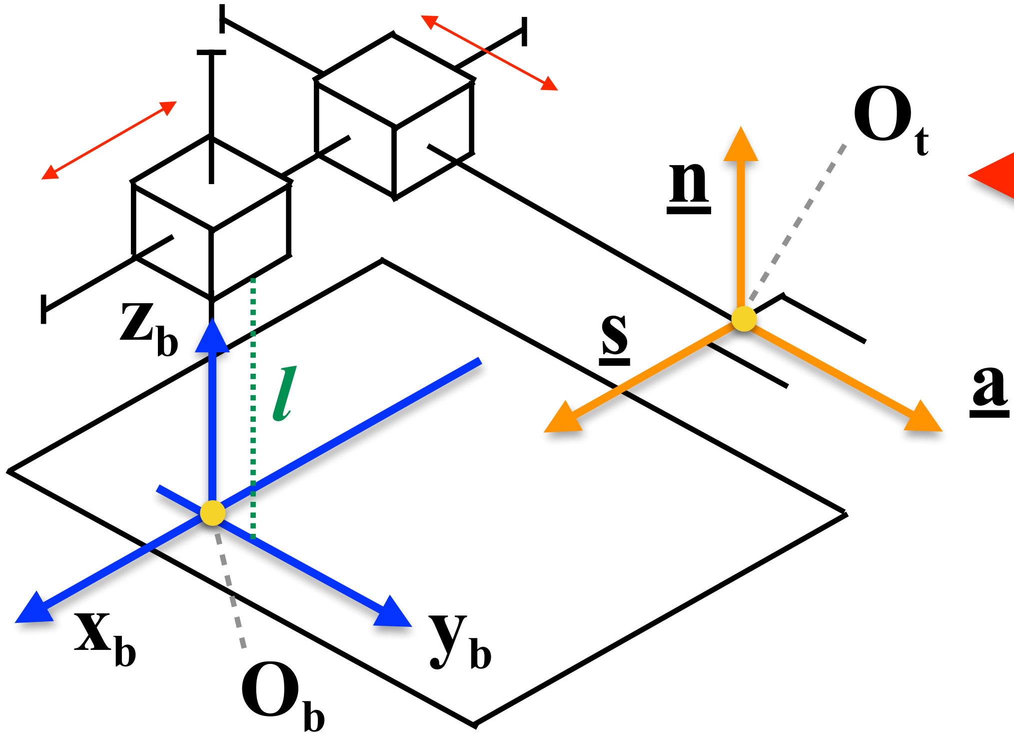


Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2]^T = [x_1,\ x_2]^T$$

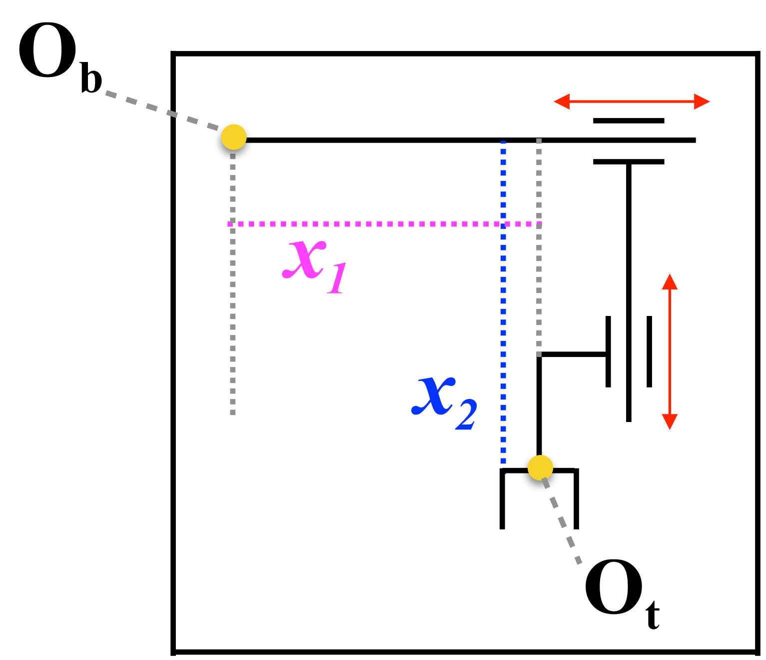
Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3



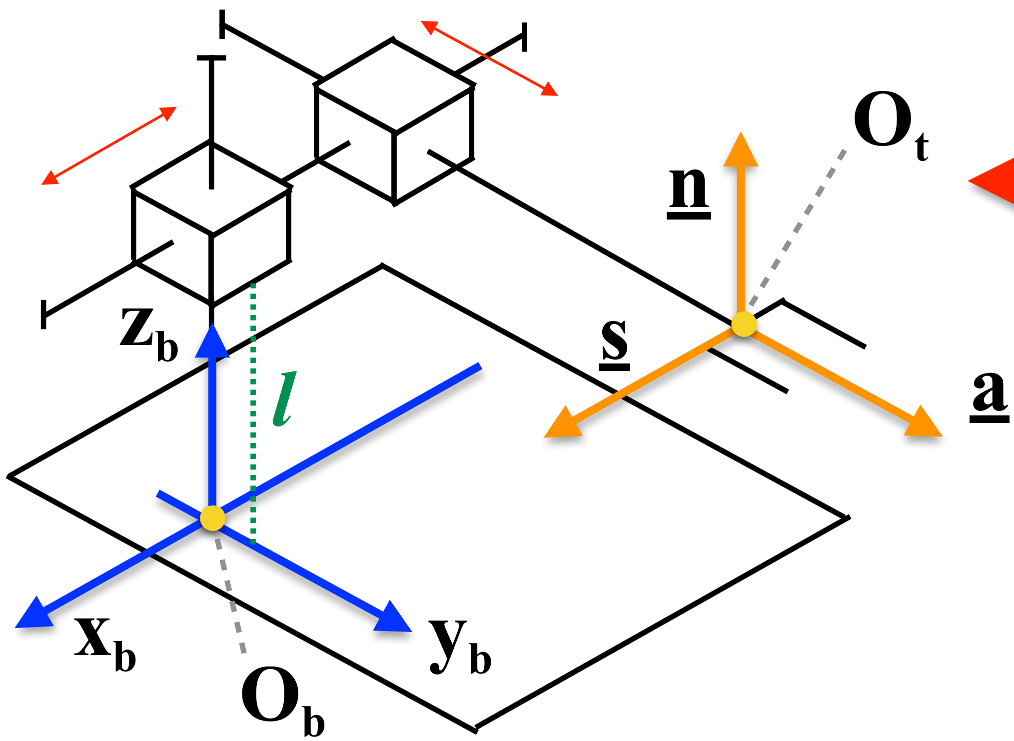
Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2]^T = [x_1,\, x_2]^T$$



Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3



Assign the Base Reference Frame and the Tool Reference

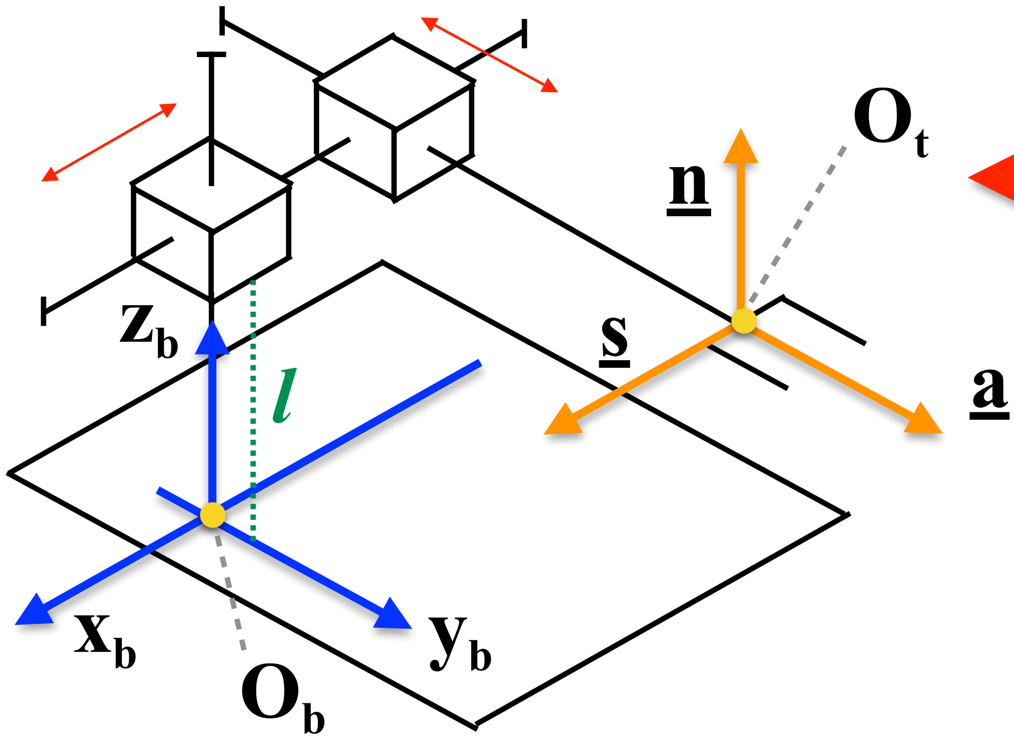
Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2]^T = [x_1,\, x_2]^T$$

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 3



Assign the Base Reference Frame and the Tool Reference

Define the vector of the joint variables

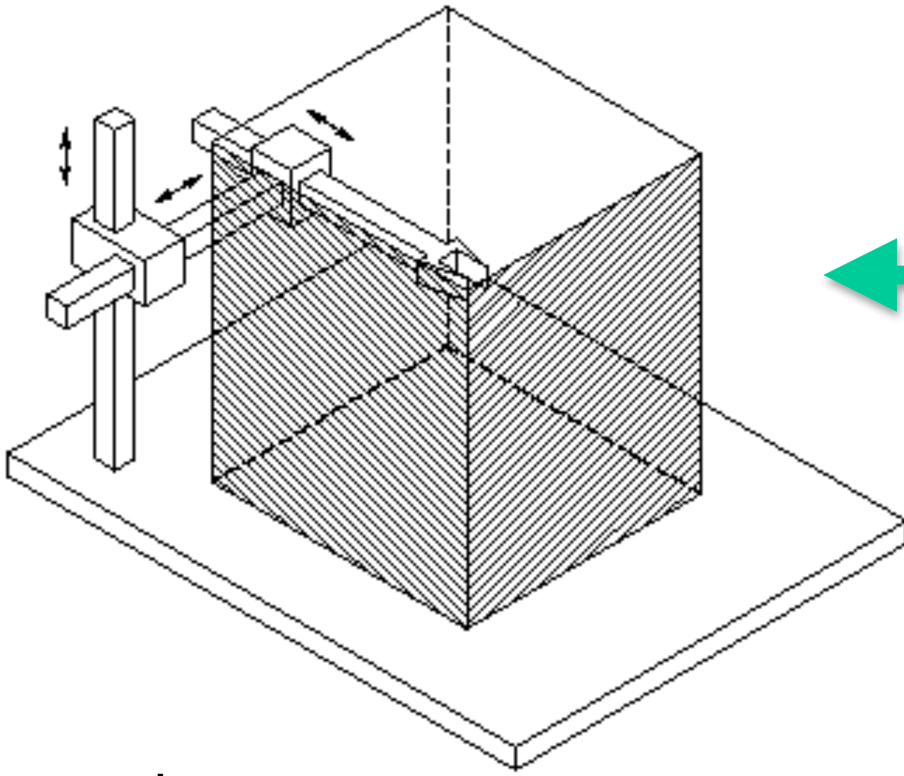
$$\underline{q} = [q_1,\, q_2]^T = [x_1,\, x_2]^T$$

Determine the homogeneous transformation matrix

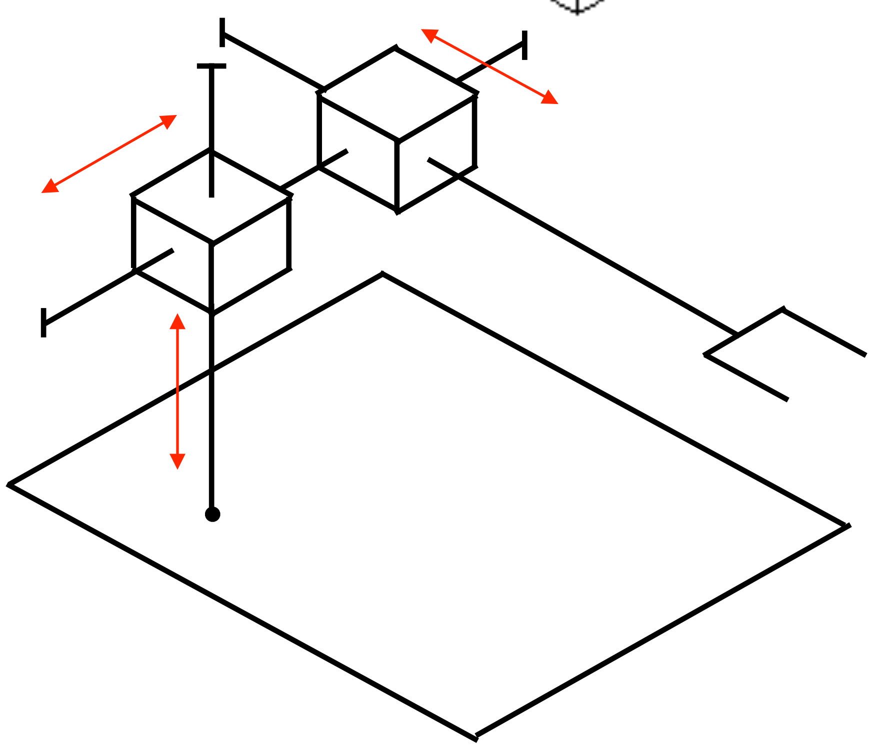
$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

$$T_t^b(\underline{q}) = \begin{bmatrix} 0 & 1 & 0 & -x_1 \\ 0 & 0 & 1 & x_2 \\ 1 & 0 & 0 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 4



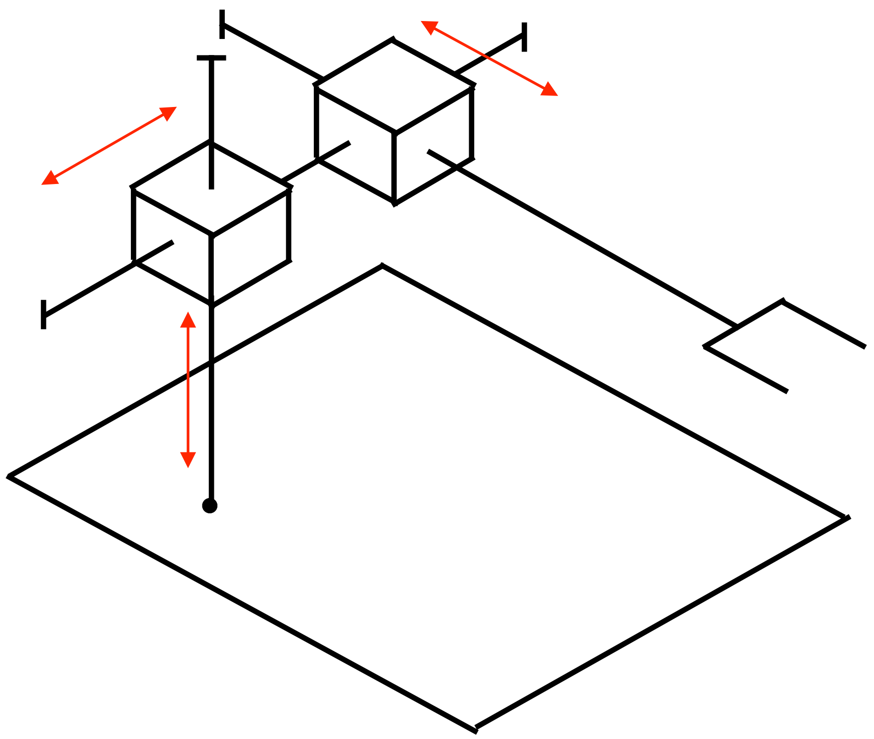
Cartesian Robot (3 DoM)



Cartesian Robot (3 DoM)

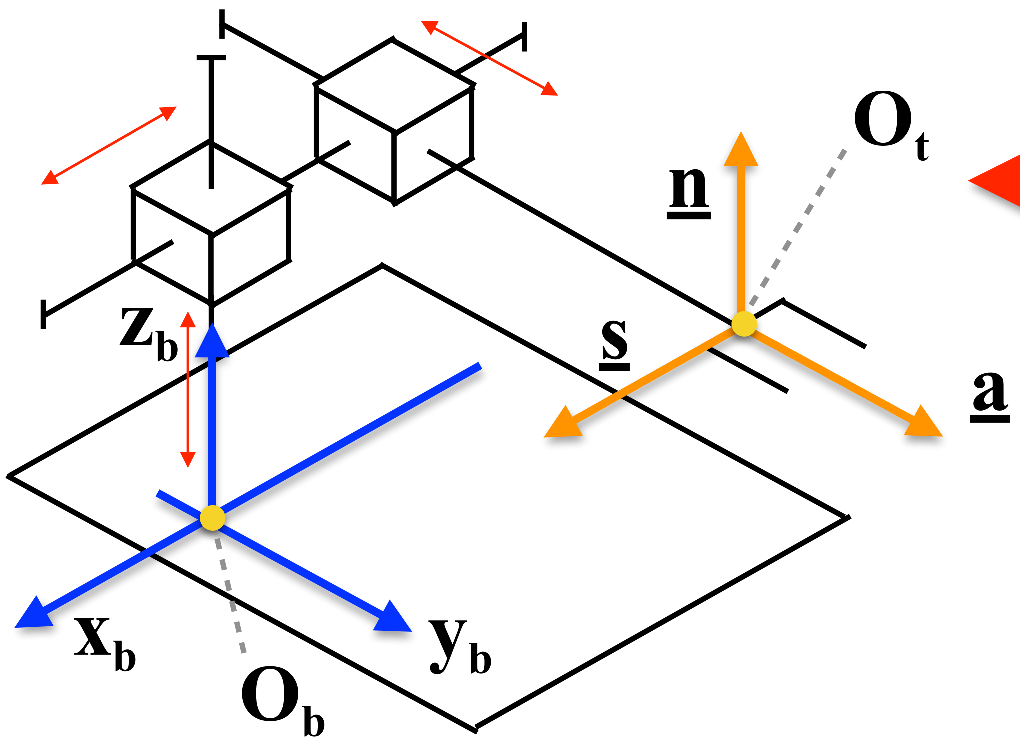
Note: neglect the prismatic joint height, length and depth

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 4



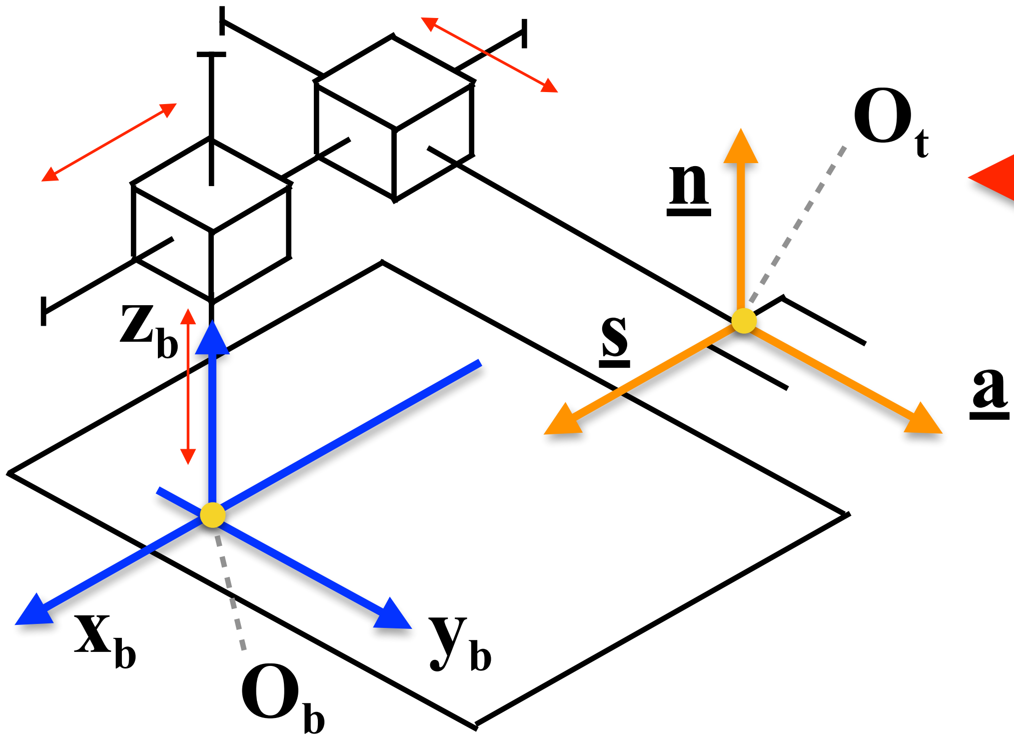
Assign the Base Reference Frame and the Tool Reference Frame

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 4



Assign the Base Reference Frame and the Tool Reference Frame

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 4

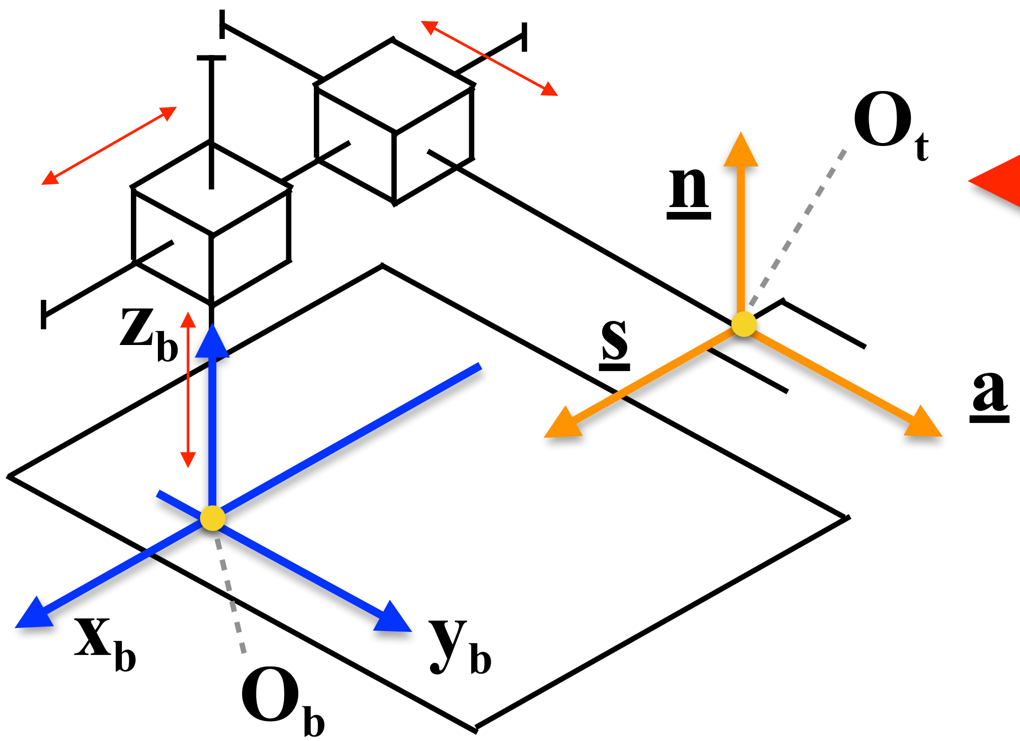


Assign the Base Reference Frame and the Tool Reference Frame

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2,\, q_3]^T = [x_1,\, x_2,\, x_3]^T$$

Description of the Direct Kinematics of Cartesian Robot Manipulators: Case 4



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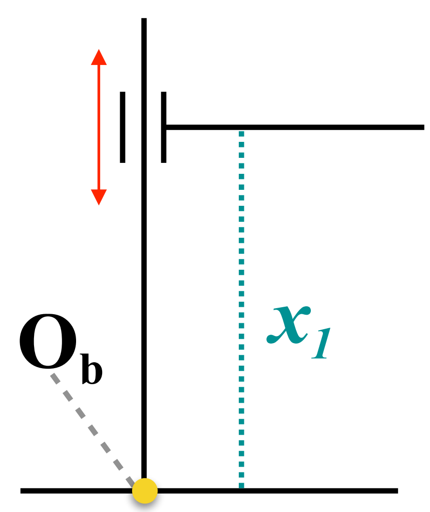

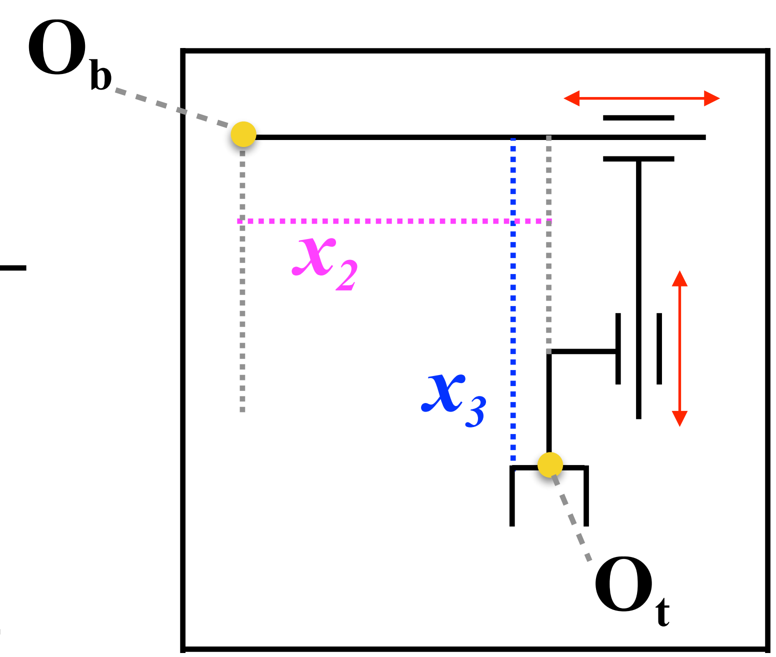

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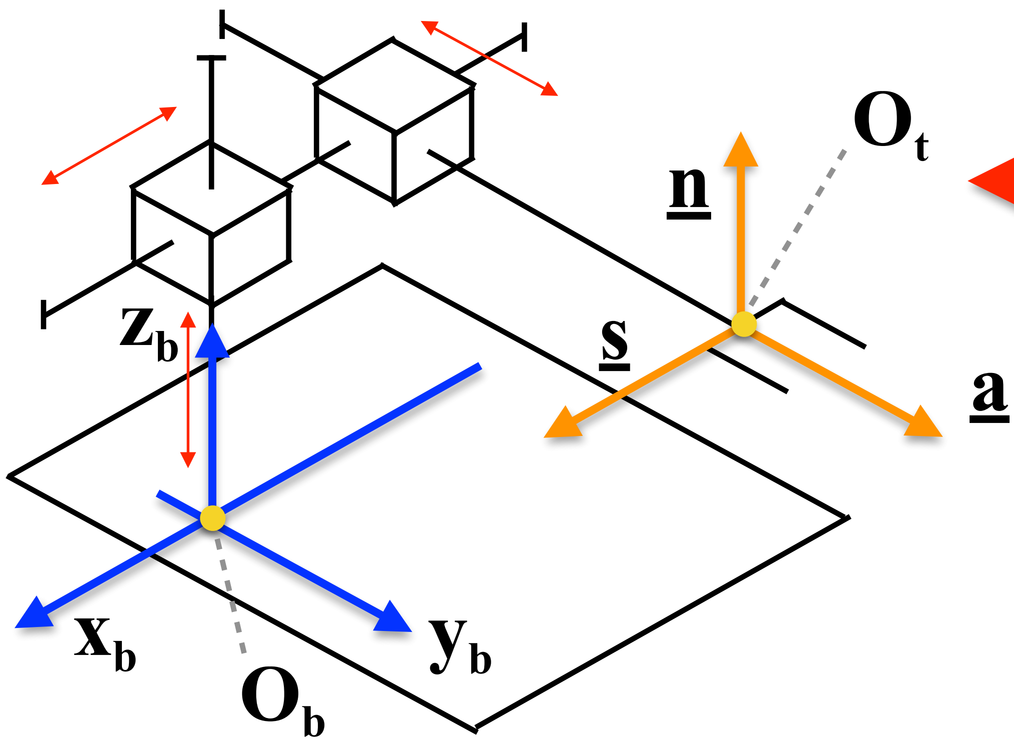

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Define the vector of the joint variables

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Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

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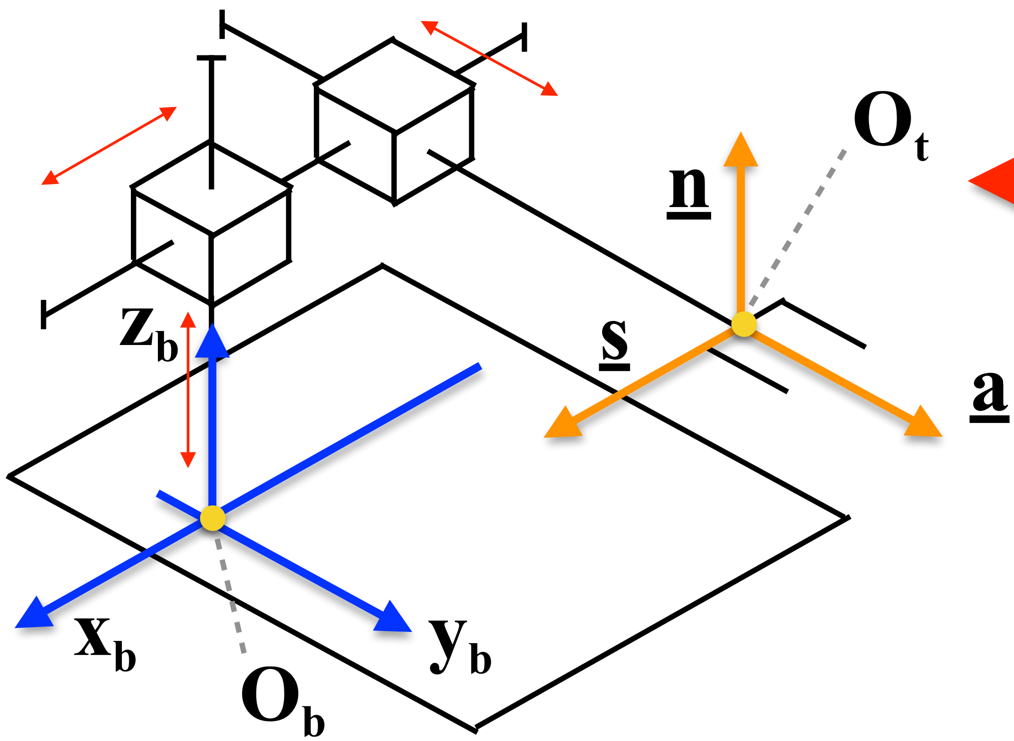

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$$T_t^b(\underline{q}) = \begin{bmatrix} 0 & 1 & 0 & -x_2 \\ 0 & 0 & 1 & x_3 \\ 1 & 0 & 0 & x_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara

Direct Kinematics - Cylindrical Robots

Cylindrical Robot Manipulators

Source: https://www.synchronlab.com/work-category/robotic-arms/
Hudson PlateCrane EX

Cylindrical Robot Manipulators: Applications

Source: https://www.synchronlab.com/work-category/robotic-arms/
Hudson PlateCrane EX

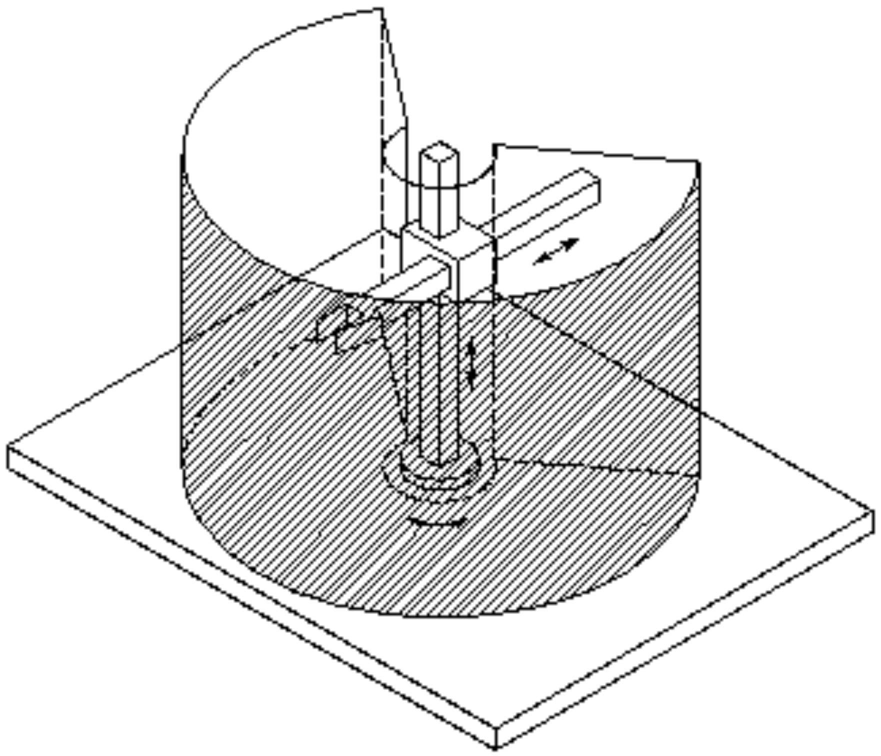
Cylindrical Robots were used most frequently two or three decades ago.

Now they are mainly used for customized solutions. In particular they are used for simple assembly tasks, coating applications or *machine tending* in tight workspaces due to their compact design.

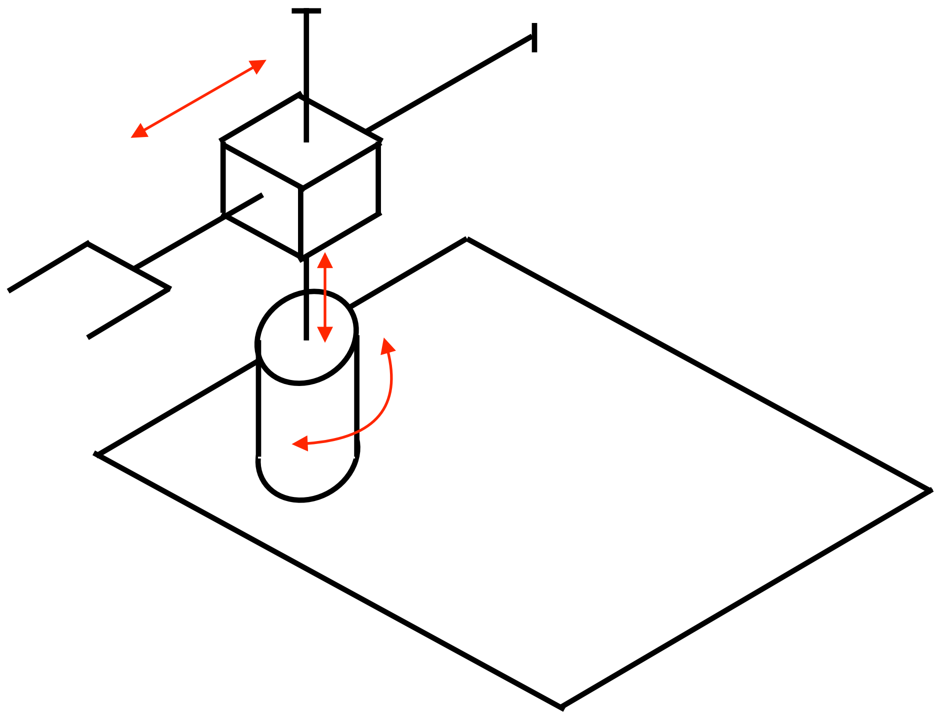
Machine tending means to load and/or unload a given machine with parts or material.

The robot in the picture is optimized for loading and unloading lab instruments, such as washers or reagent dispensers.

Description of the Direct Kinematics of Cylindrical Robot Manipulators



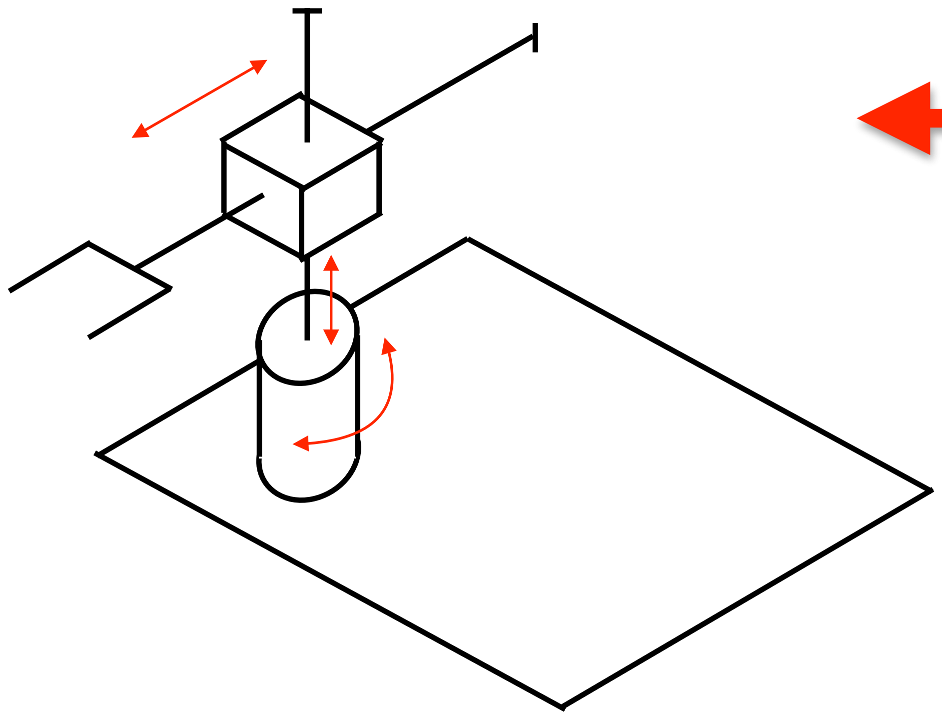
Cylindrical Robot
(3 DoM)



Cylindrical Robot
(3 DoM)

Note: the joints height, length and depth are negligible

Description of the Direct Kinematics of Cylindrical Robot Manipulators



Assign the Base Reference Frame and the Tool Reference Frame

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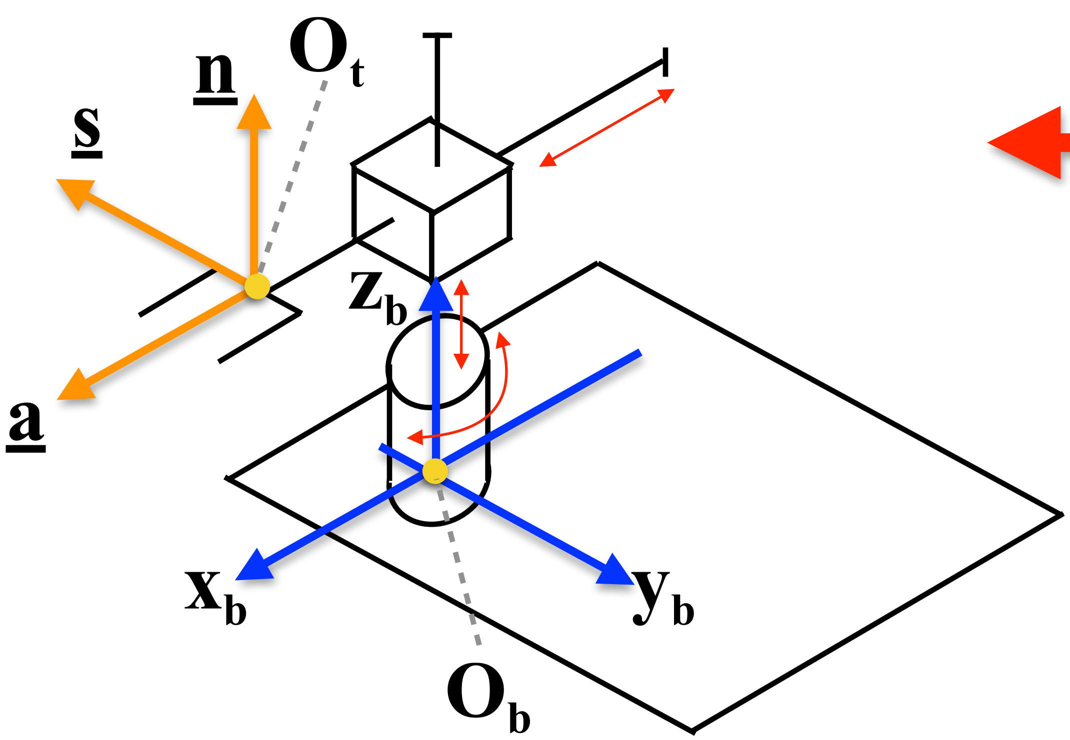

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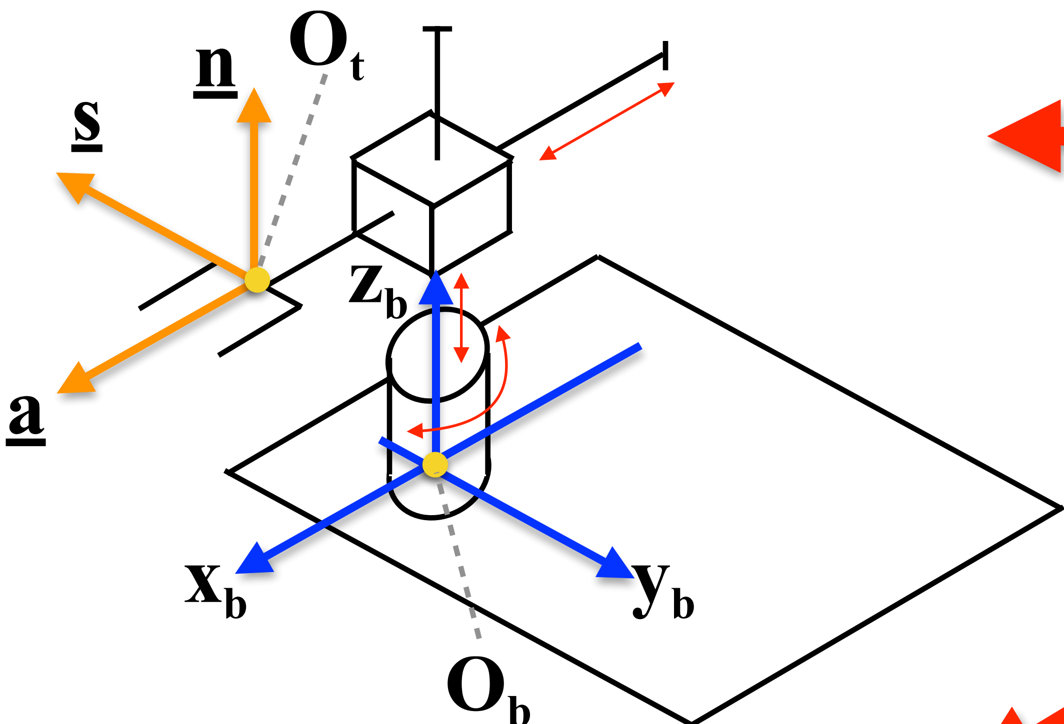
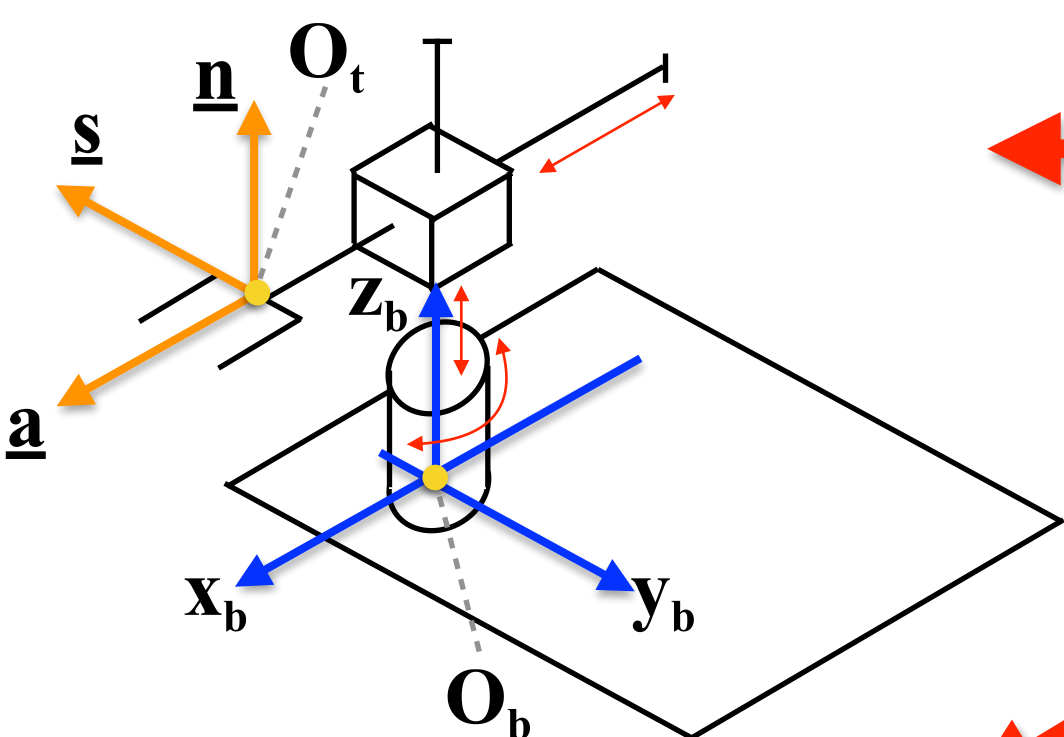


Assign the Base Reference Frame and the Tool Reference Frame

Define the vector of the joint variables

$$\underline{q} = [q_1,\ q_2,\ q_3]^T = [\theta_1,\ x_2,\ x_3]^T$$

Description of the Direct Kinematics of Cylindrical Robot Manipulators



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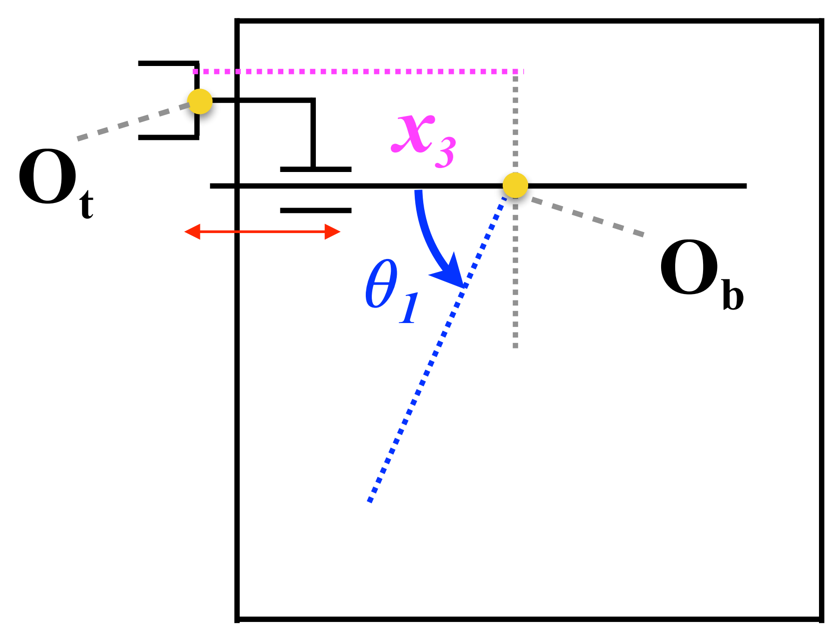

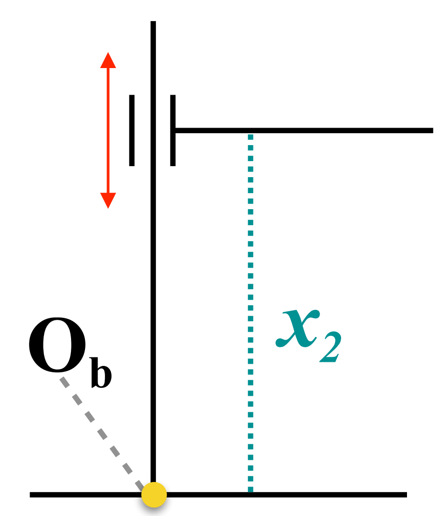

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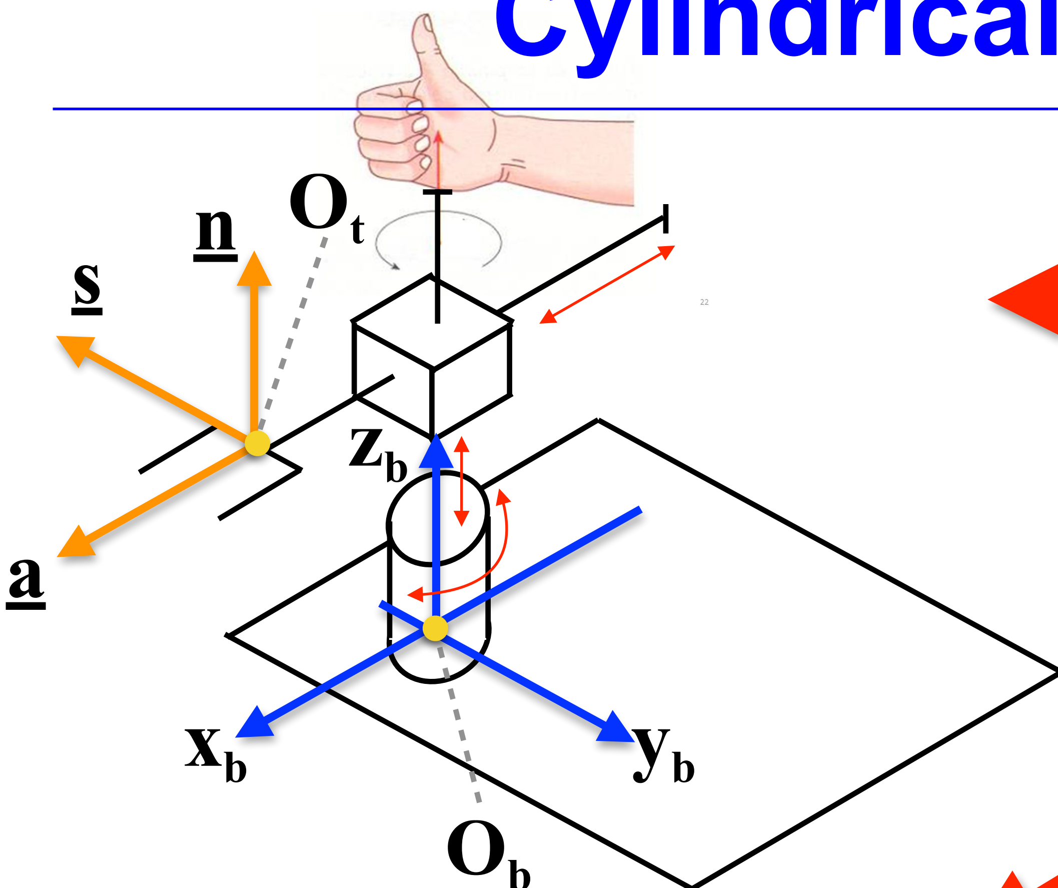

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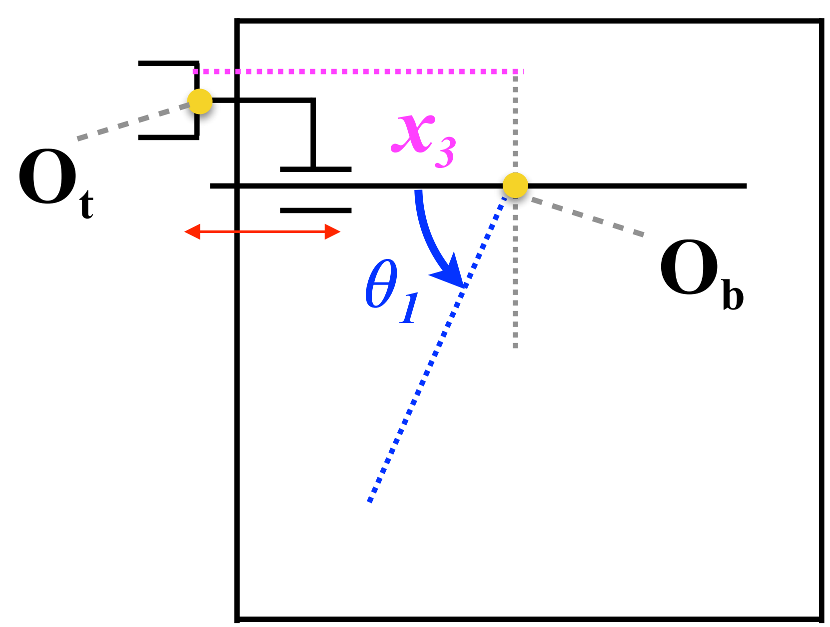

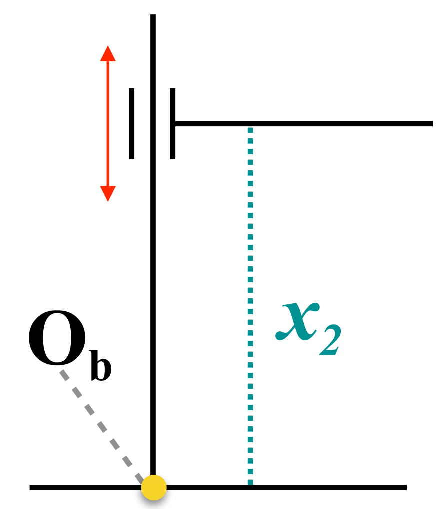

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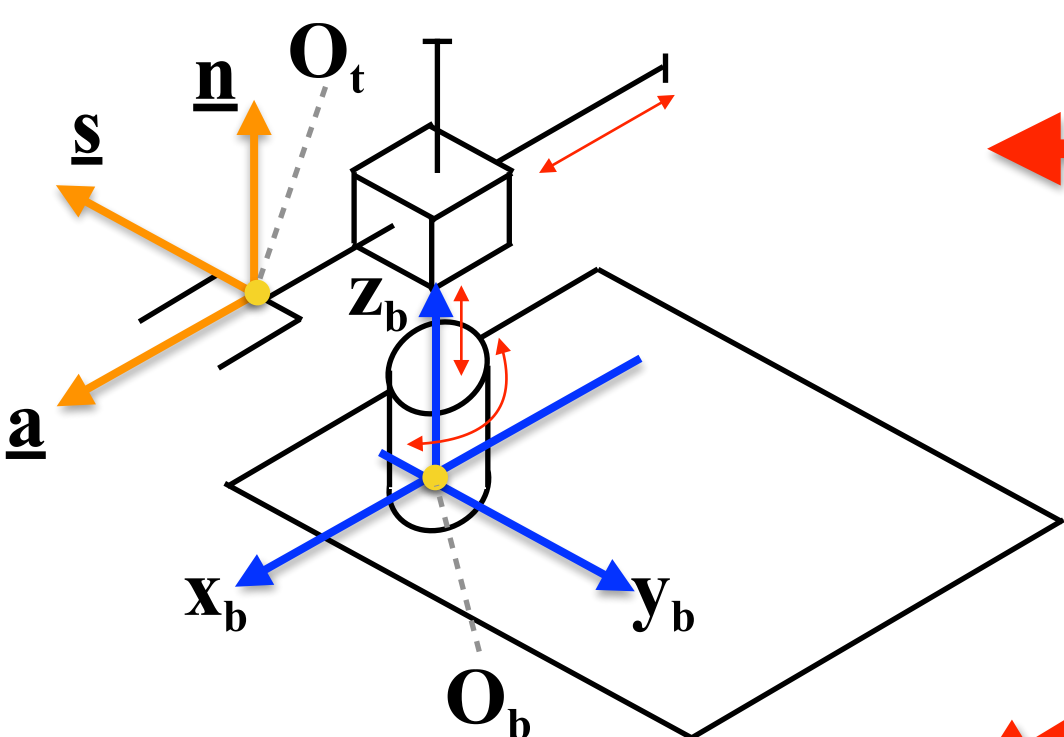

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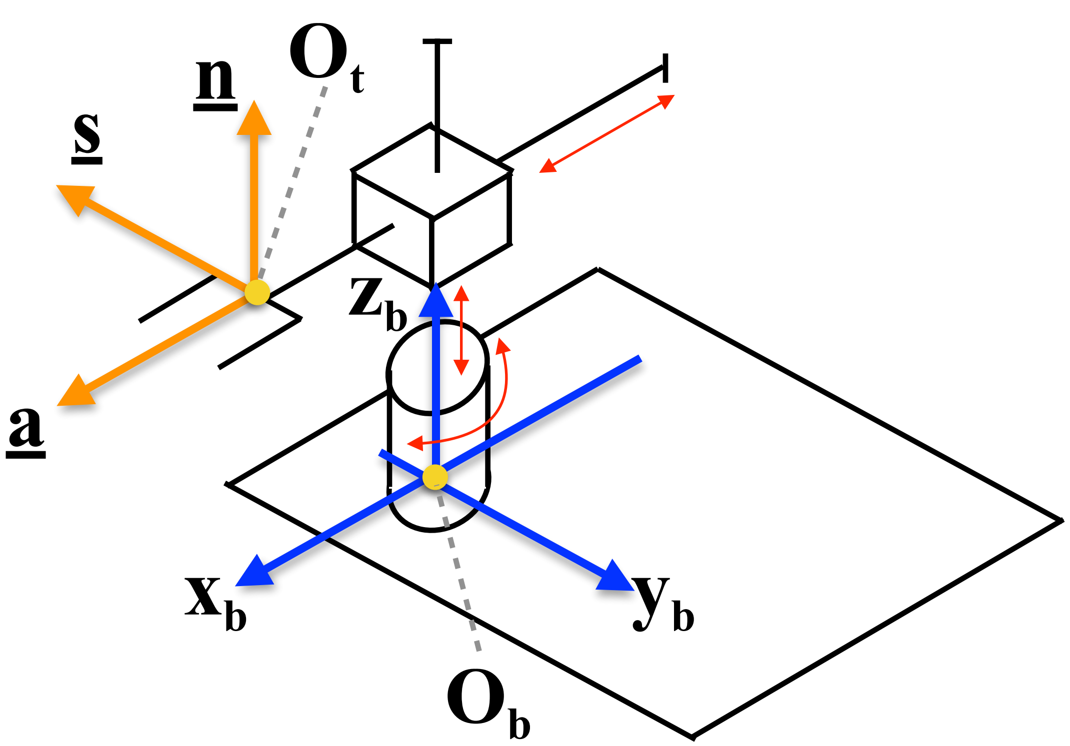
Define the vector of the joint variables

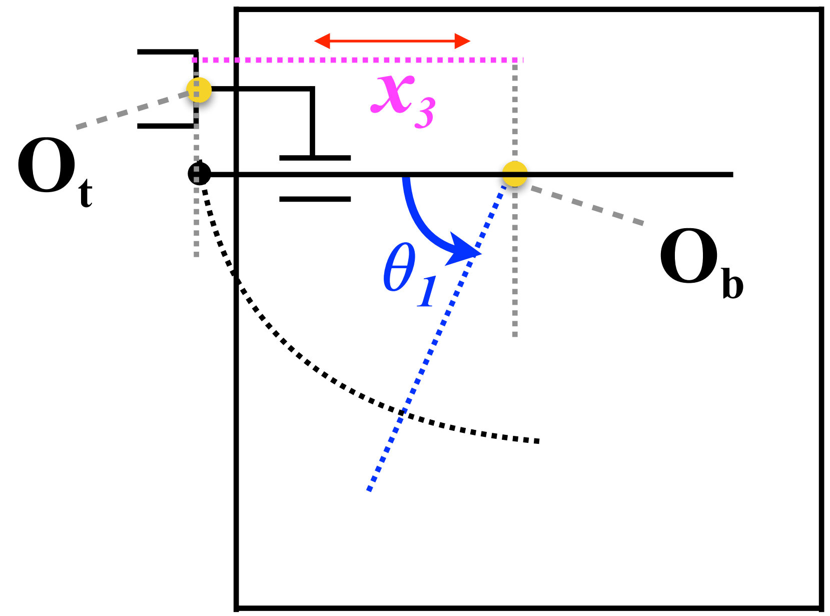
$$\underline{q} = [q_1,\ q_2,\ q_3]^T = [\theta_1,\ x_2,\ x_3]^T$$

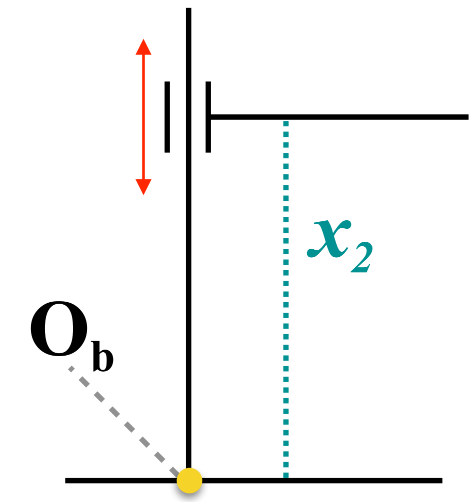
Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Cylindrical Robot Manipulators







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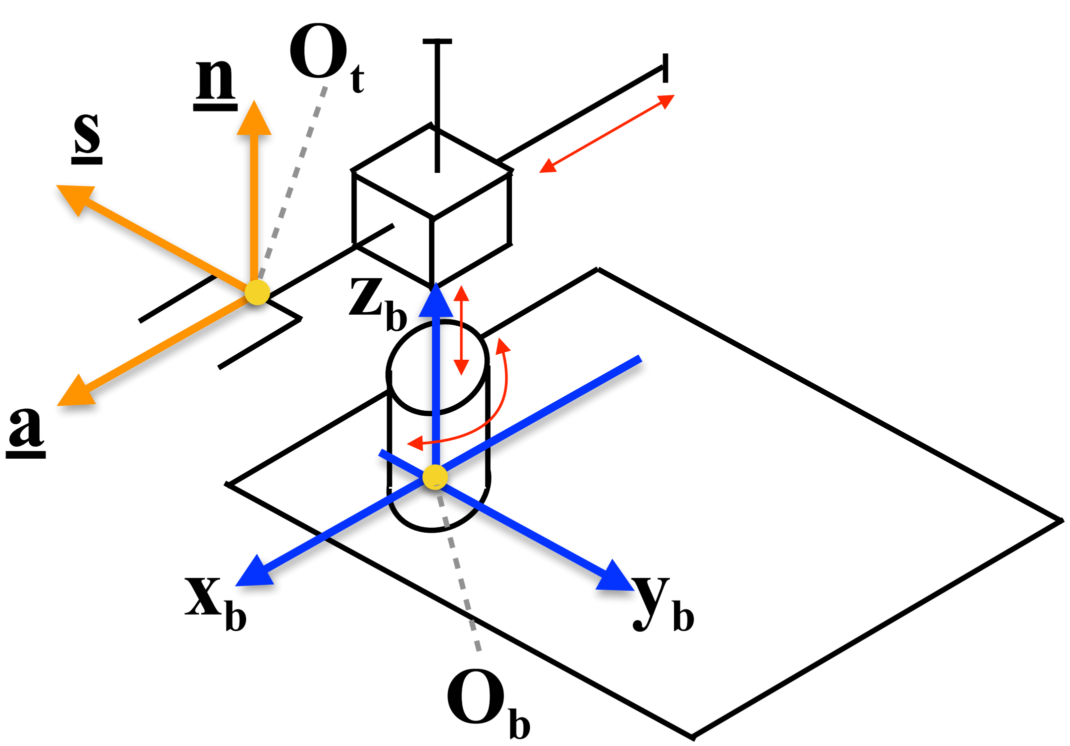

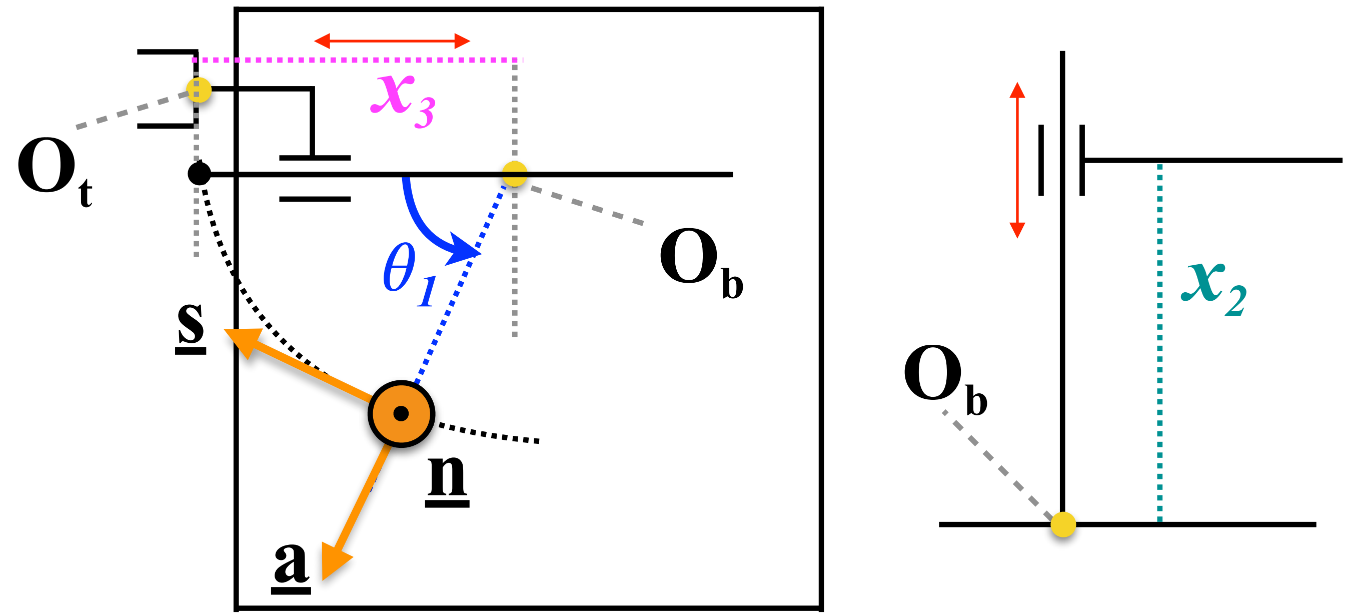

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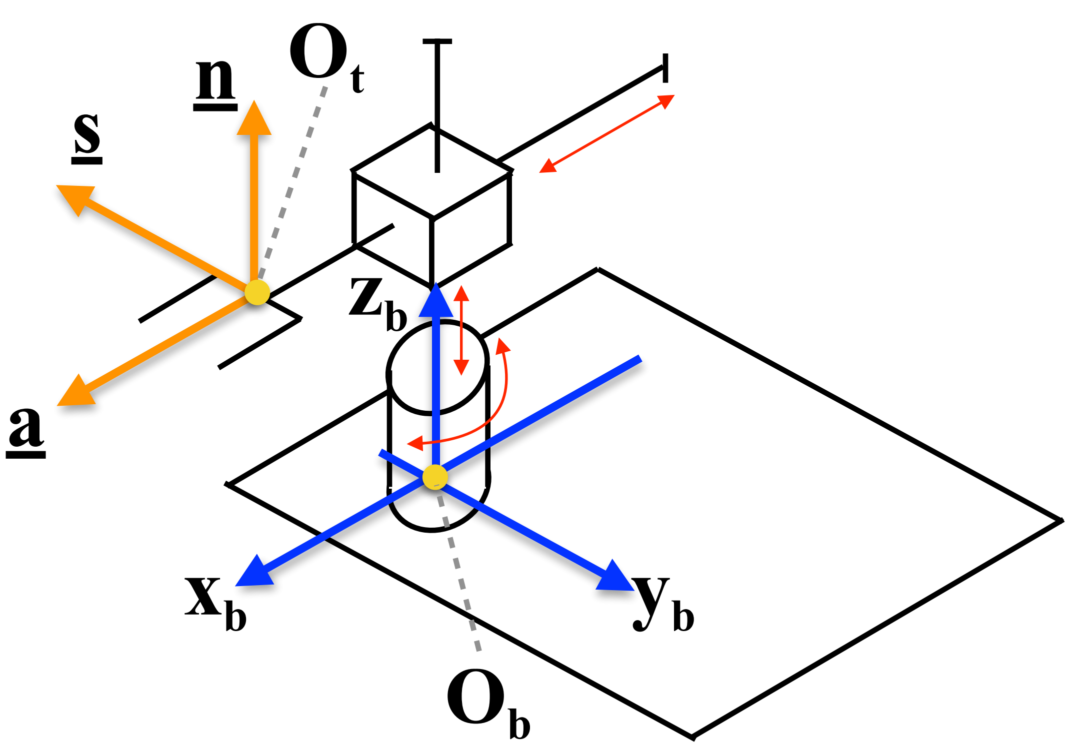

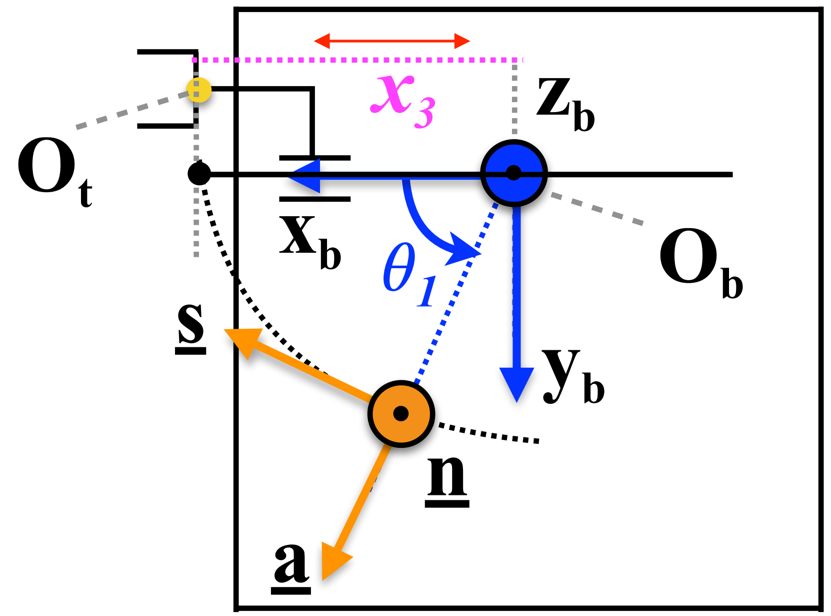

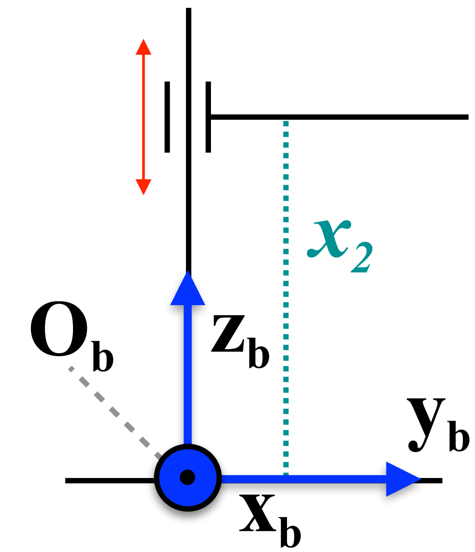

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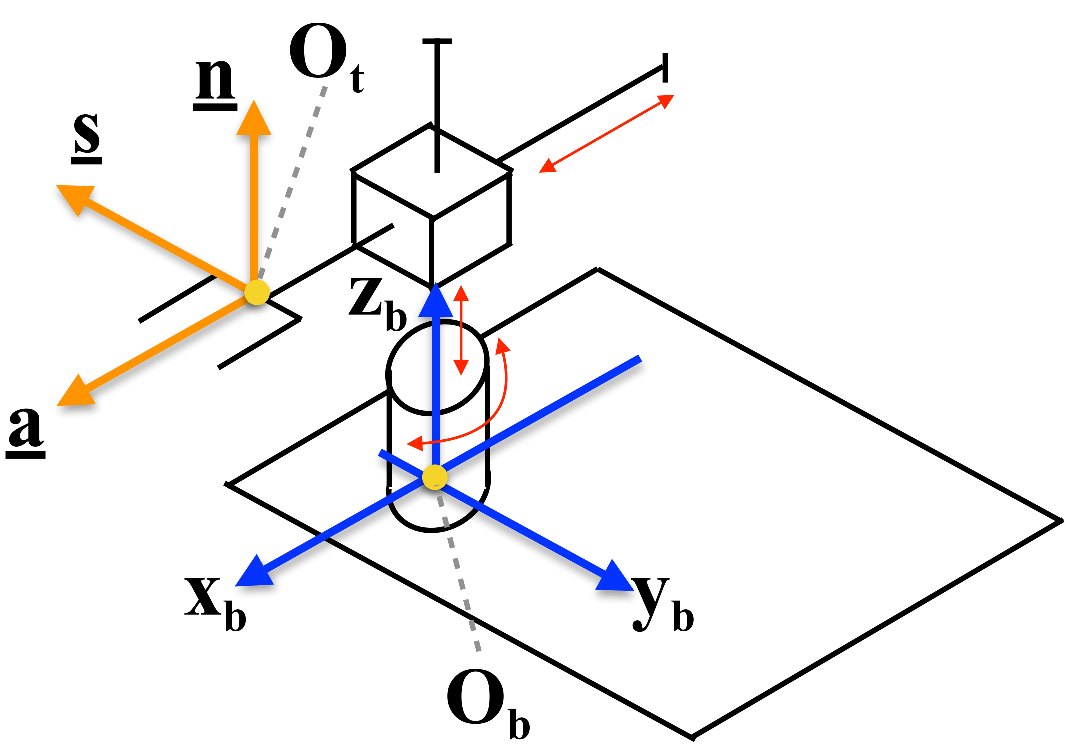

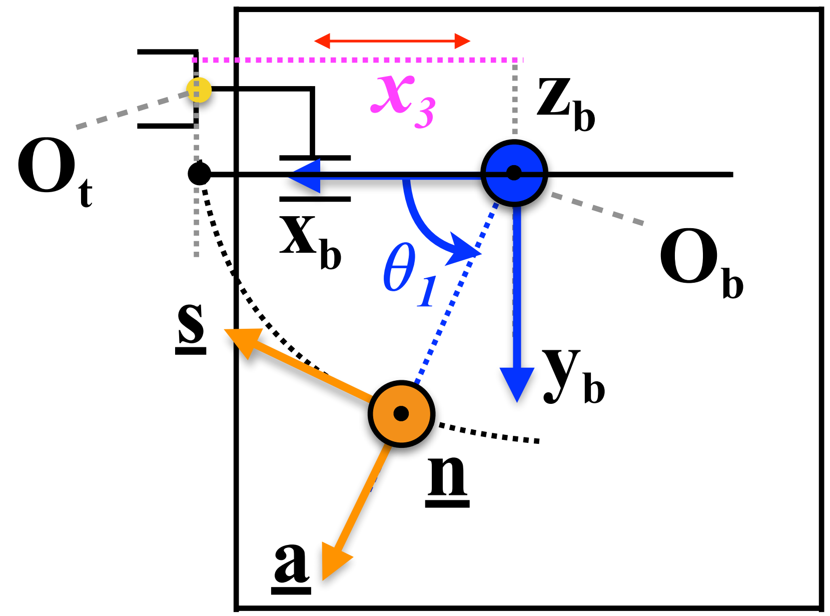

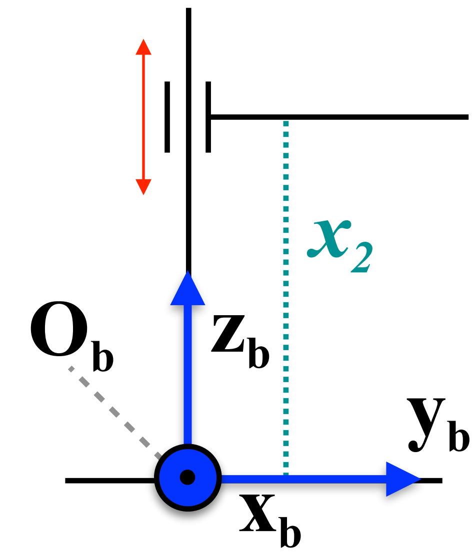

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$$T_t^b(\underline{q}) = \begin{bmatrix} 0 & cos(90^0 - \theta_1) & cos(\theta_1) & x_3 \cdot cos(\theta_1) \\ 0 & -sin(90^0 - \theta_1) & sin(\theta_1) & x_3 \cdot sin(\theta_1) \\ 1 & 0 & 0 & x_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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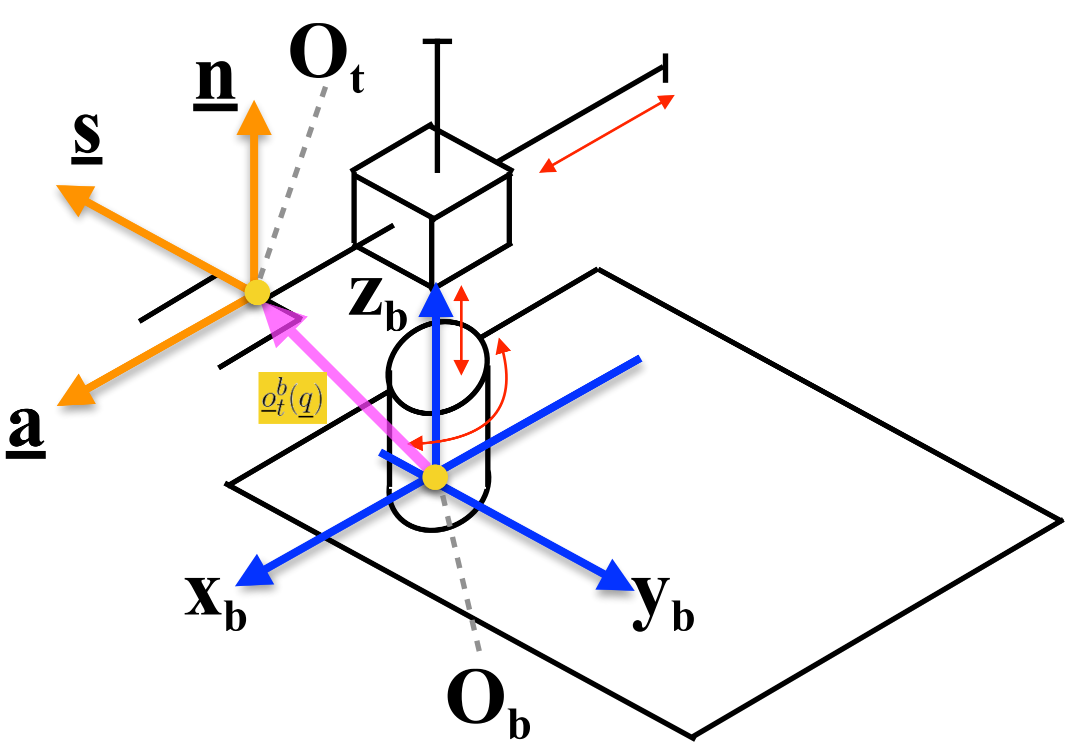

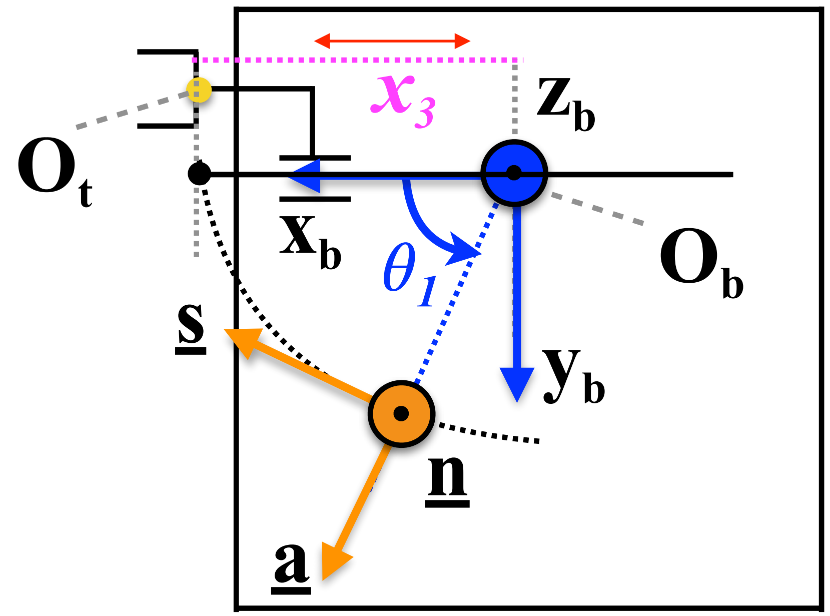

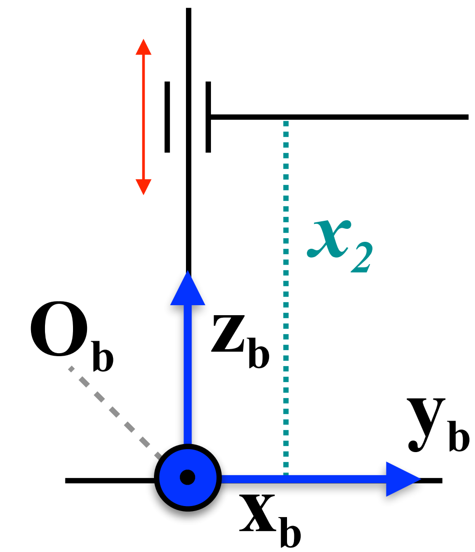

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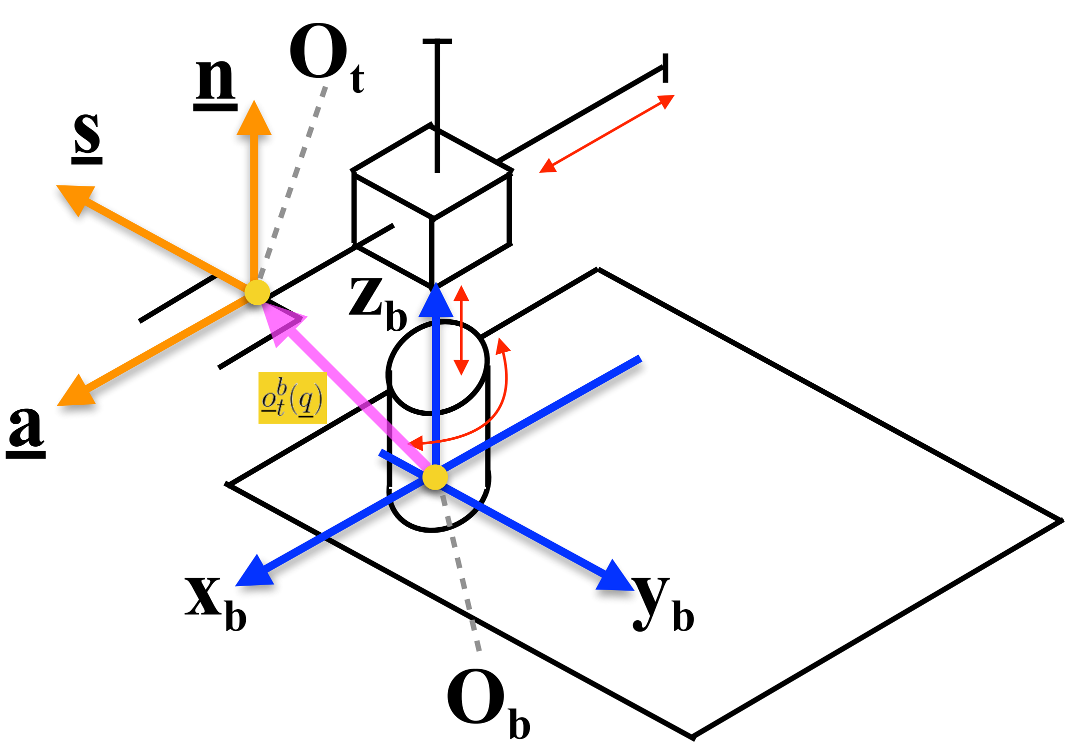

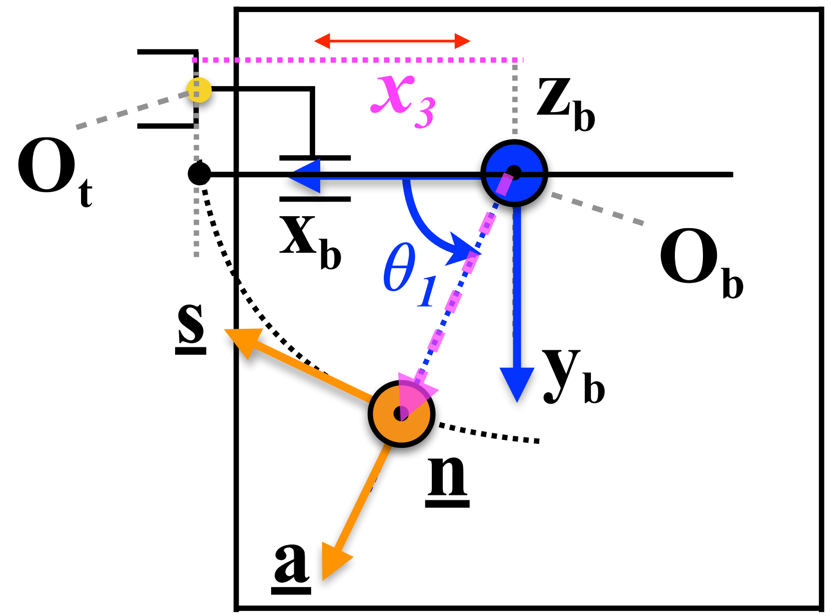

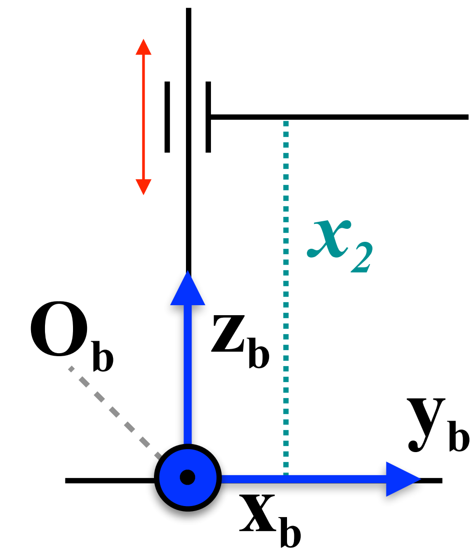

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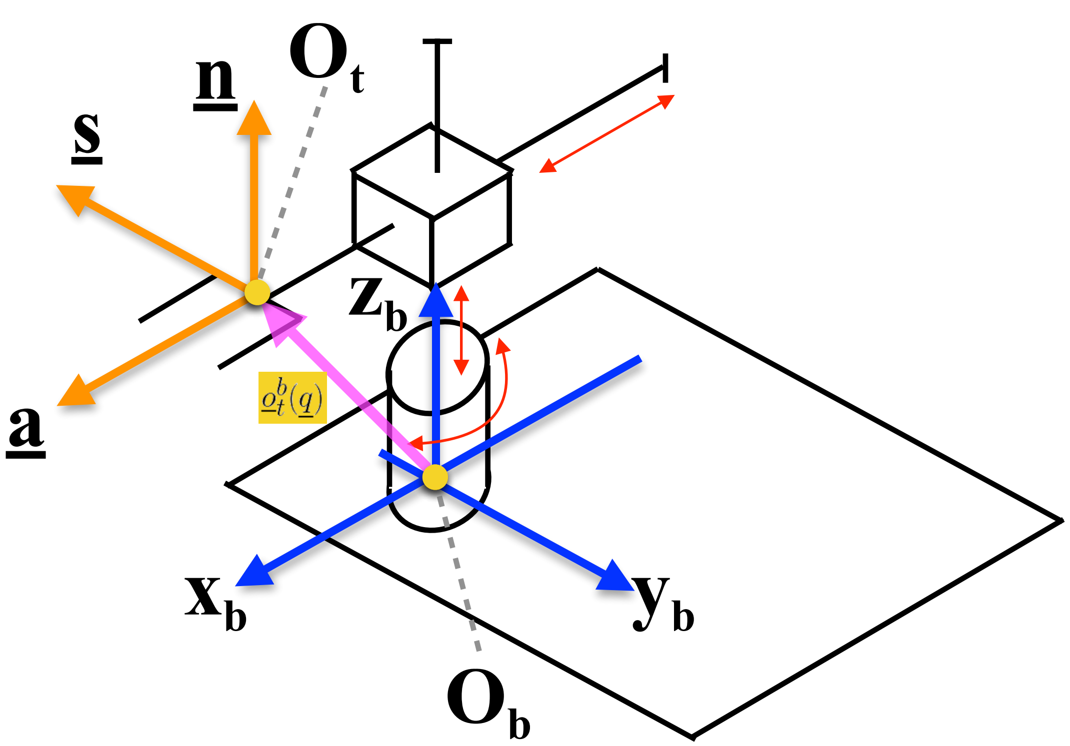

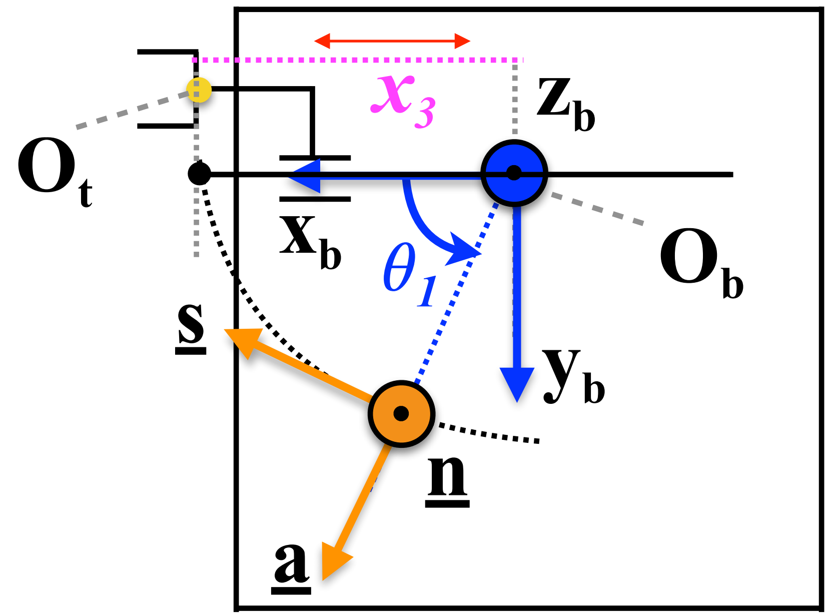

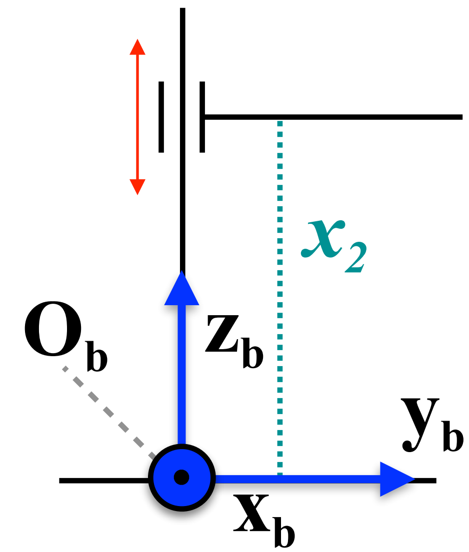

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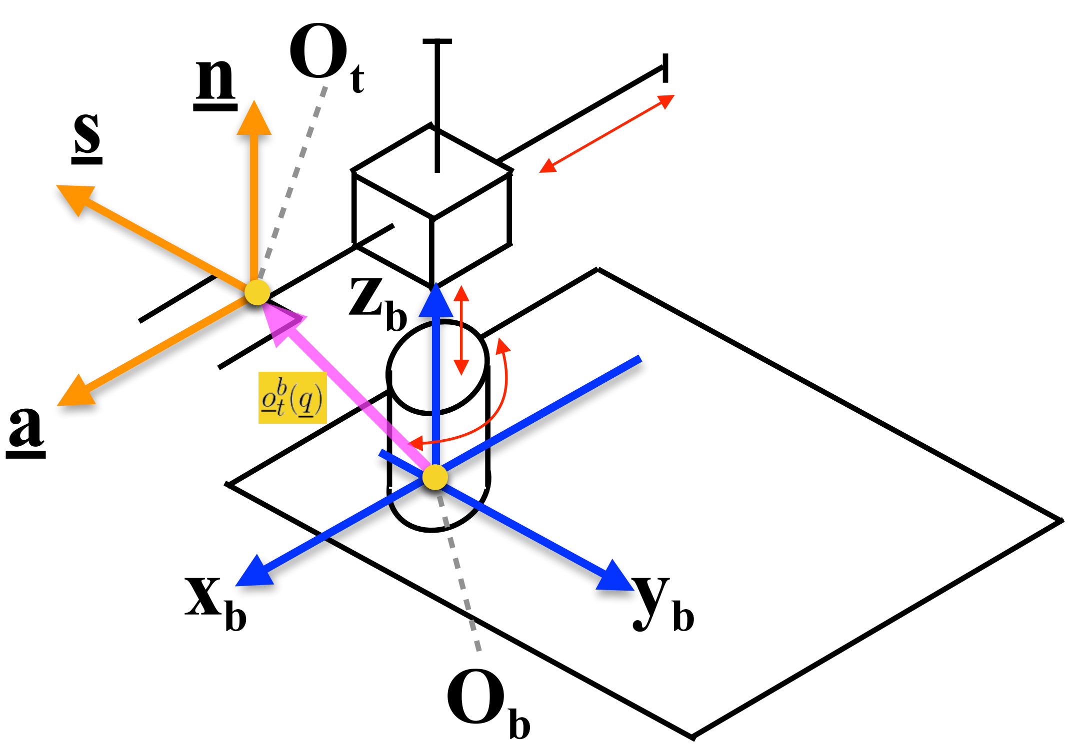

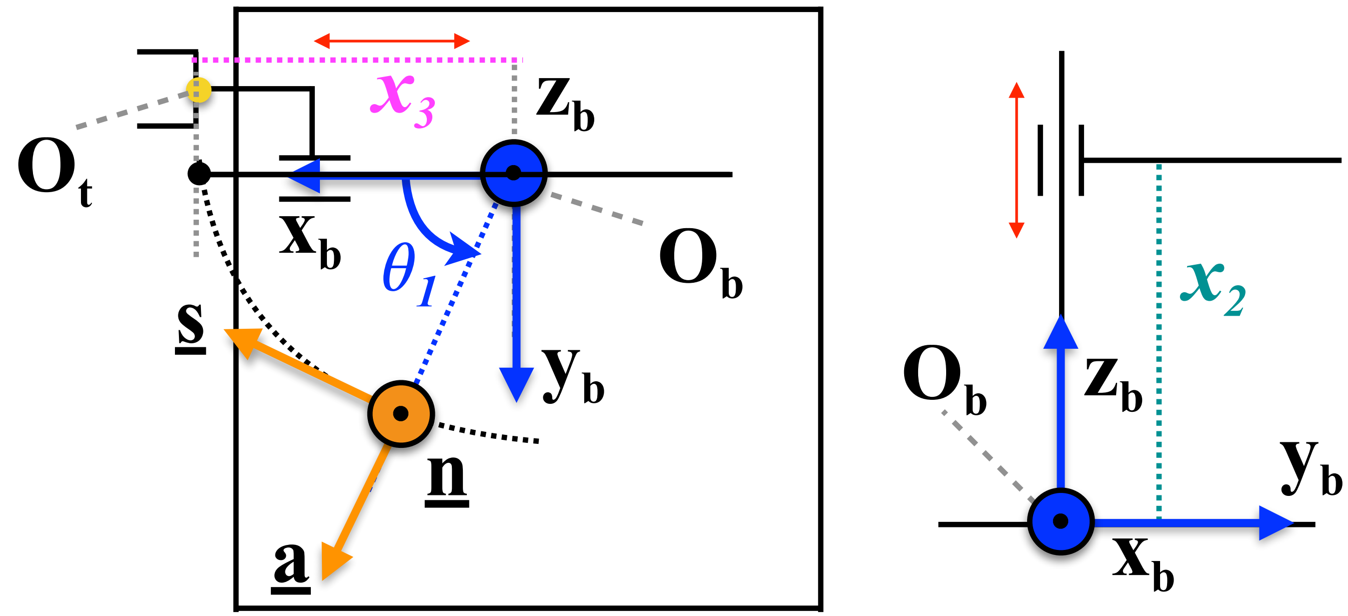

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$$T_t^b(\underline{q}) = \begin{bmatrix} 0 & s_1 & c_1 & x_3\, c_1 \\ 0 & -c_1 & s_1 & x_3\, s_1 \\ 1 & 0 & 0 & x_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

UNIVERSITÀ DI PAVIA
Dipartimento di
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e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara

Direct Kinematics - S.C.A.R.A.

S.C.A.R.A.

Large type SCARA robots YK-XG
by YAMAHA

Suitable for assembly and transfer of large objects or heavy objects.

Arm length:700mm to 1200mm
Maximum payload:10kg to 50kg

Small type SCARA robots YK-XG
by YAMAHA

Suitable for assembly, transfer, pushing work, and sealing of small components.

Arm length:250mm to 400mm
Maximum payload:5kg

S.C.A.R.A.

SCARA is an acronym. Yet, the meaning of the first letter A has changed over the years. There are indeed two definitions:

- the classical one, **Selective Compliance Assembly Robot Arm**
- a second one: **Selective Compliance Articulated Robot Arm**

This second definition became more used when SCARAs started to be used as a valid alternative to cartesian or cylindrical robots in tasks which were not only of assembly type

S.C.A.R.A.

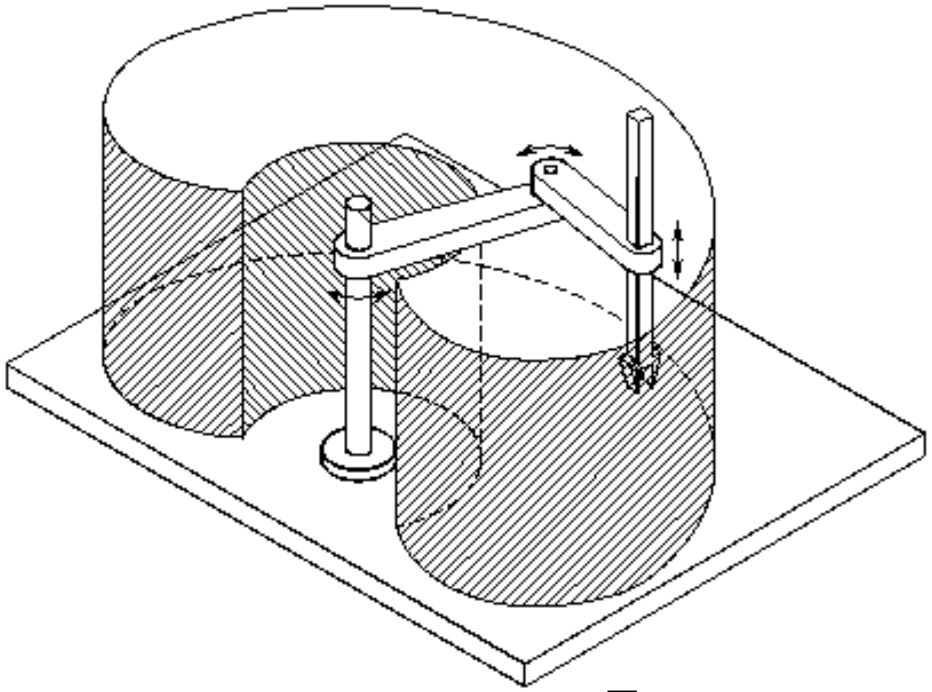
SCARA are used in a wide variety of processes and applications.

They are generally **faster than 6-axis robots**, so they are very appropriate for high-speed assembly applications.

SCARAs are **less rigid than Cartesian or gantry robots**. However, they are more rigid than 6-axis robots due to their rigid Z-axis (they are compliant in the X-Y axis but rigid in the
Z-axis)

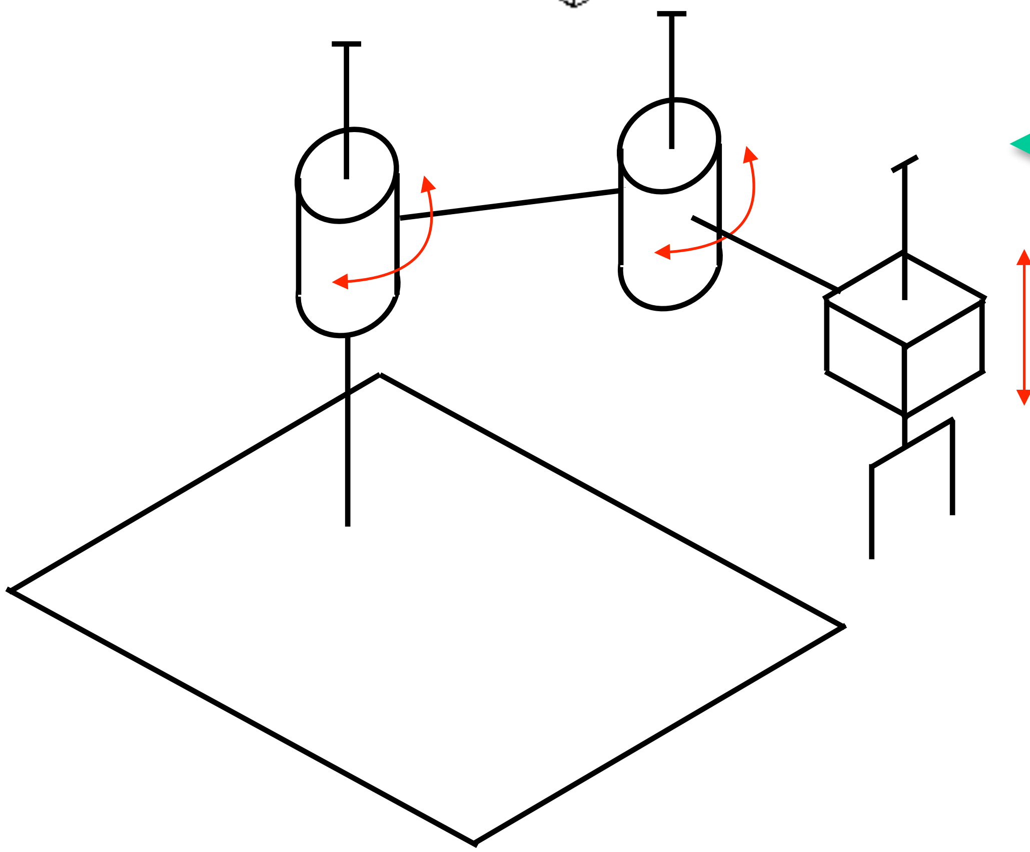
The **payload of SCARAs is generally quite low**

Description of the Direct Kinematics of a S.C.A.R.A.



S.C.A.R.A.

Selective Compliance Assembly Robot Arm



S.C.A.R.A.

Selective Compliance Assembly Robot Arm

Note: the joints height, length and depth are negligible

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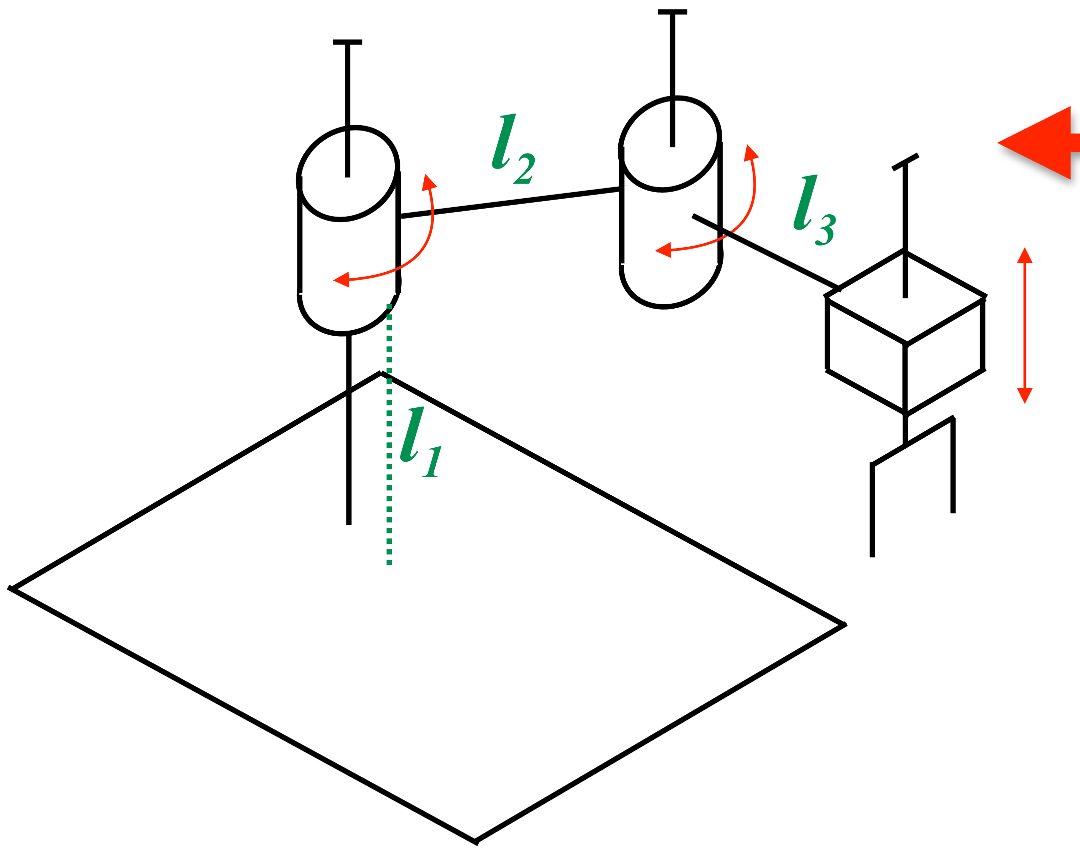

Assign the Base Reference Frame and the Tool Reference Frame

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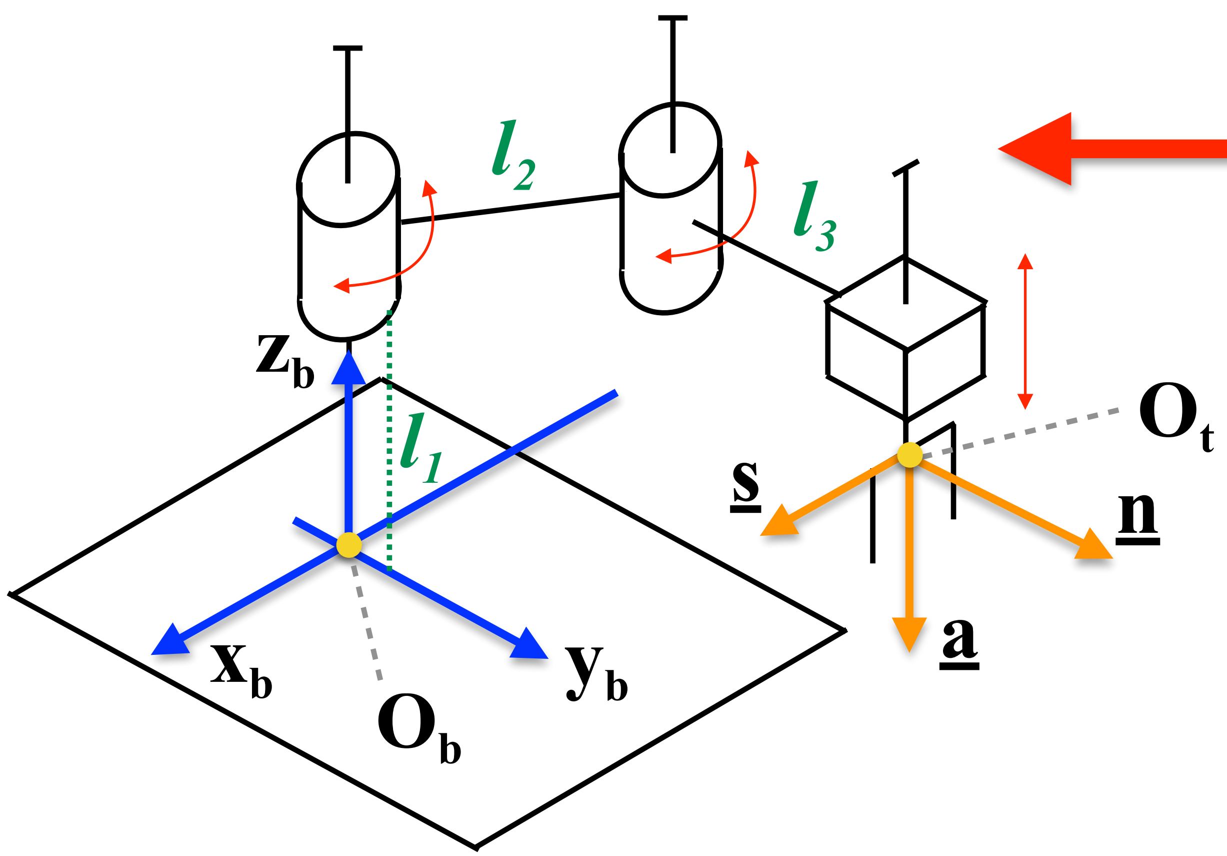

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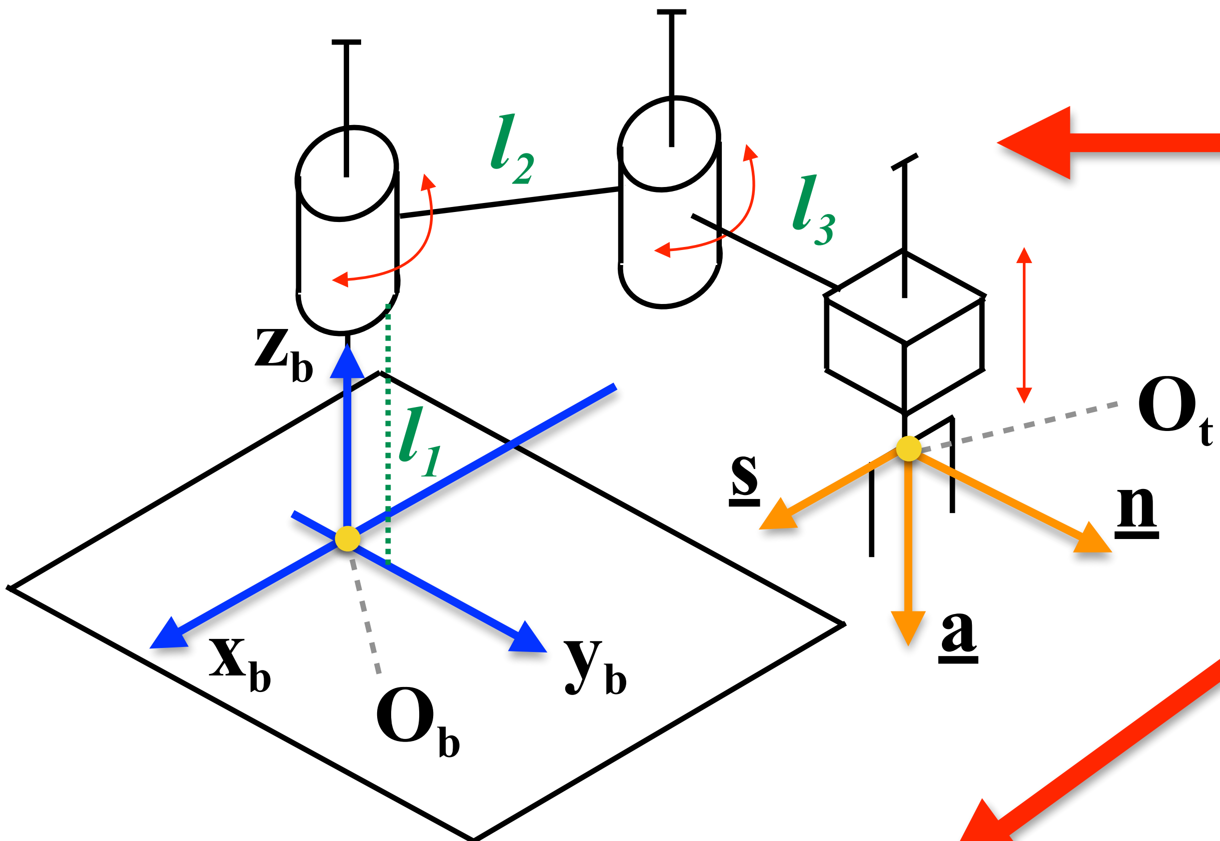

Assign the Base Reference Frame and the Tool Reference Frame

Define the vector of the joint variables

$$\underline{q} = [q_1,\, q_2,\, q_3]^T = [\theta_1,\, \theta_2,\, x_3]^T$$

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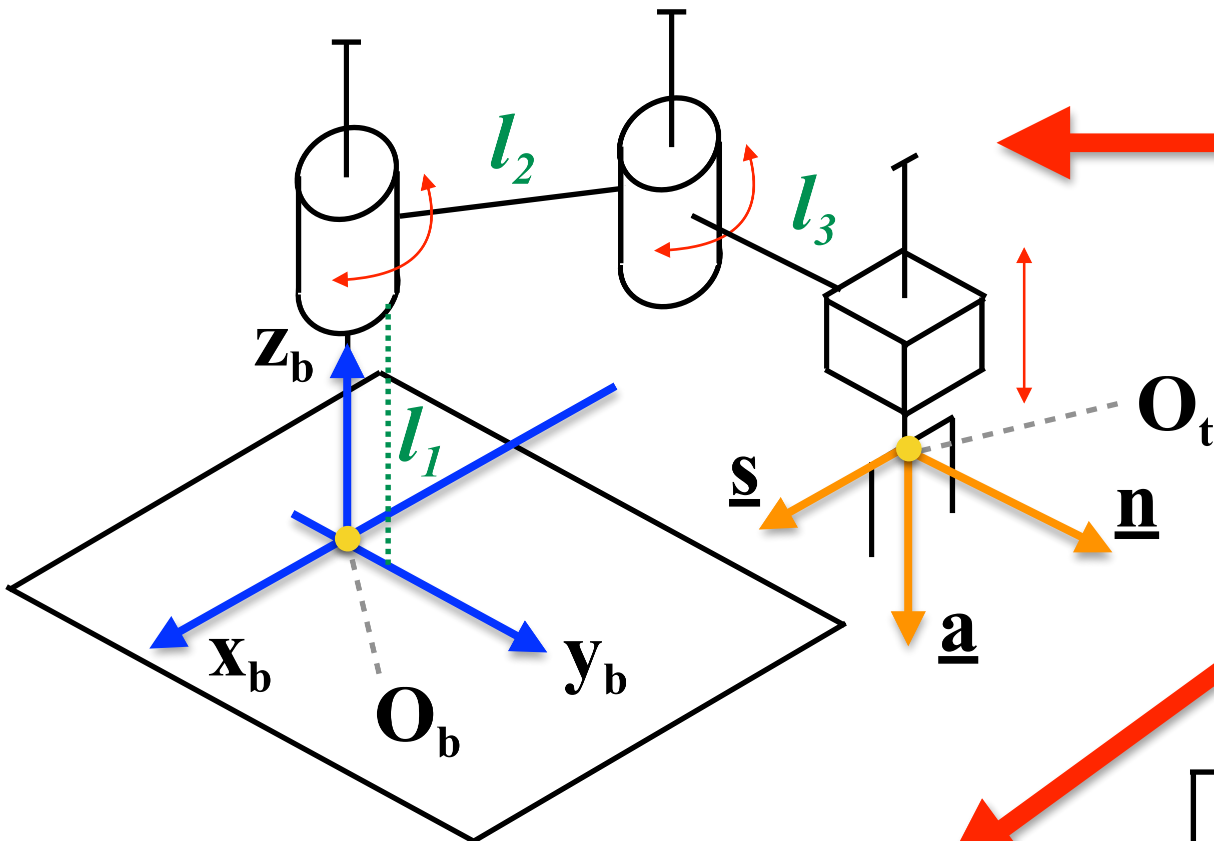

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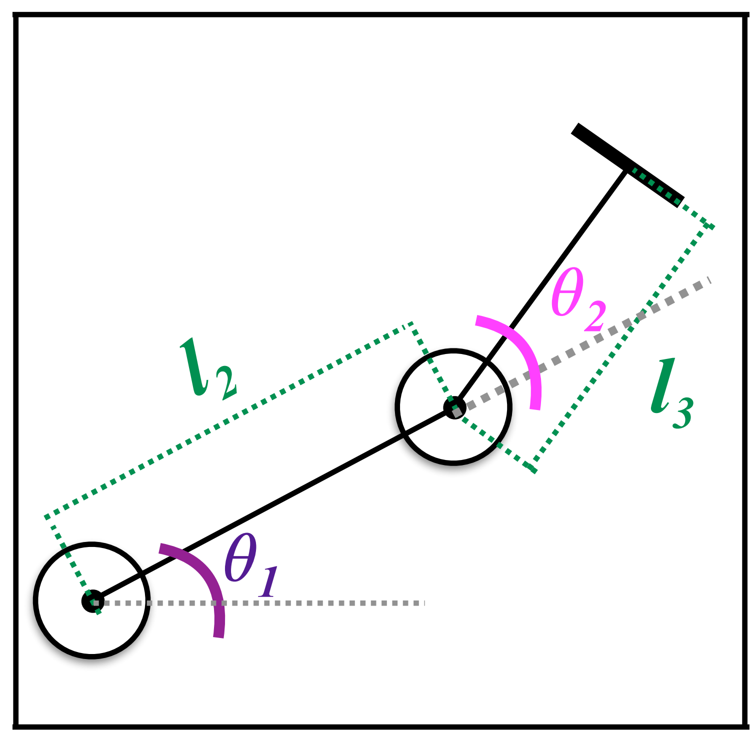

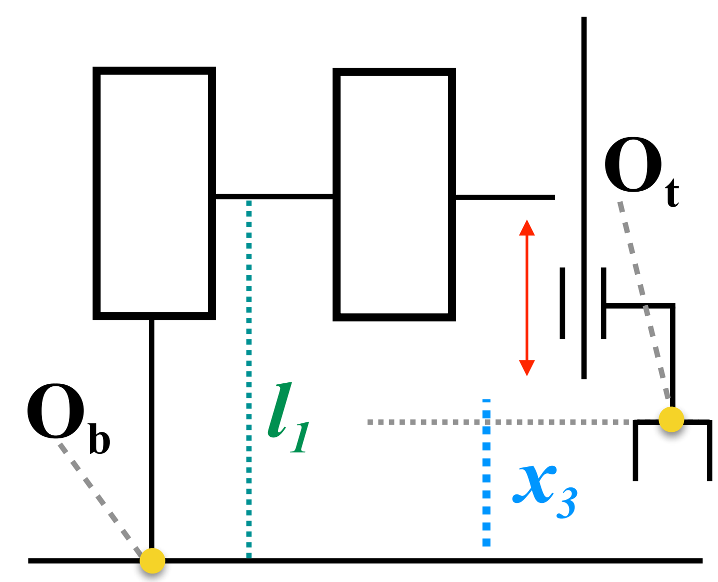

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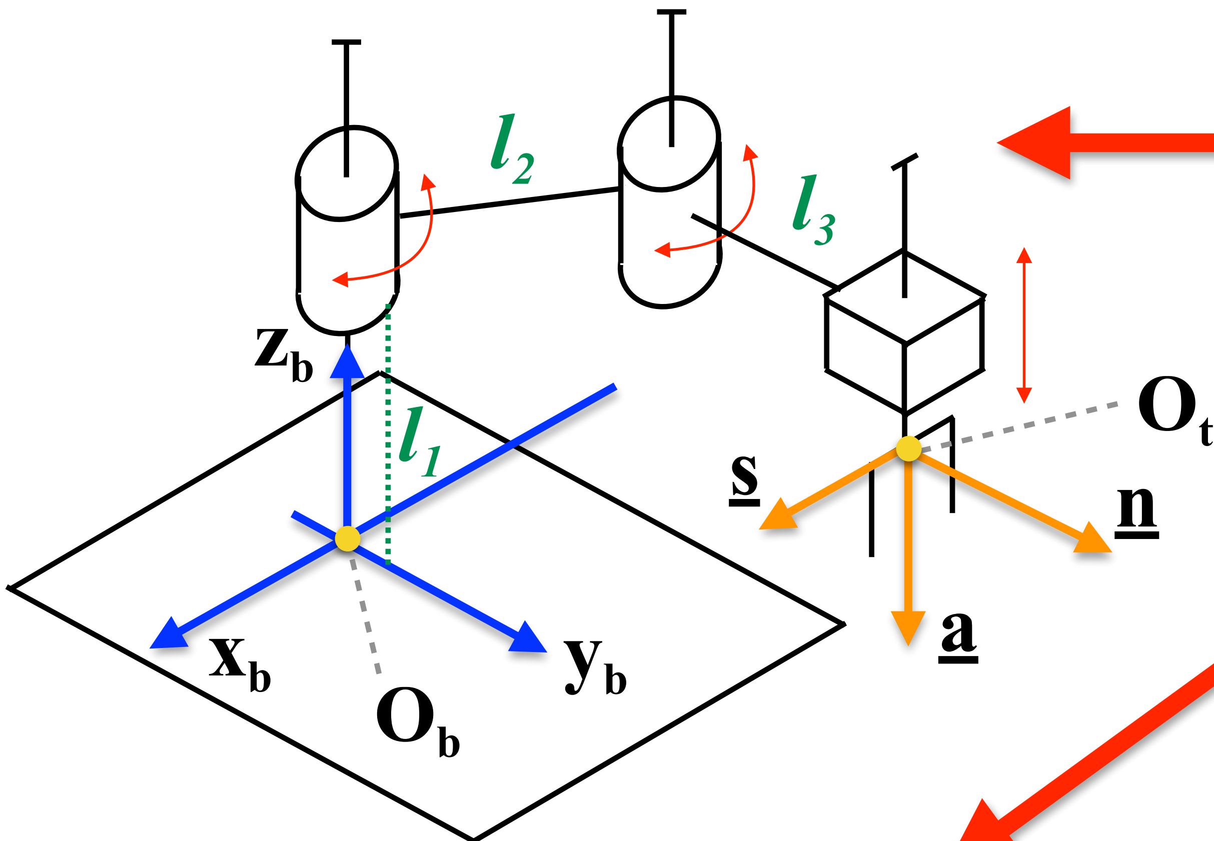

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Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

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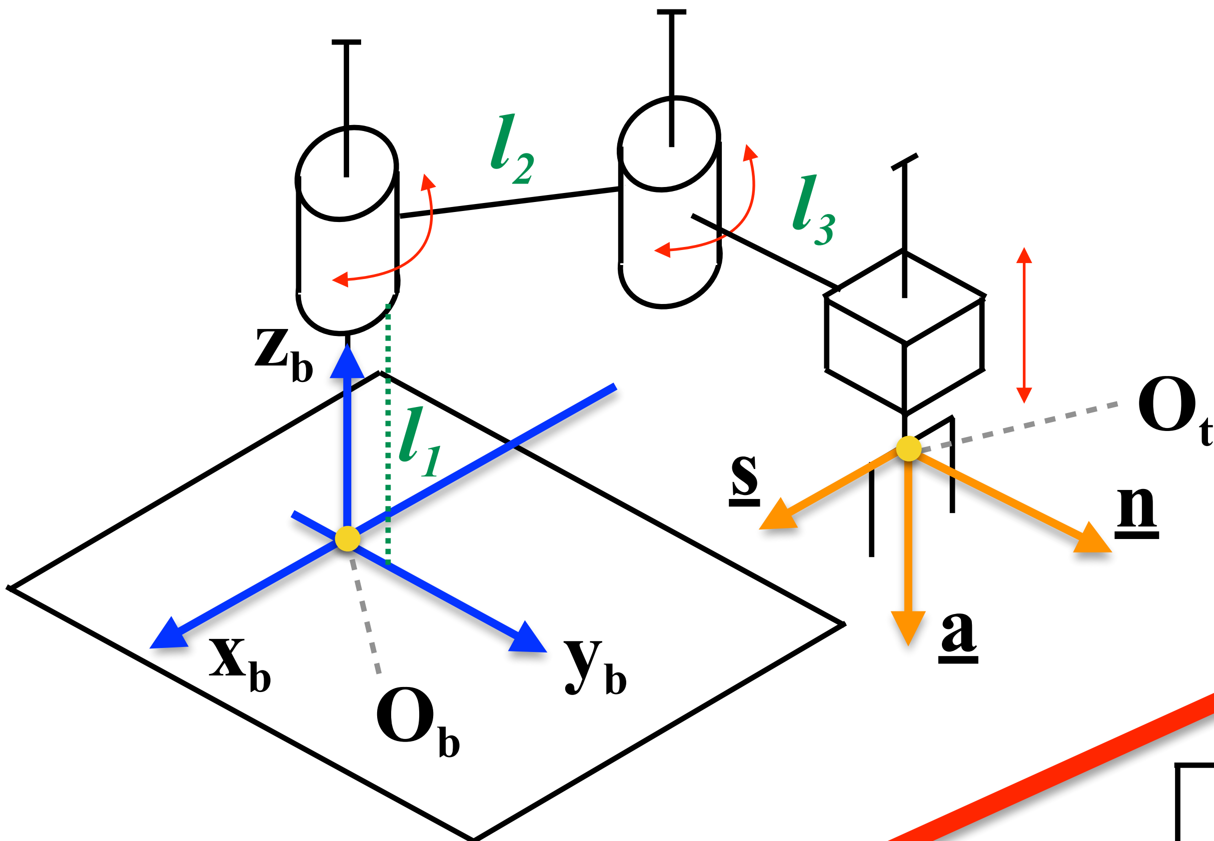

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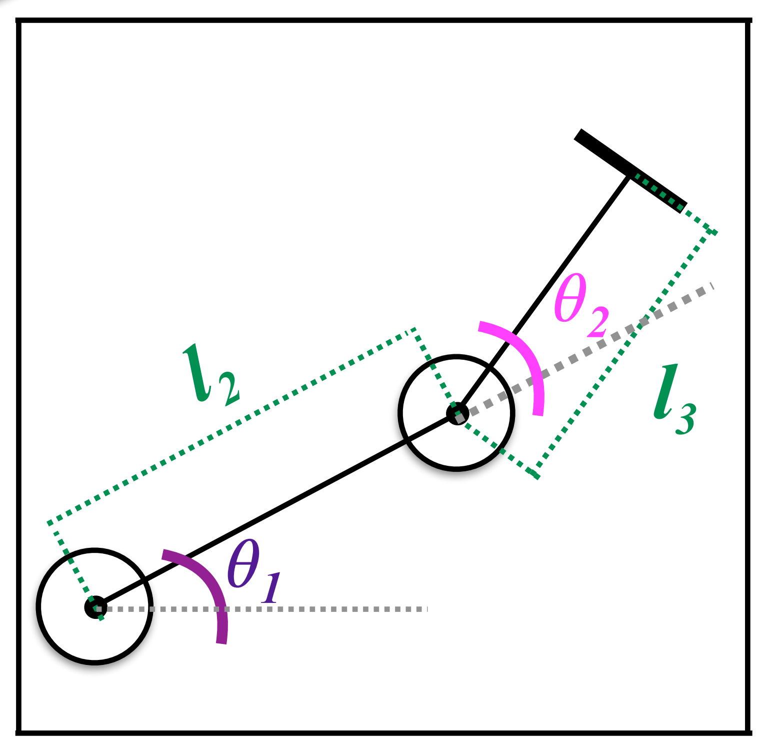

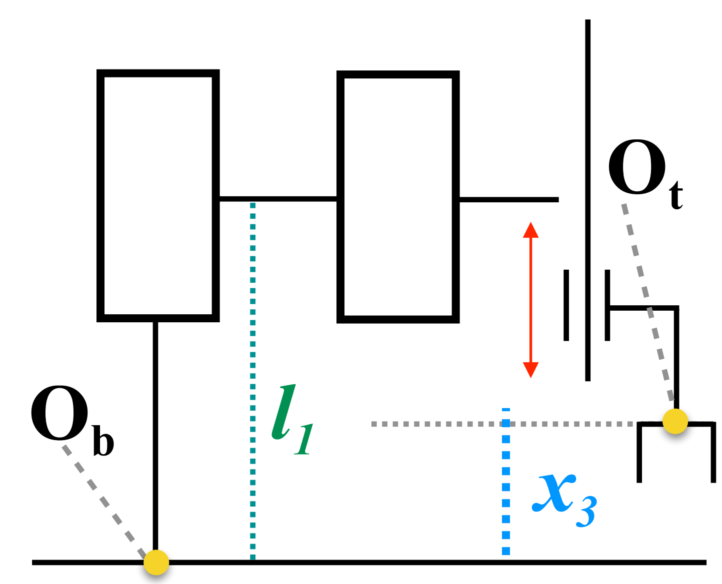

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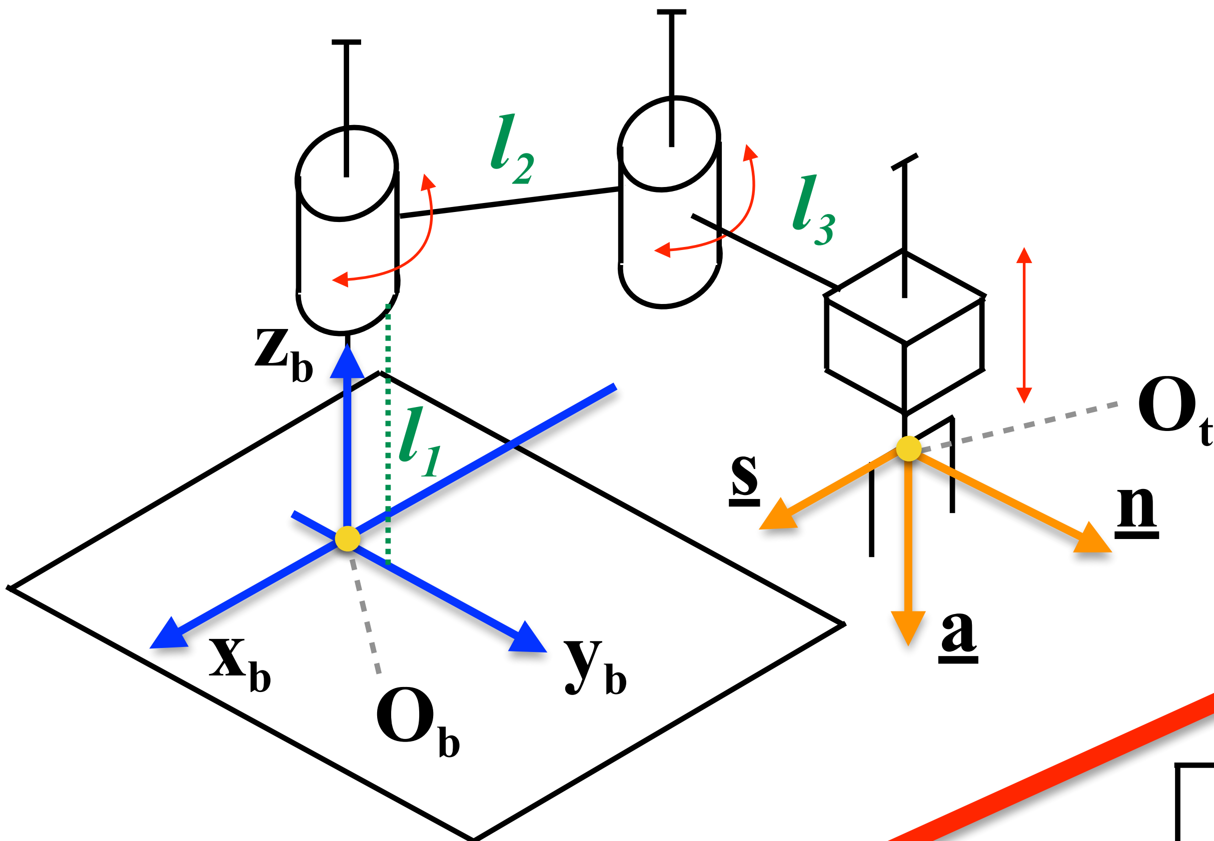

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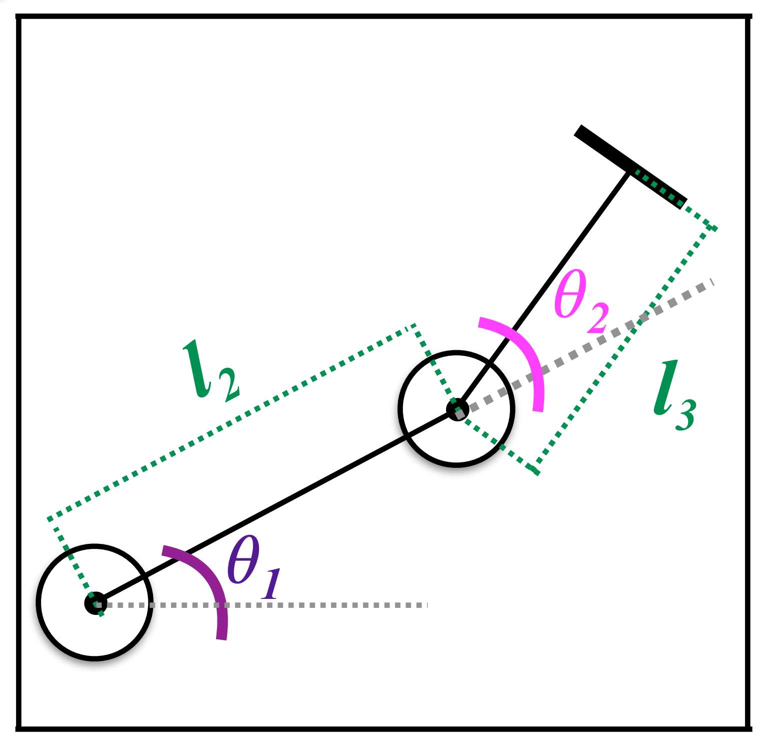

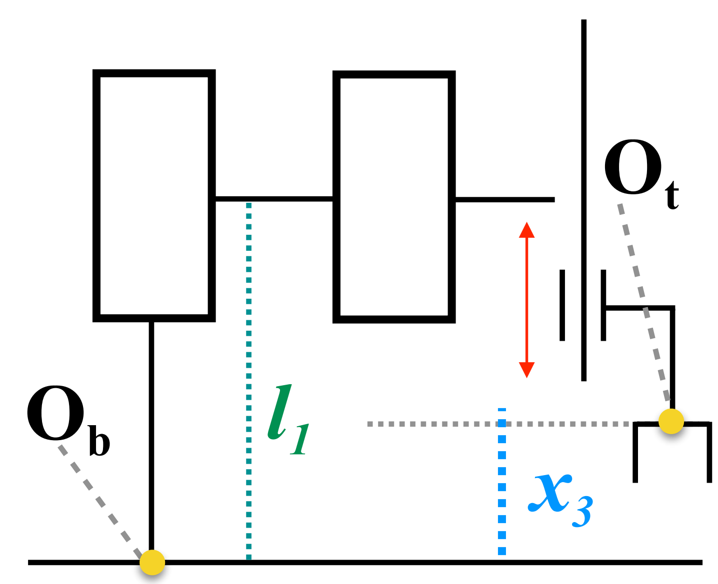

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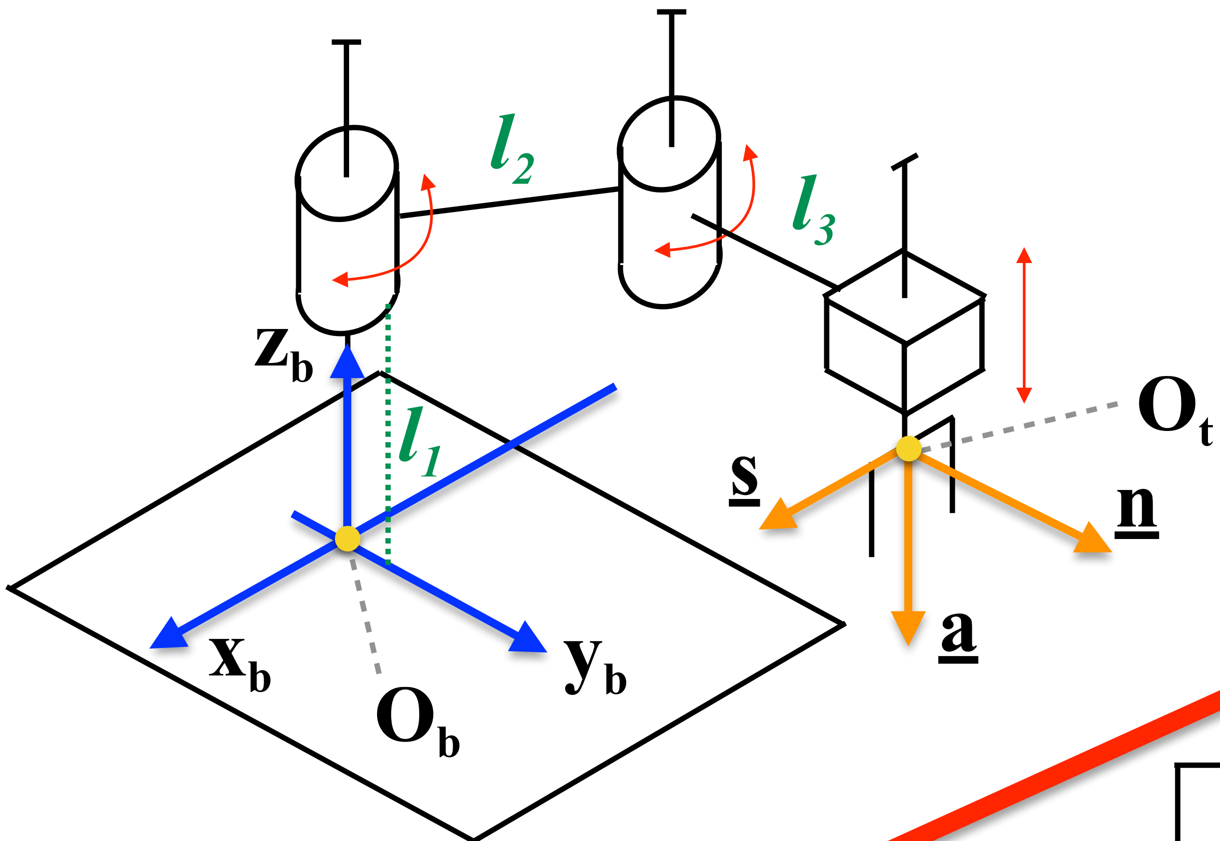

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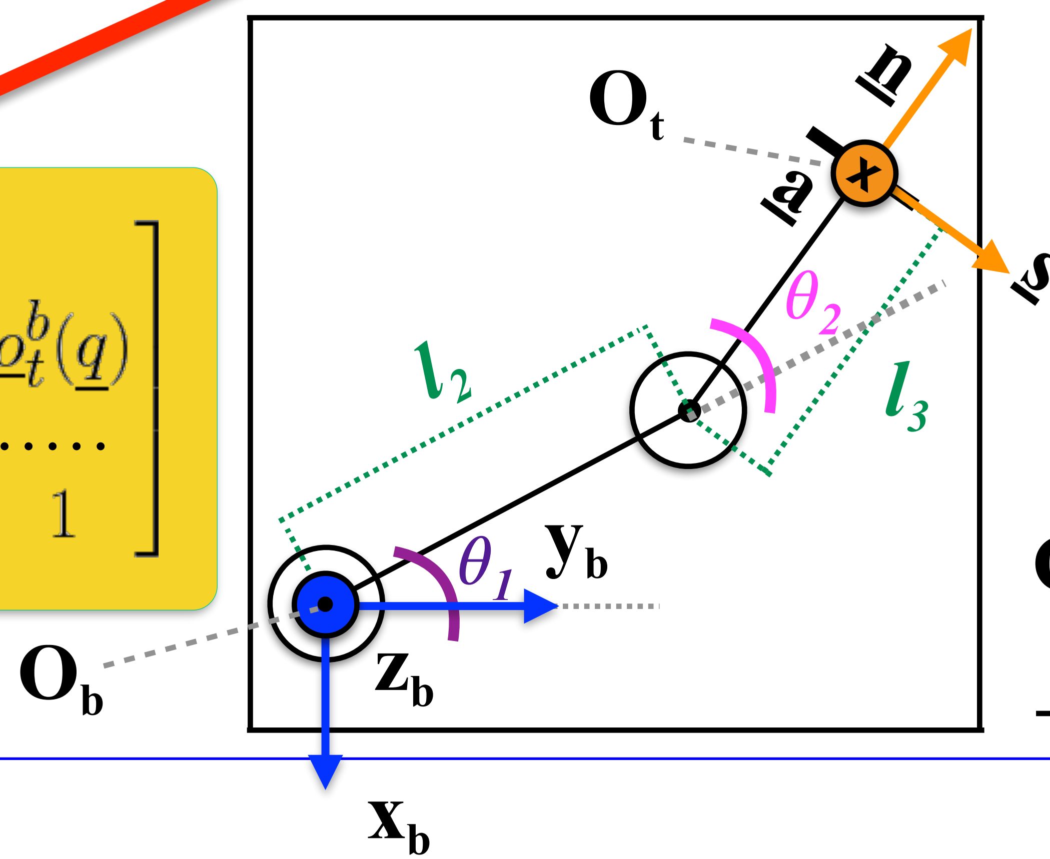

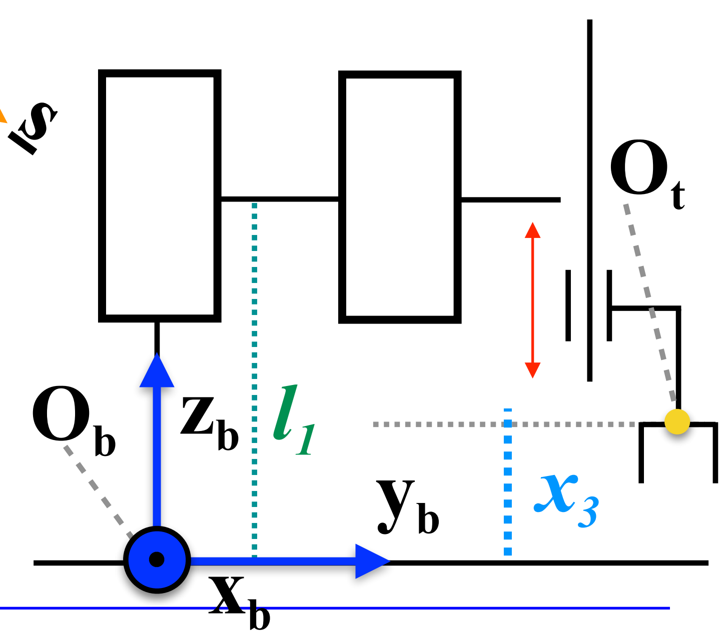

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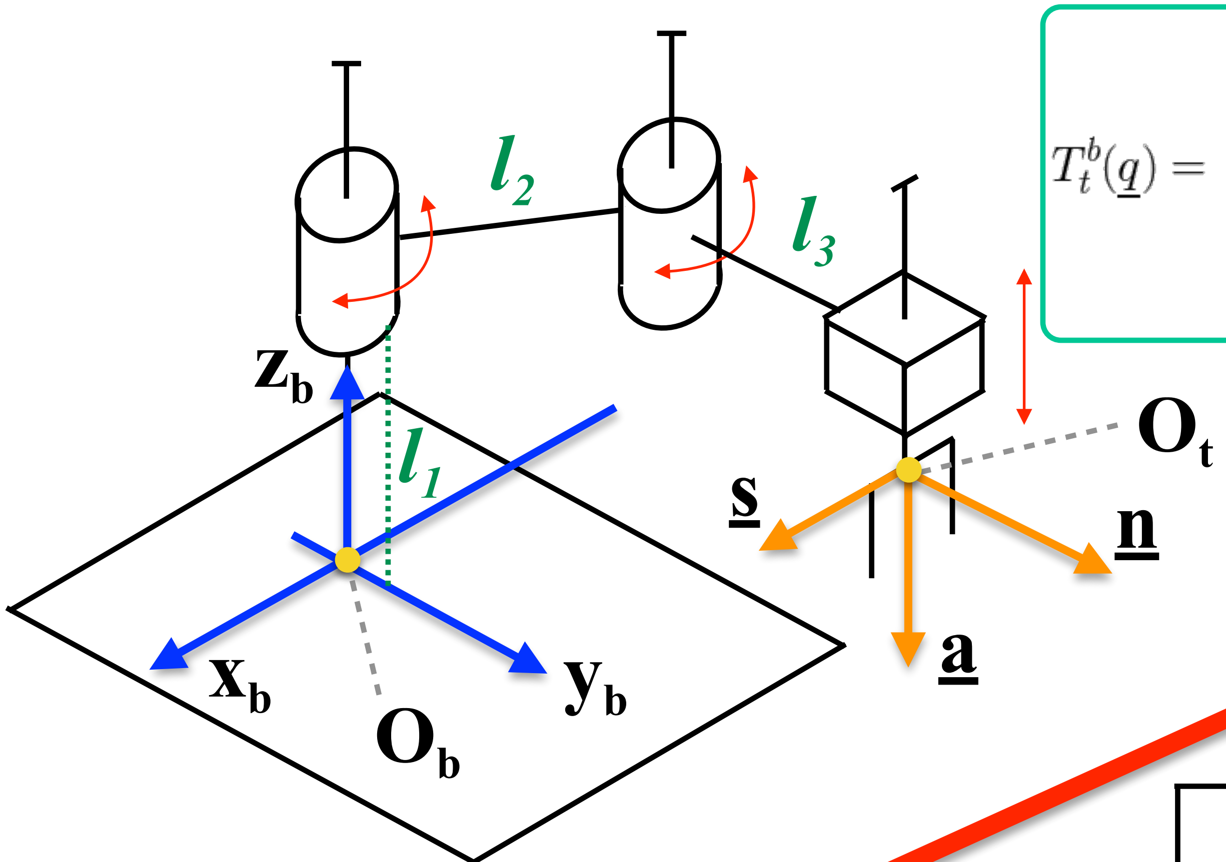

$$T_t^b(\underline{q}) = \begin{bmatrix} -sin(\theta_1+\theta_2) & sin[90^0-(\theta_1+\theta_2)] & 0 & -[l_2 sin(\theta_1)+l_3 sin(\theta_1+\theta_2)] \\ cos(\theta_1+\theta_2) & cos[90^0-(\theta_1+\theta_2)] & 0 & l_2 cos(\theta_1)+l_3 cos(\theta_1+\theta_2) \\ 0 & 0 & -1 & x_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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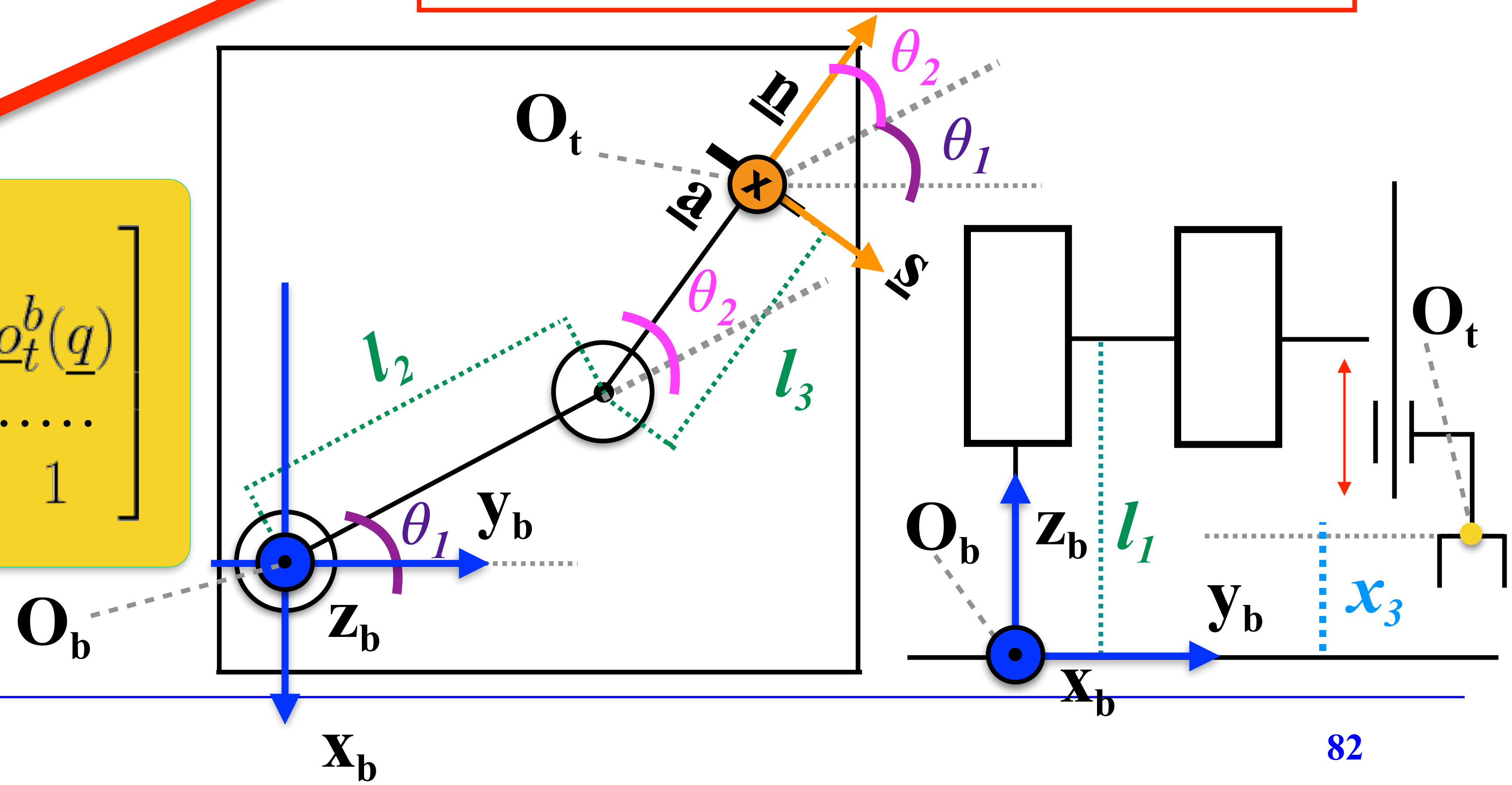

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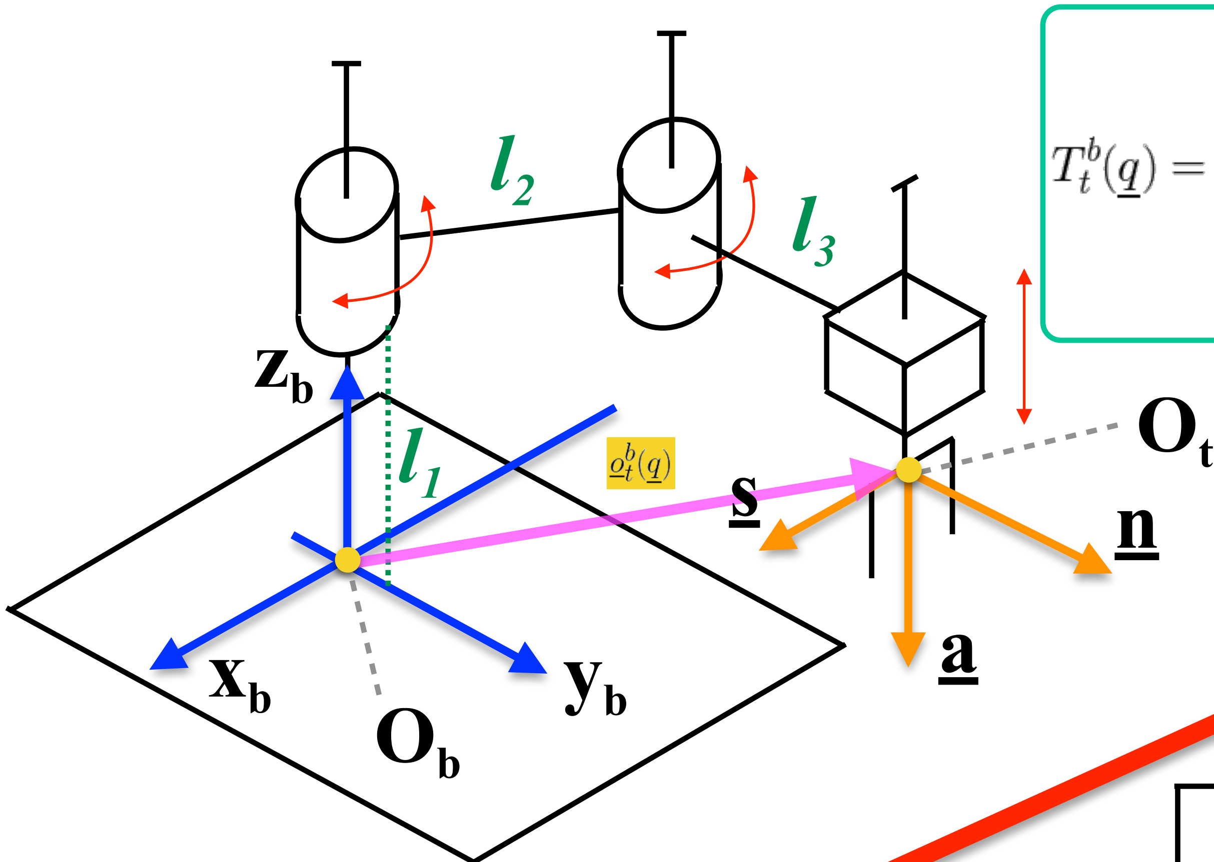

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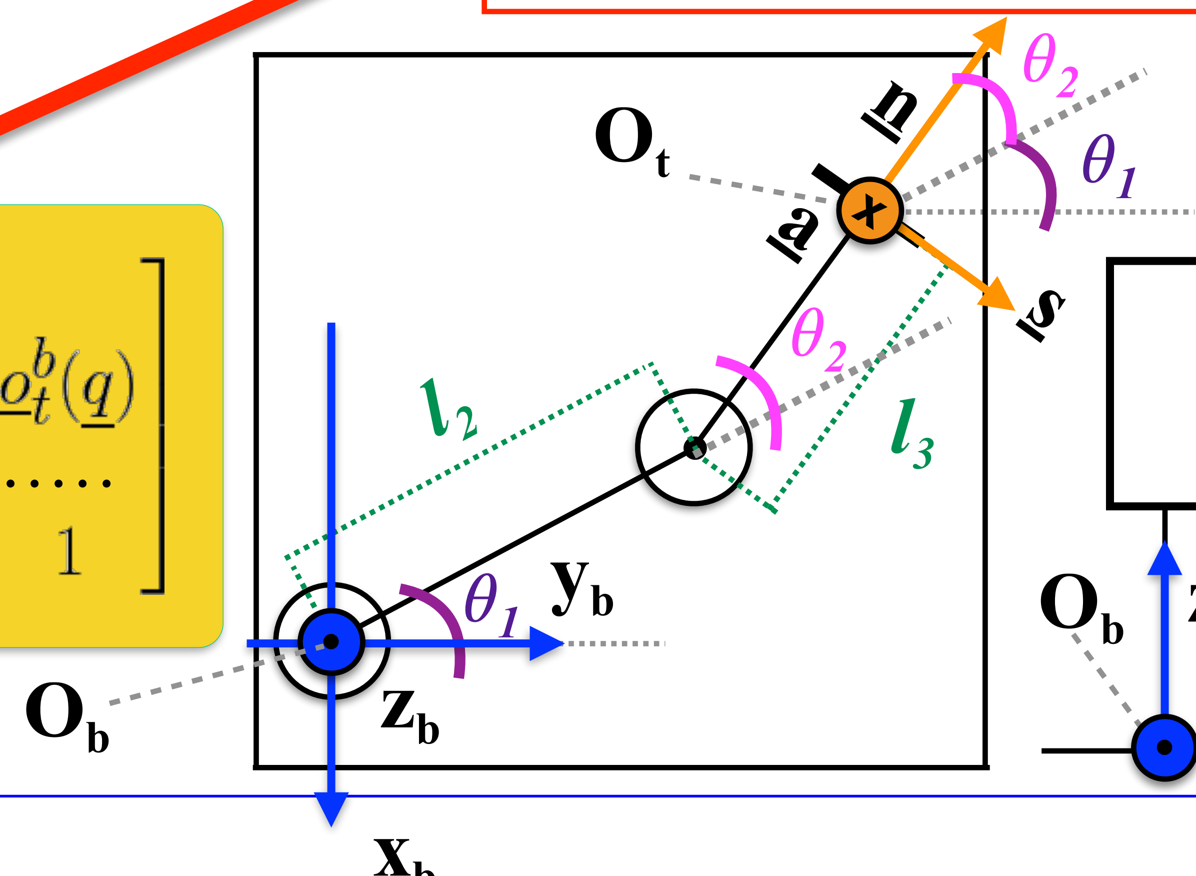

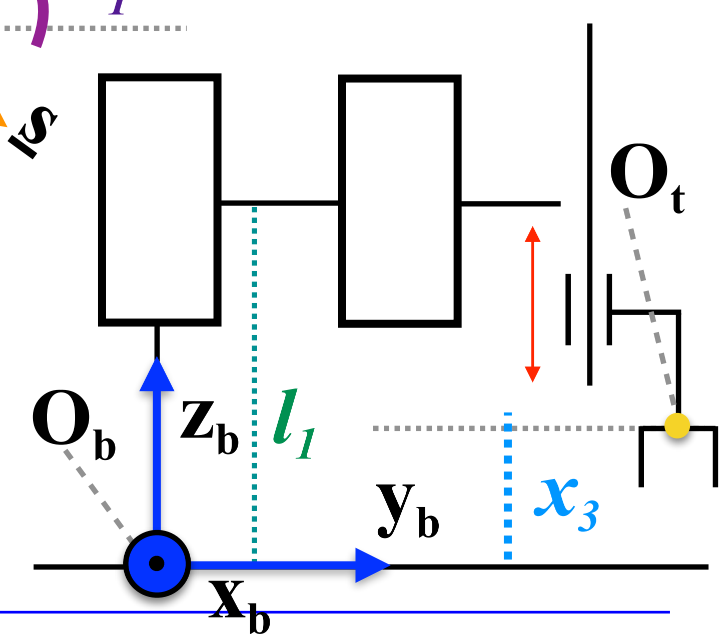

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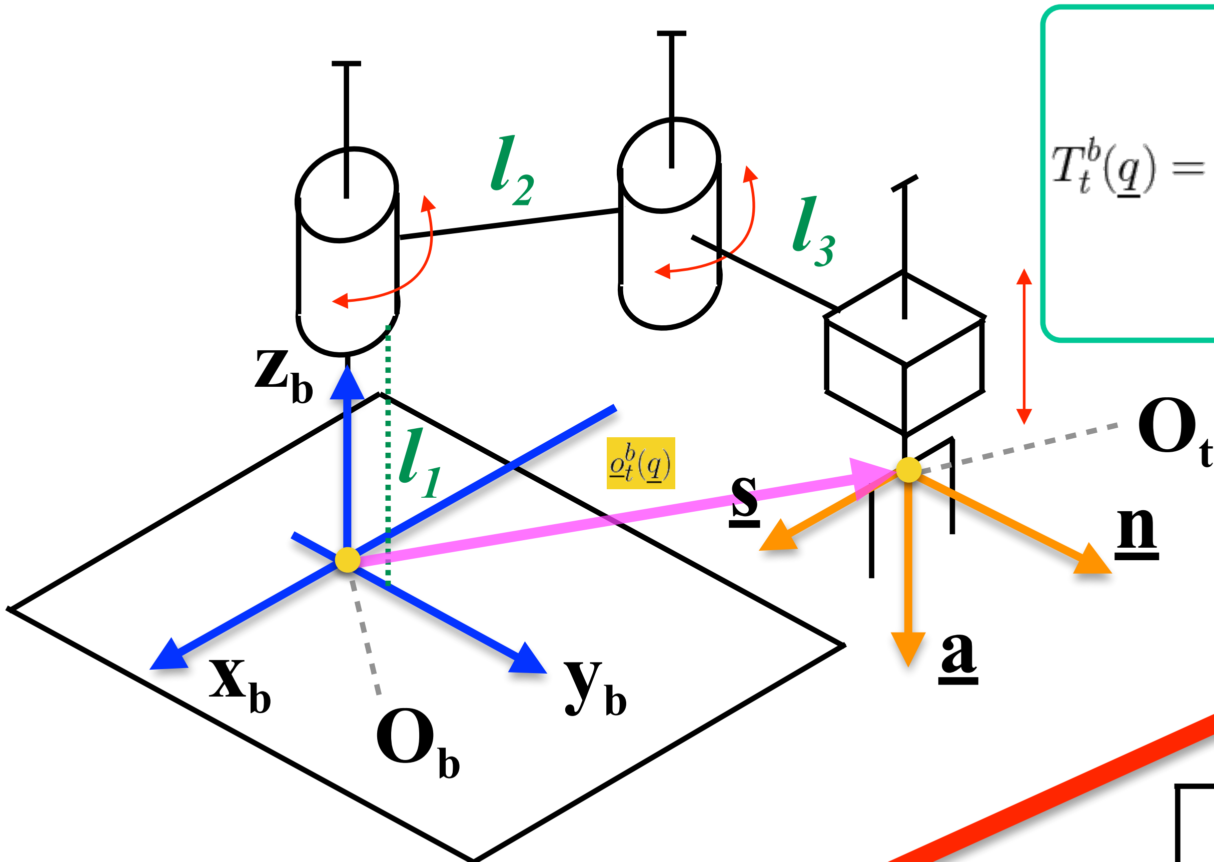

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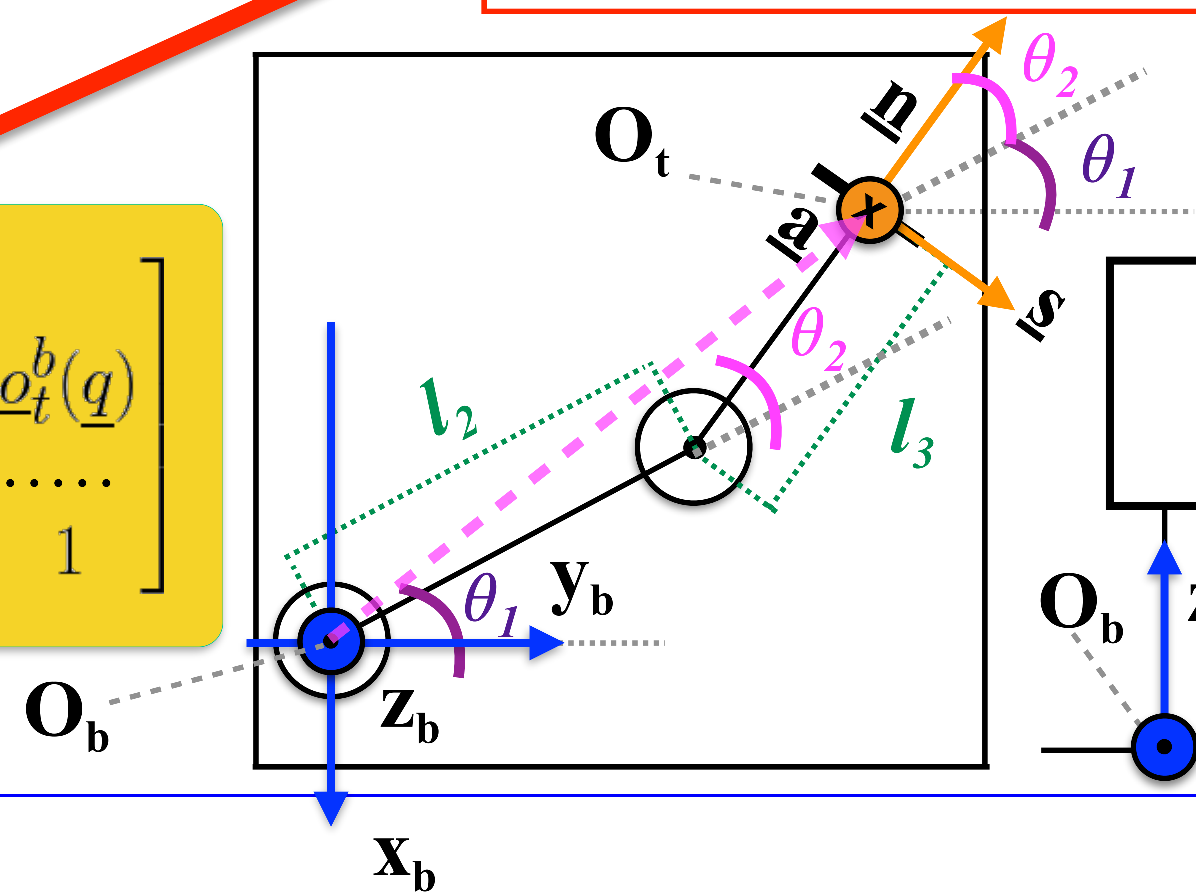

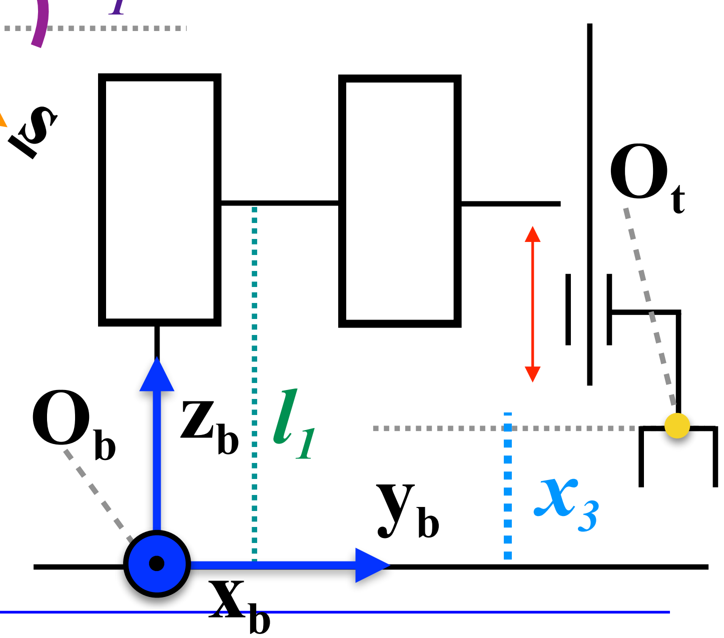

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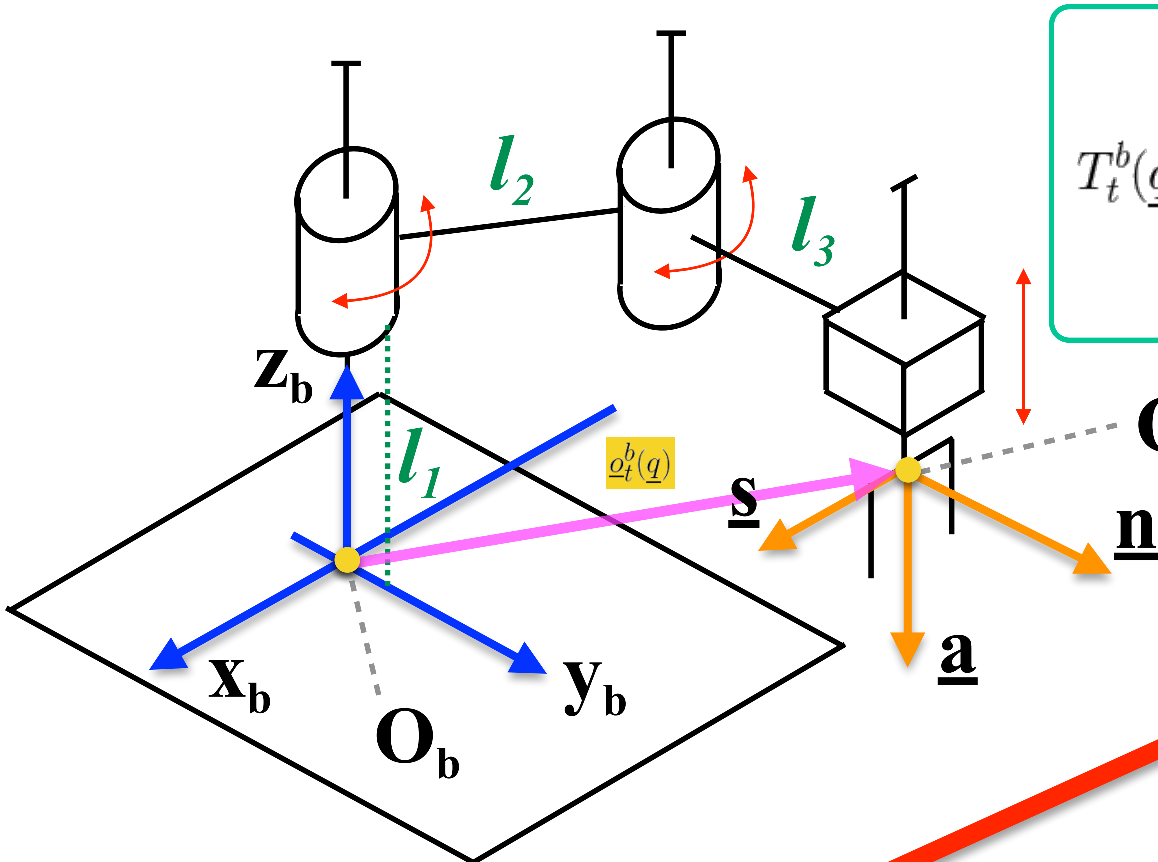

$$T_t^b(q) = \begin{bmatrix} -sin(\theta_1+\theta_2) & cos(\theta_1+\theta_2) & 0 & -[l_2 sin(\theta_1) + l_3 sin(\theta_1+\theta_2)] \\ cos(\theta_1+\theta_2) & sin(\theta_1+\theta_2) & 0 & l_2 cos(\theta_1) + l_3 cos(\theta_1+\theta_2) \\ 0 & 0 & -1 & x_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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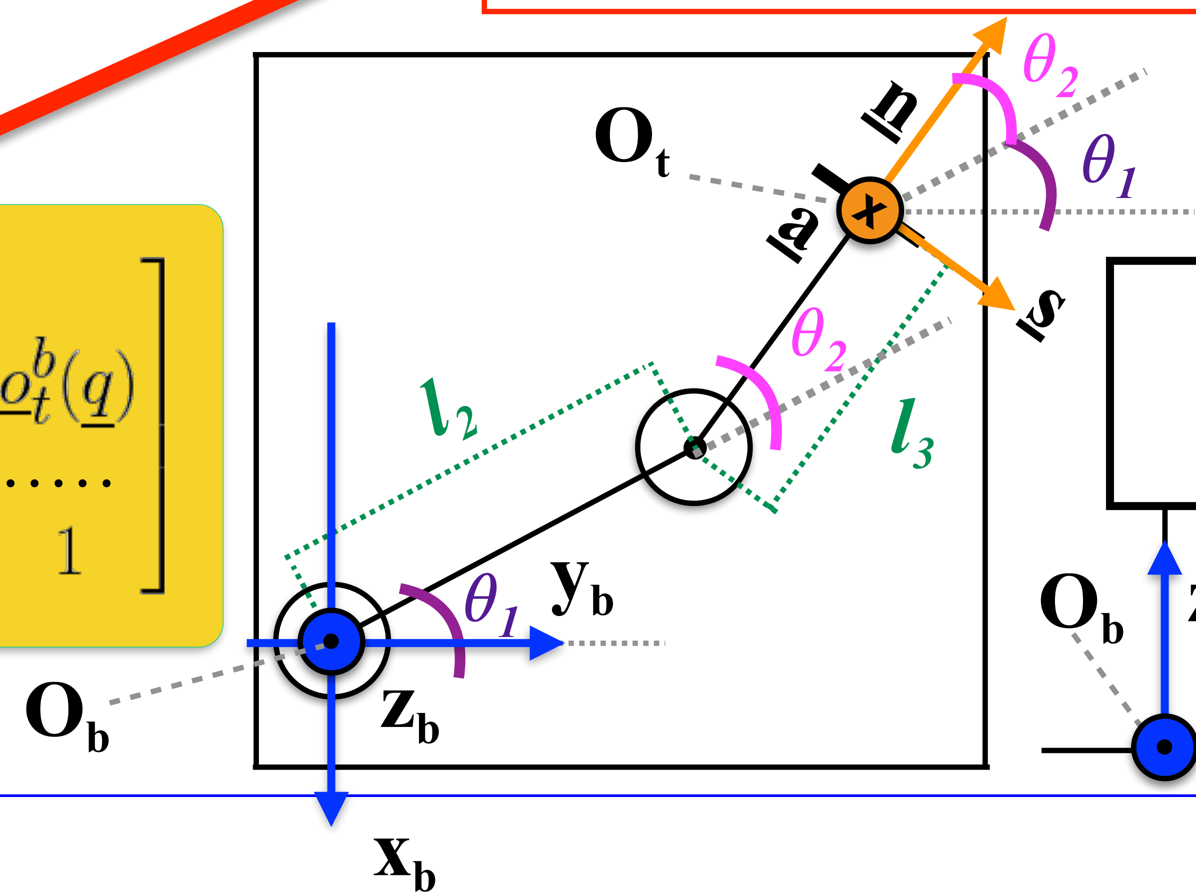

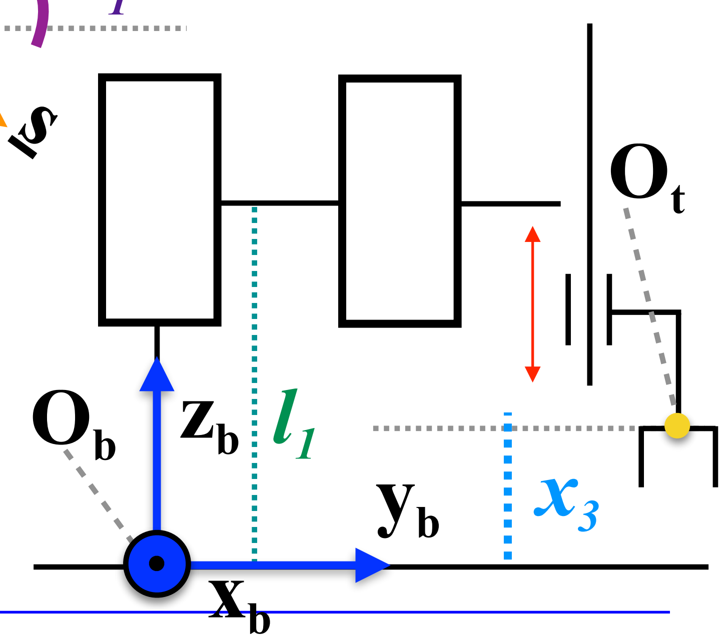

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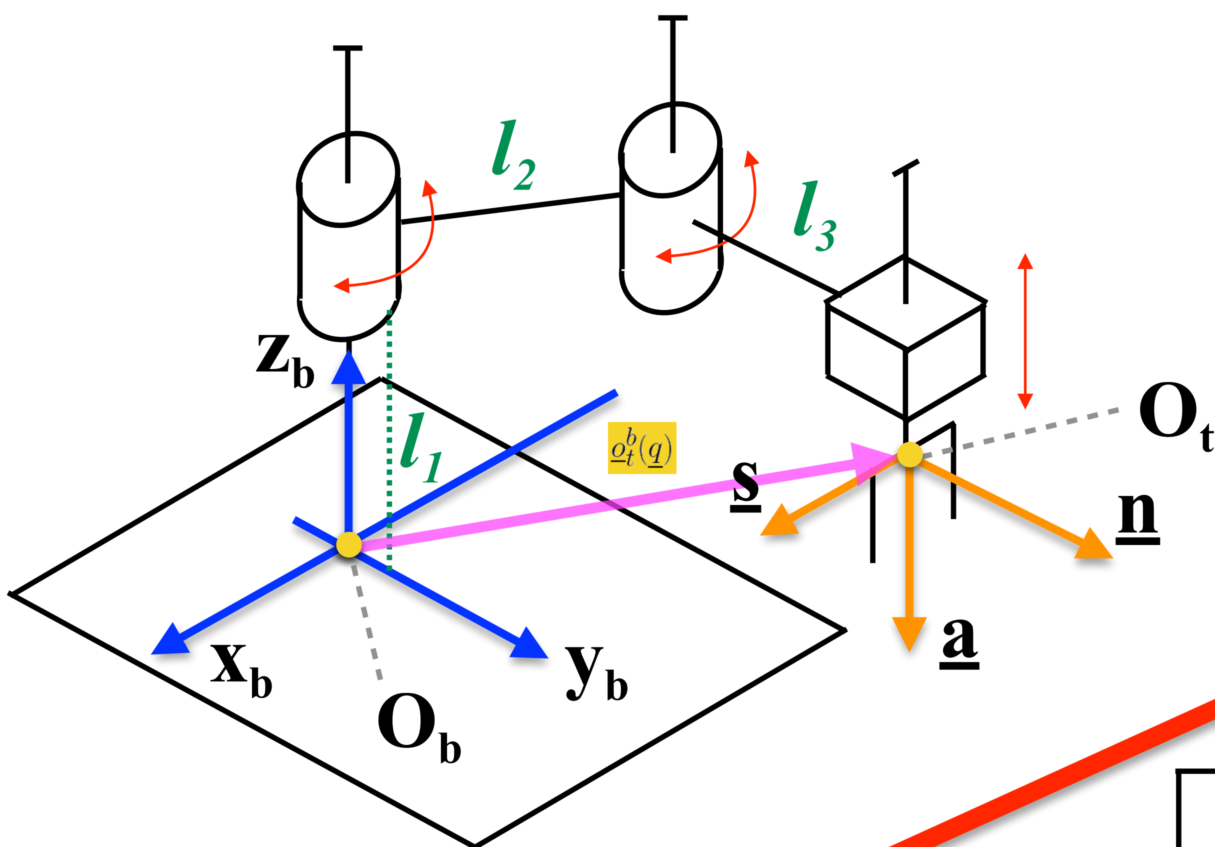
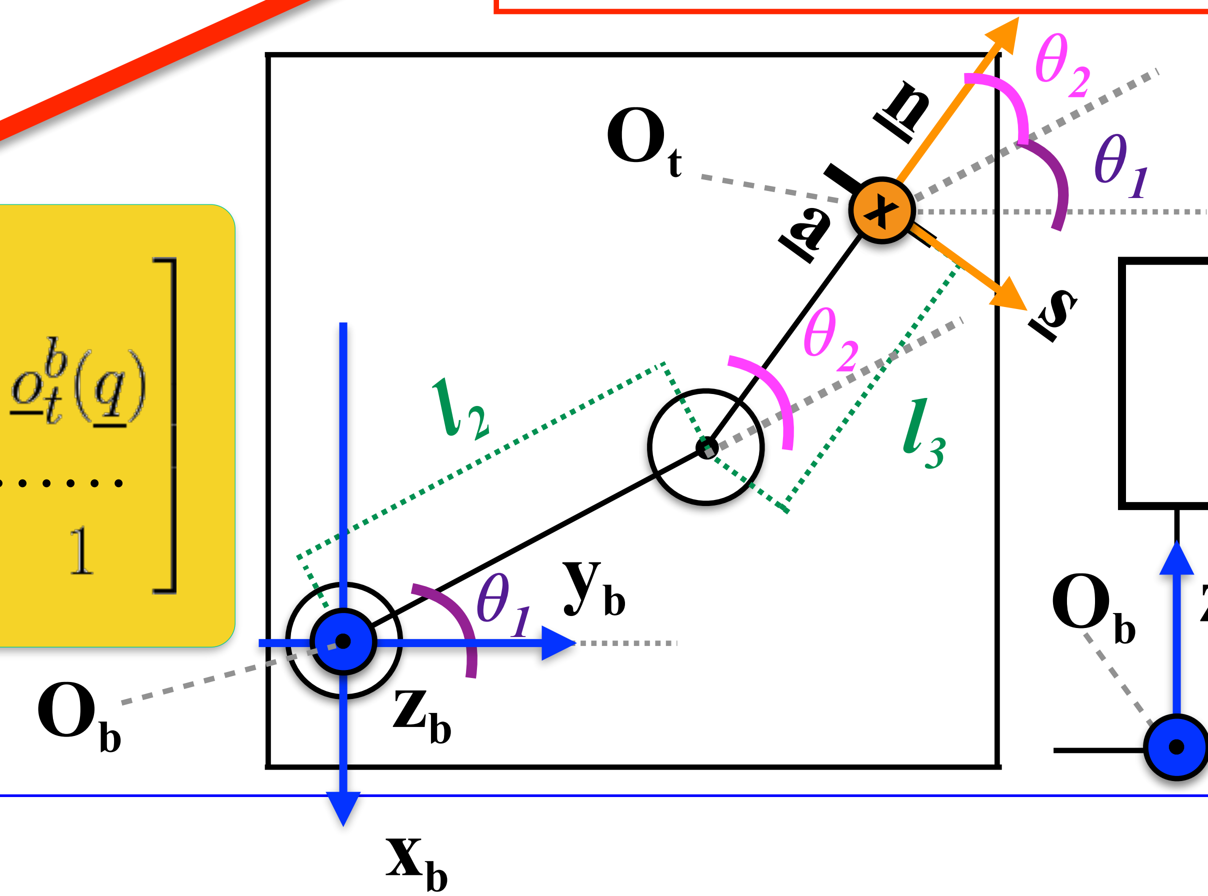
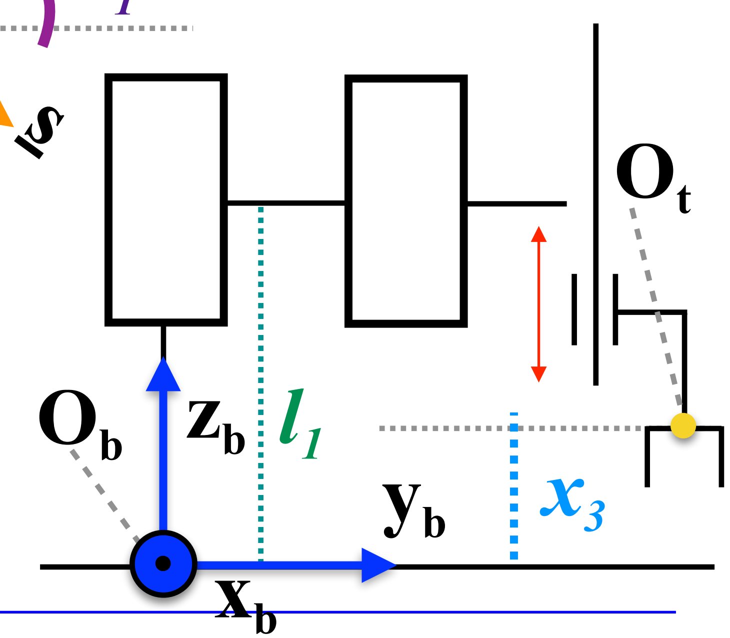


$$T_t^b(\underline{q}) = \begin{bmatrix} -s_{12} & c_{12} & 0 & -[l_2 s_1 + l_3 s_{12}] \\ c_{12} & s_{12} & 0 & l_2 c_1 + l_3 c_{12} \\ 0 & 0 & -1 & x_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$





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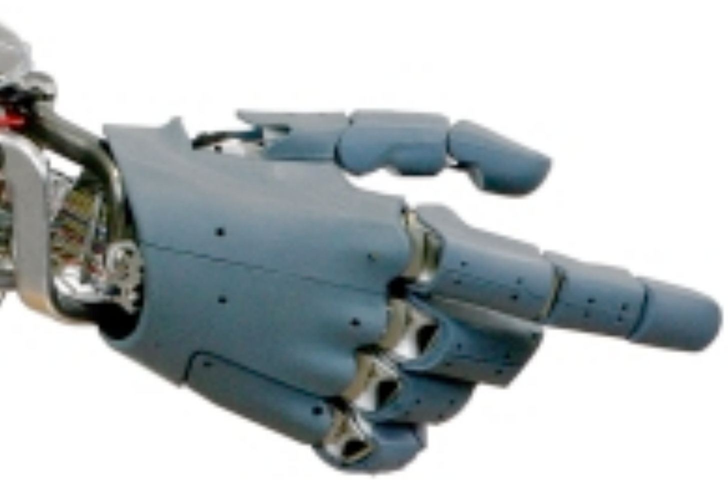


Anthropomorphic Robots

Anthropomorphic Robot Manipulators

Kawasaki anthropomorphic robot MX350L

Degrees of Freedom: 6 axes
Payload: 350 kg
Horizontal Reach: 3,018 mm
Vertical Reach: 3,502 mm
Repeatability: ±0.5 mm
Max. Linear Speed: 2,000 mm/s

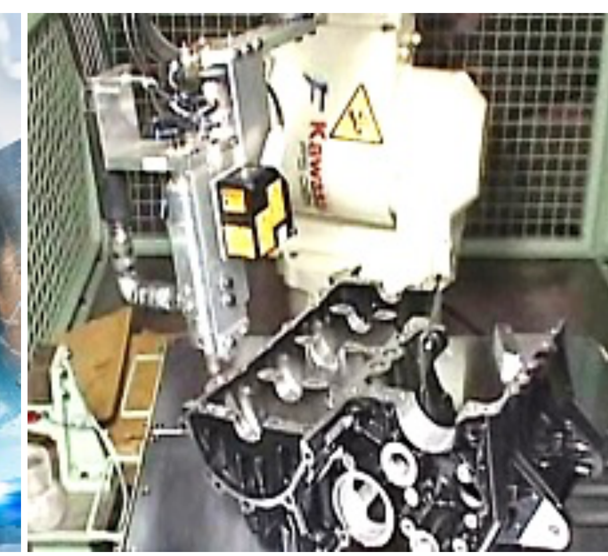
Mitsubishi Electric articulated robot RV-FR

Degrees of Freedom: 6 axes
Payload: Max.: 70 kg (154.324 lb)
Min.: 2 kg (4.409 lb)
Reach: Max.: 2,050 mm (80.71 in)
Min.: 504 mm (19.84 in)

EPSON VT6

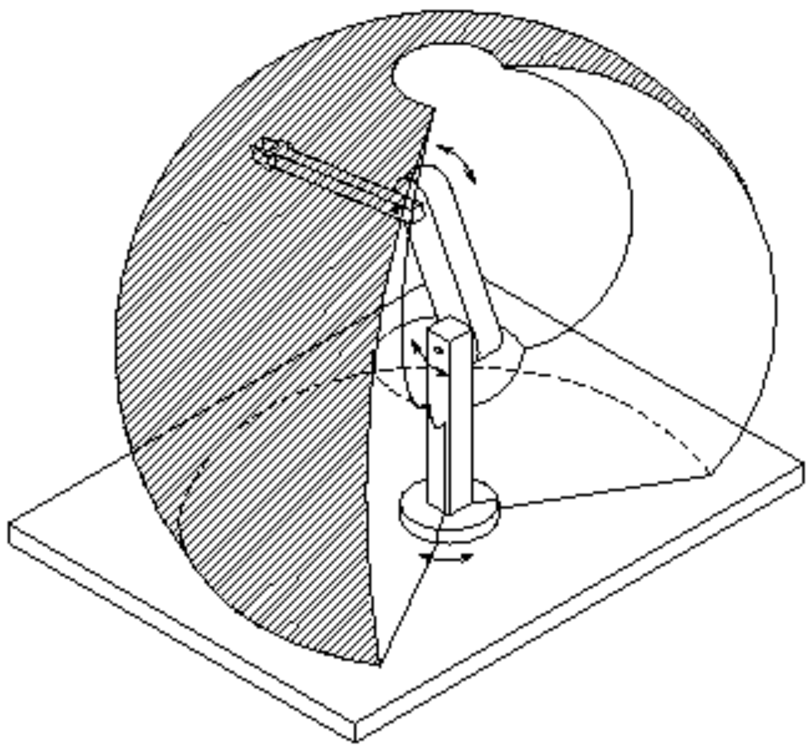
Motors	Servo with absolute encoders and brake
Operating Voltage	230V
No. Of Axes	6
Pay Load	6kg max
H-Reach	900mm
Horizontal Travel	900mm
Rotation Angle	270

Anthropomorphic Robot Manipulators: Applications

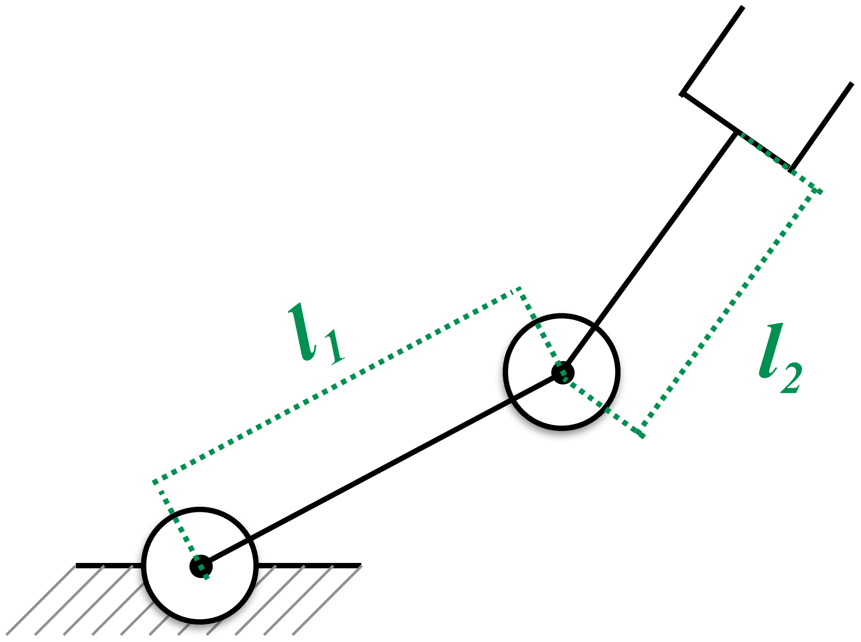


Source: https://robotics.kawasaki.com/en1/index.html?language_id=2

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 1

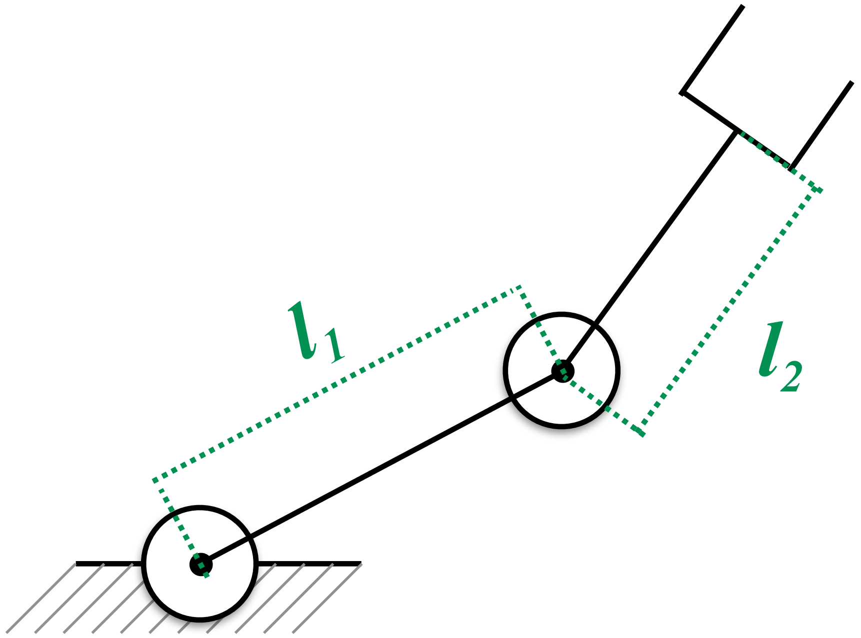


Anthropomorphic Robot (3 DoM)



Planar Anthropomorphic Robot (2 DoM)

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 1



Assign the Base Reference Frame and the Tool Reference Frame

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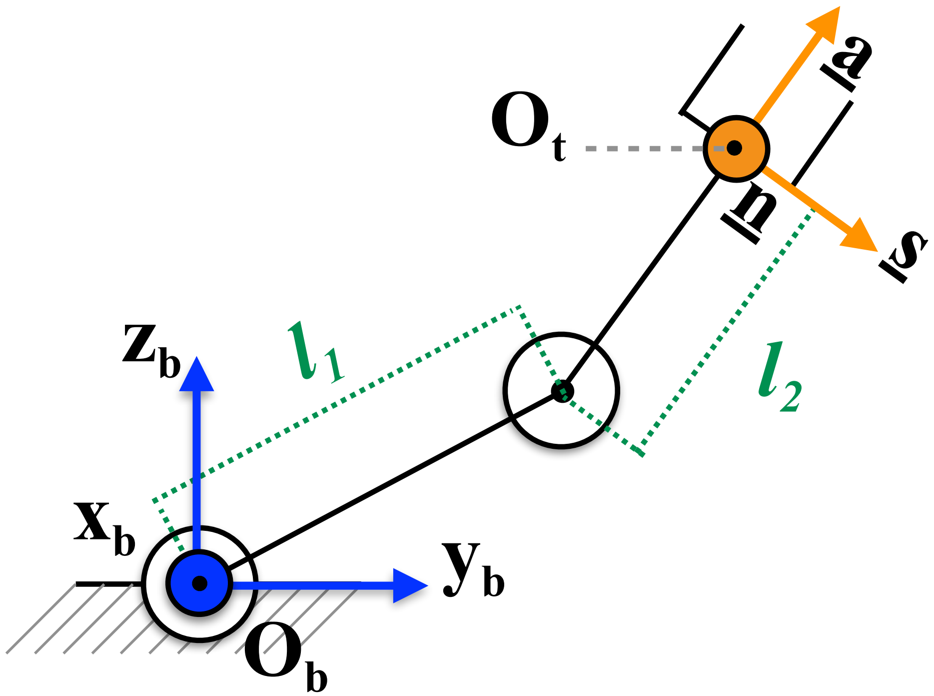

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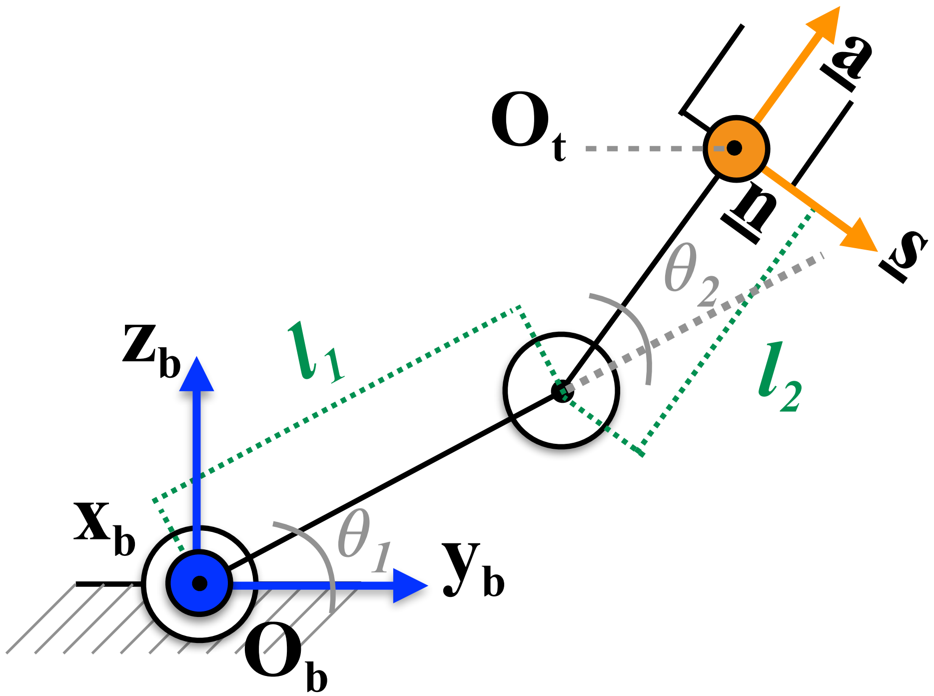

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Define the vector of the joint variables

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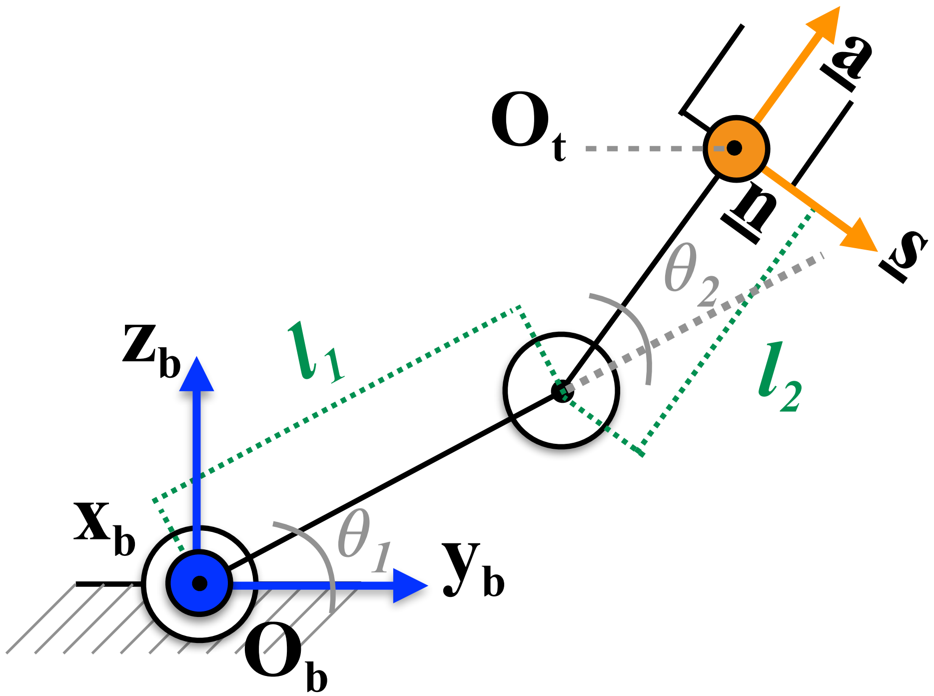

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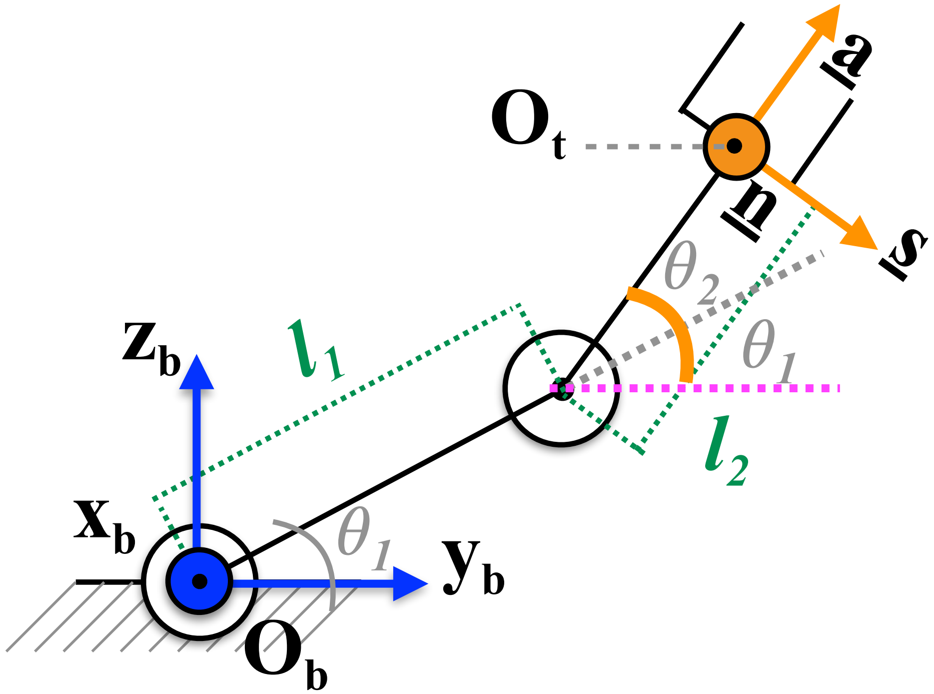

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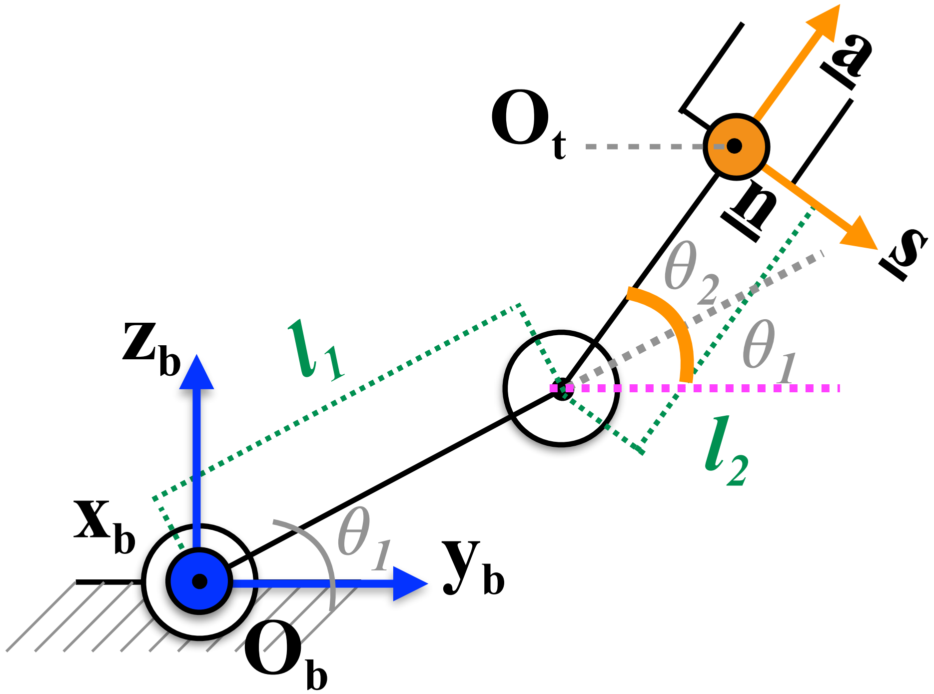
$$T_t^b(\underline{q}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & cos[90^0 - (\theta_1 + \theta_2)] & cos(\theta_1 + \theta_2) & l_1 cos(\theta_1) + l_2 cos(\theta_1 + \theta_2) \\ 0 & -sin[90^0 - (\theta_1 + \theta_2)] & sin(\theta_1 + \theta_2) & l_1 sin(\theta_1) + l_2 sin(\theta_1 + \theta_2) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 1



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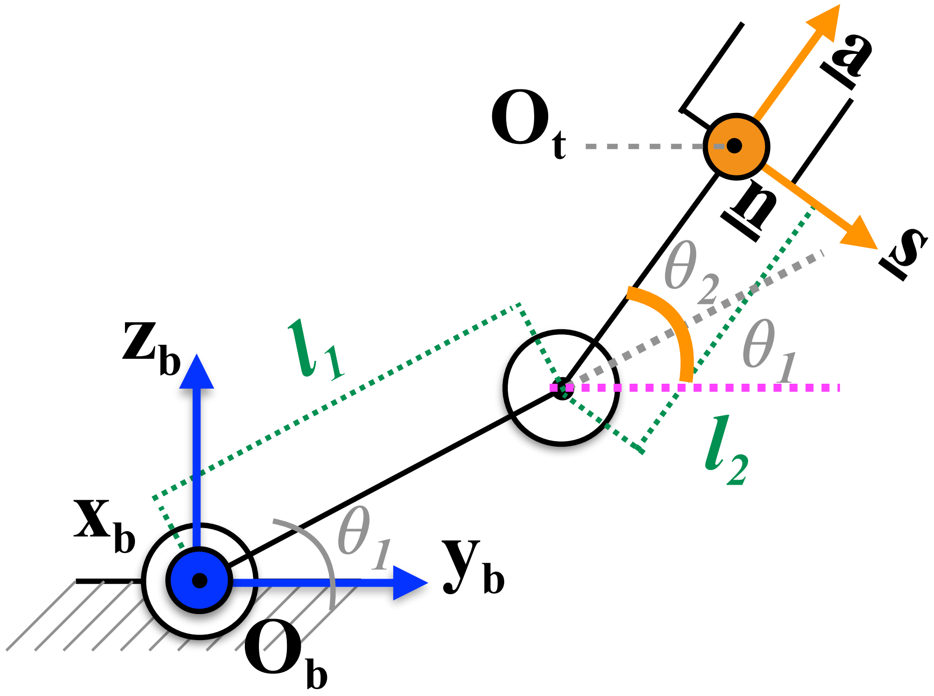
$$\underline{q} = [q_1,\ q_2]^T = [\theta_1,\ \theta_2]^T$$

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \begin{bmatrix} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_t^b(\underline{q}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & sin(\theta_1+\theta_2) & cos(\theta_1+\theta_2) & l_1 cos(\theta_1) + l_2 cos(\theta_1+\theta_2) \\ 0 & -cos(\theta_1+\theta_2) & sin(\theta_1+\theta_2) & l_1 sin(\theta_1) + l_2 sin(\theta_1+\theta_2) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 1



Assign the Base Reference Frame and the Tool Reference Frame

Define the vector of the joint variables

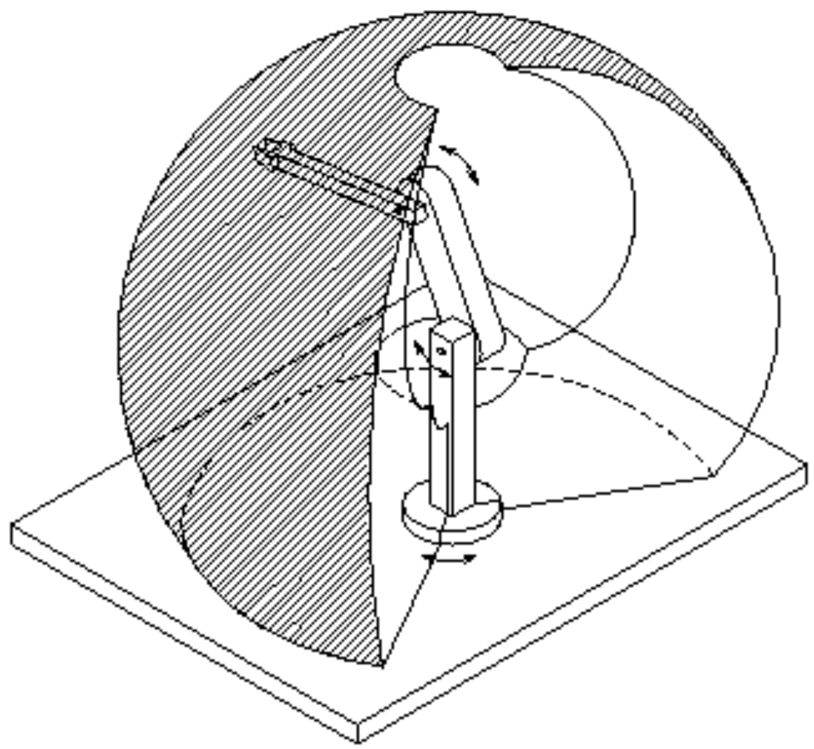
$$\underline{q} = [q_1,\, q_2]^T = [\theta_1,\, \theta_2]^T$$

Determine the homogeneous transformation matrix

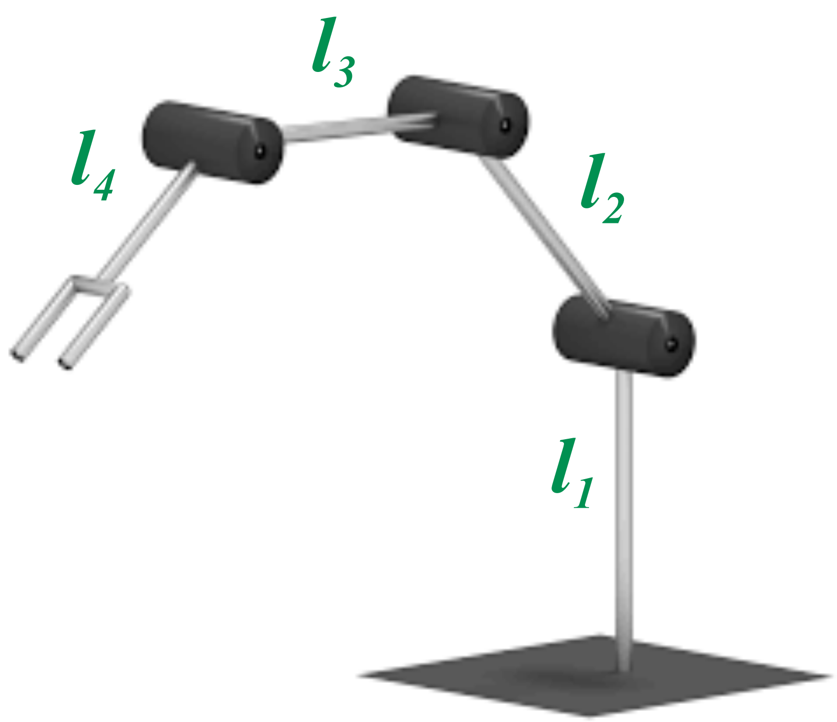
$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

$$T_t^b(\underline{q}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & s_{12} & c_{12} & l_1 c_1 + l_2 c_{12} \\ 0 & -c_{12} & s_{12} & l_1 s_1 + l_2 s_{12} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2

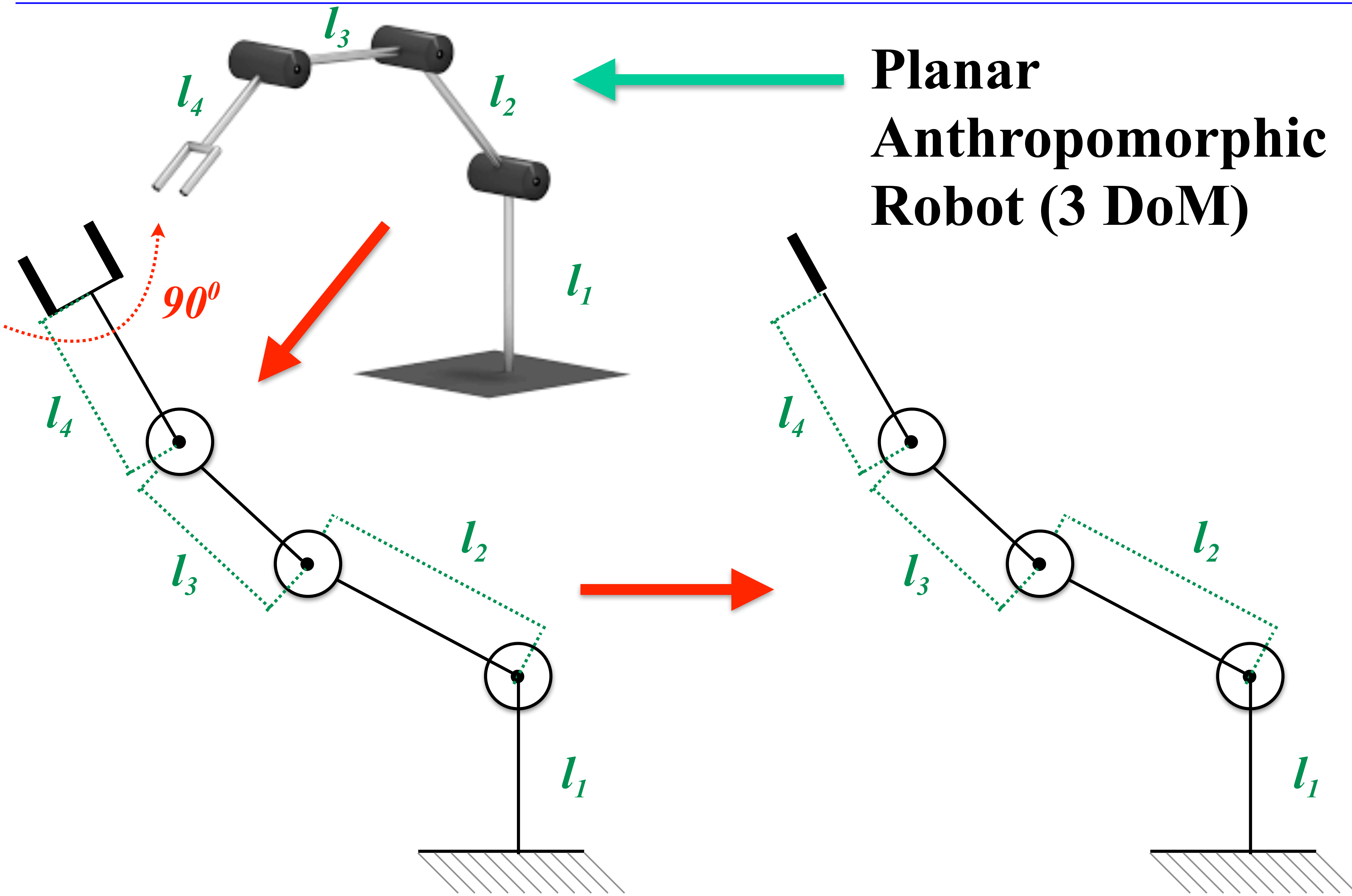


Anthropomorphic Robot (3 DoM)



Planar Anthropomorphic Robot (3 DoM)

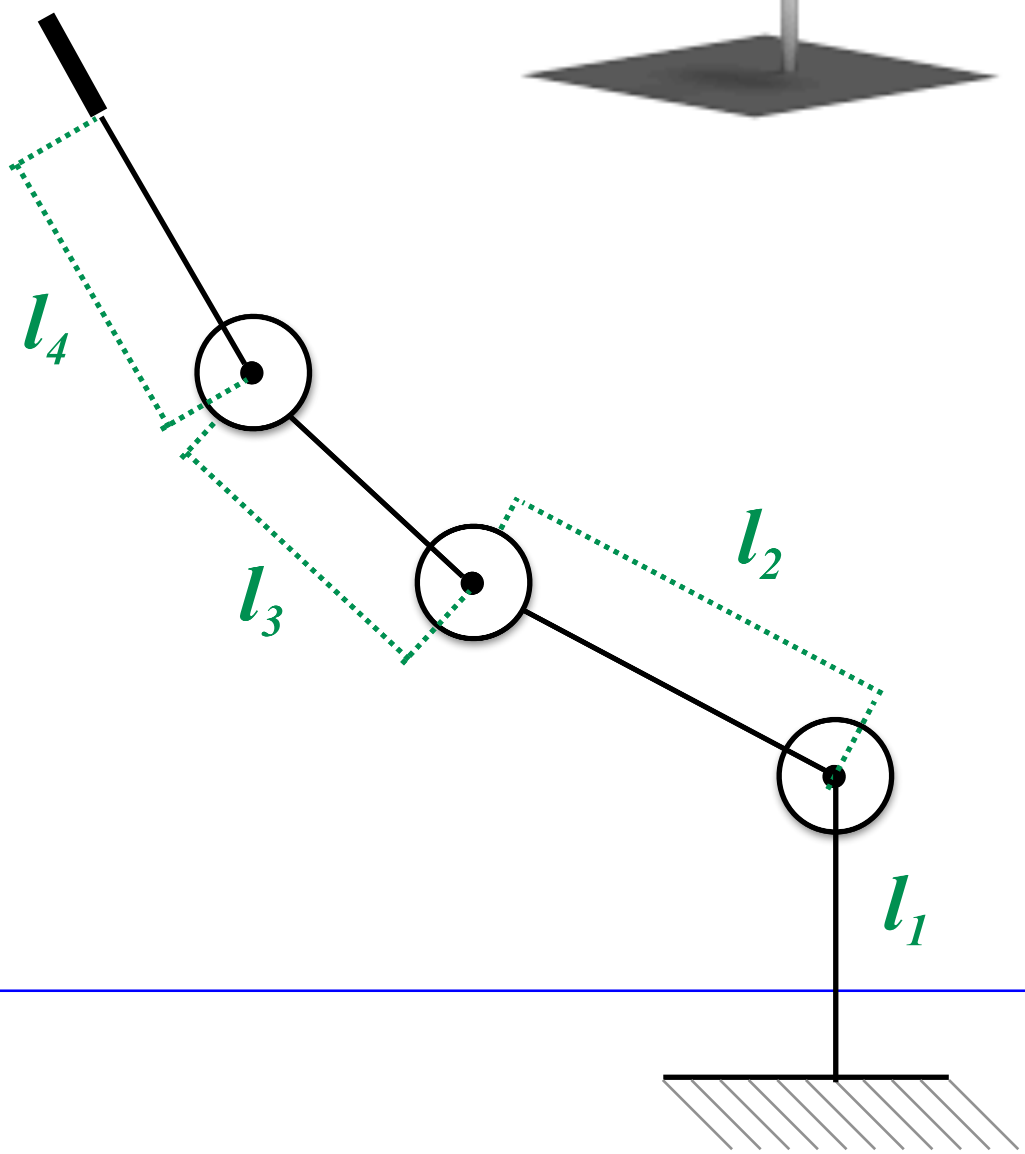
Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2



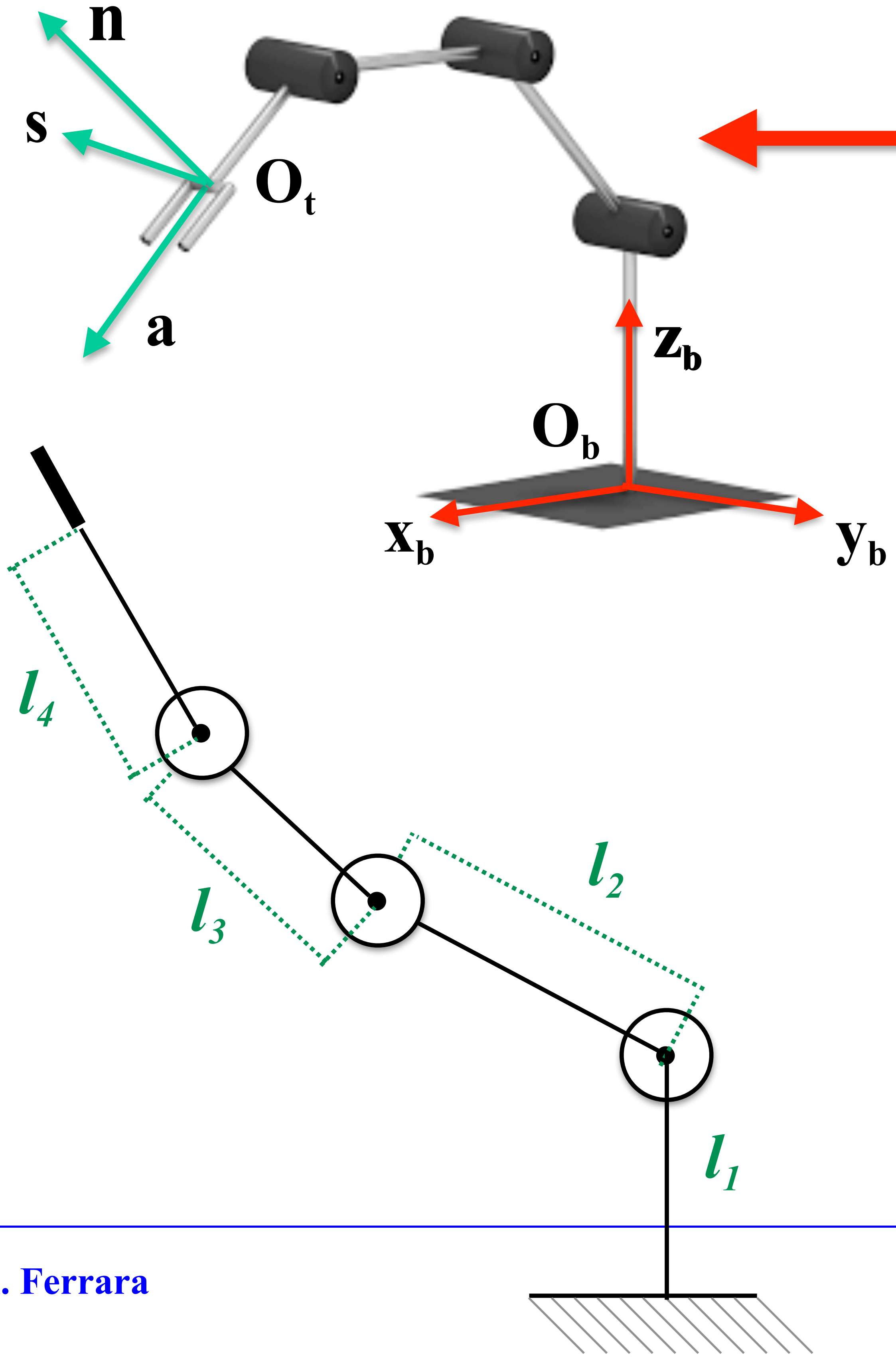
Planar Anthropomorphic Robot (3 DoM)

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2

Assign the Base Reference Frame and the Tool Reference Frame

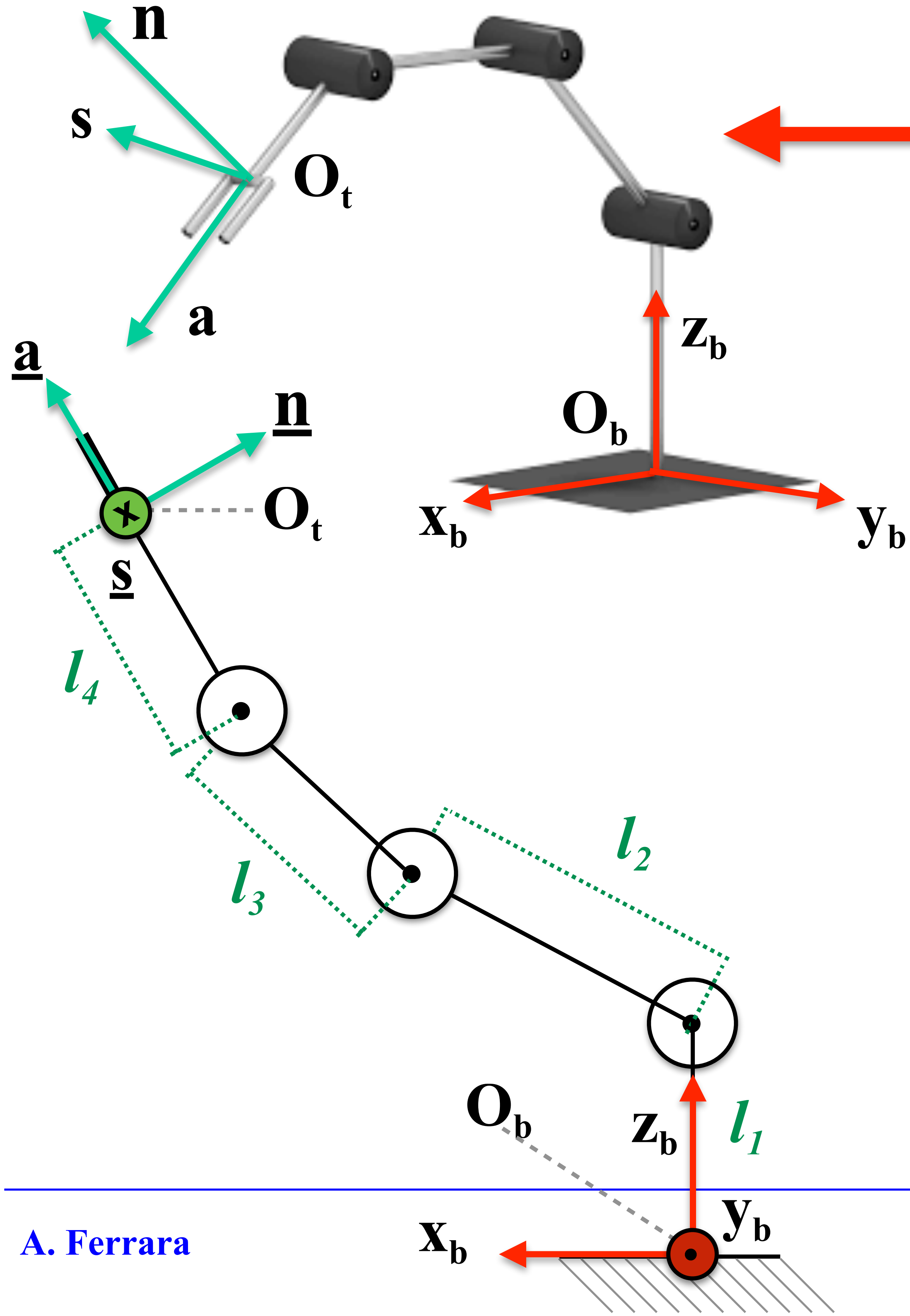


Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2



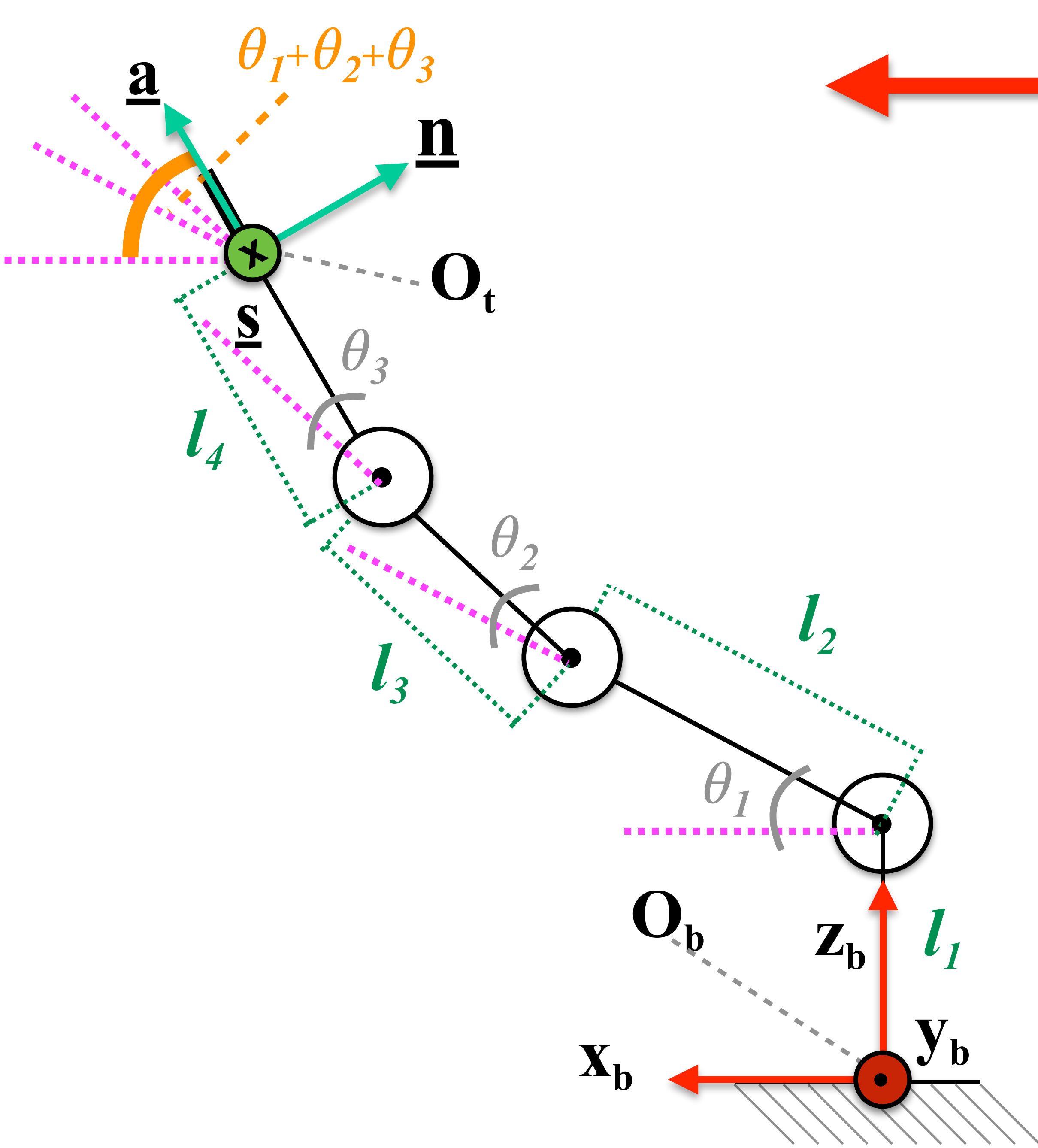
Assign the Base Reference Frame and the Tool Reference Frame

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2



Assign the Base Reference Frame and the Tool Reference Frame

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2

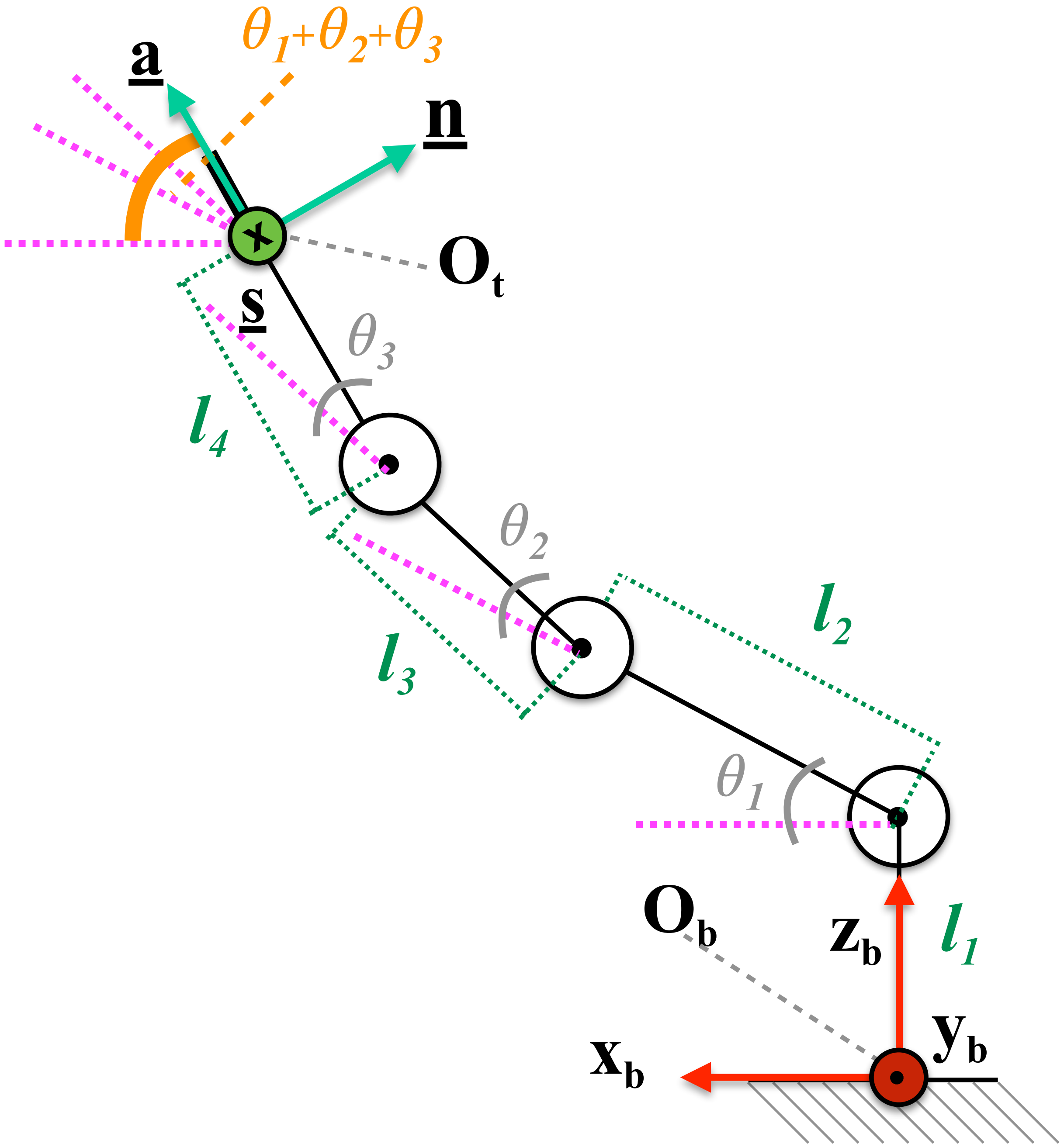


Assign the Base Reference Frame and the Tool Reference Frame

Define the vector of the joint variables

$$\underline{q} = [q_1, q_2, q_3]^T = [\theta_1, \theta_2, \theta_3]^T$$

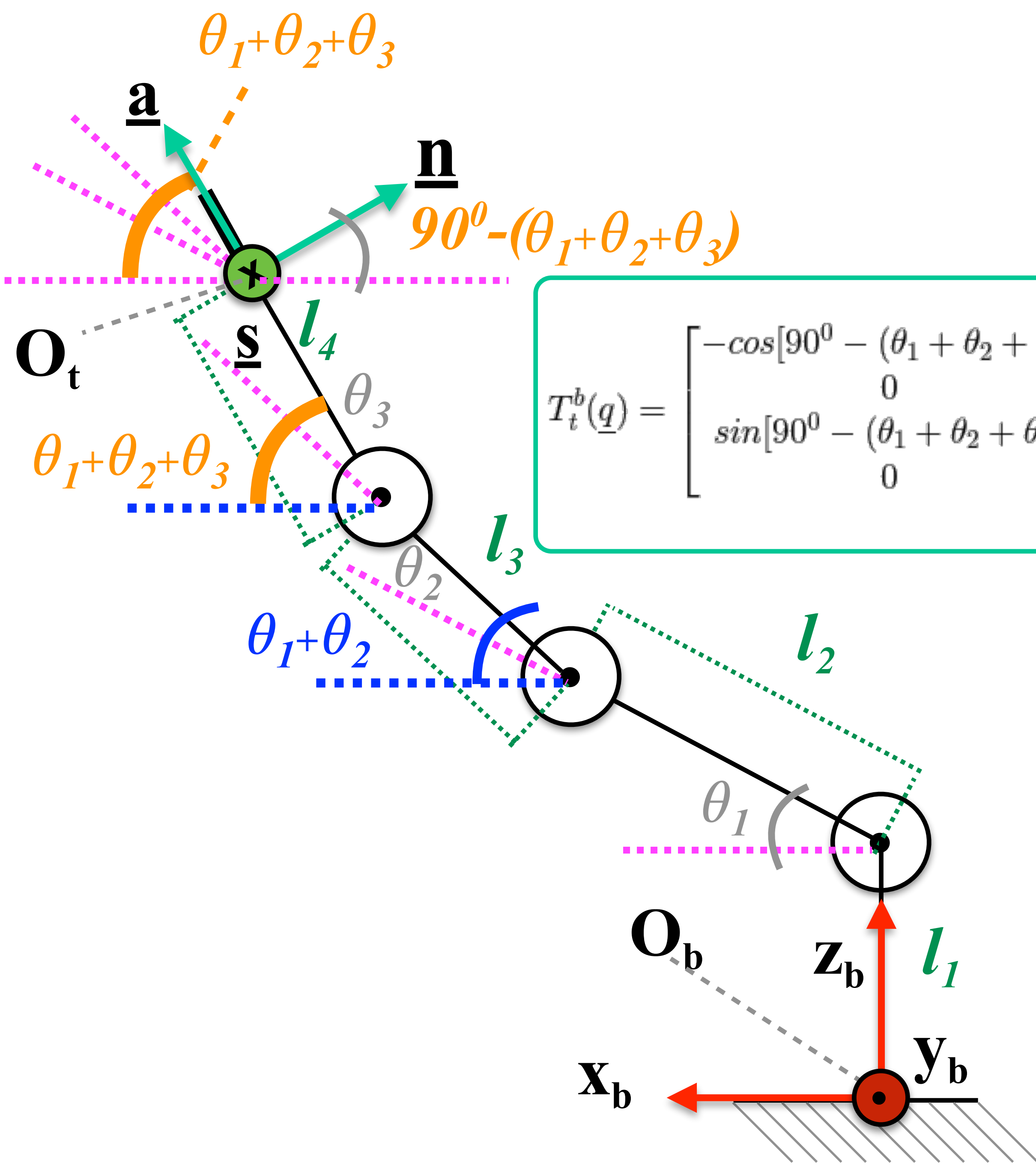
Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2



Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2

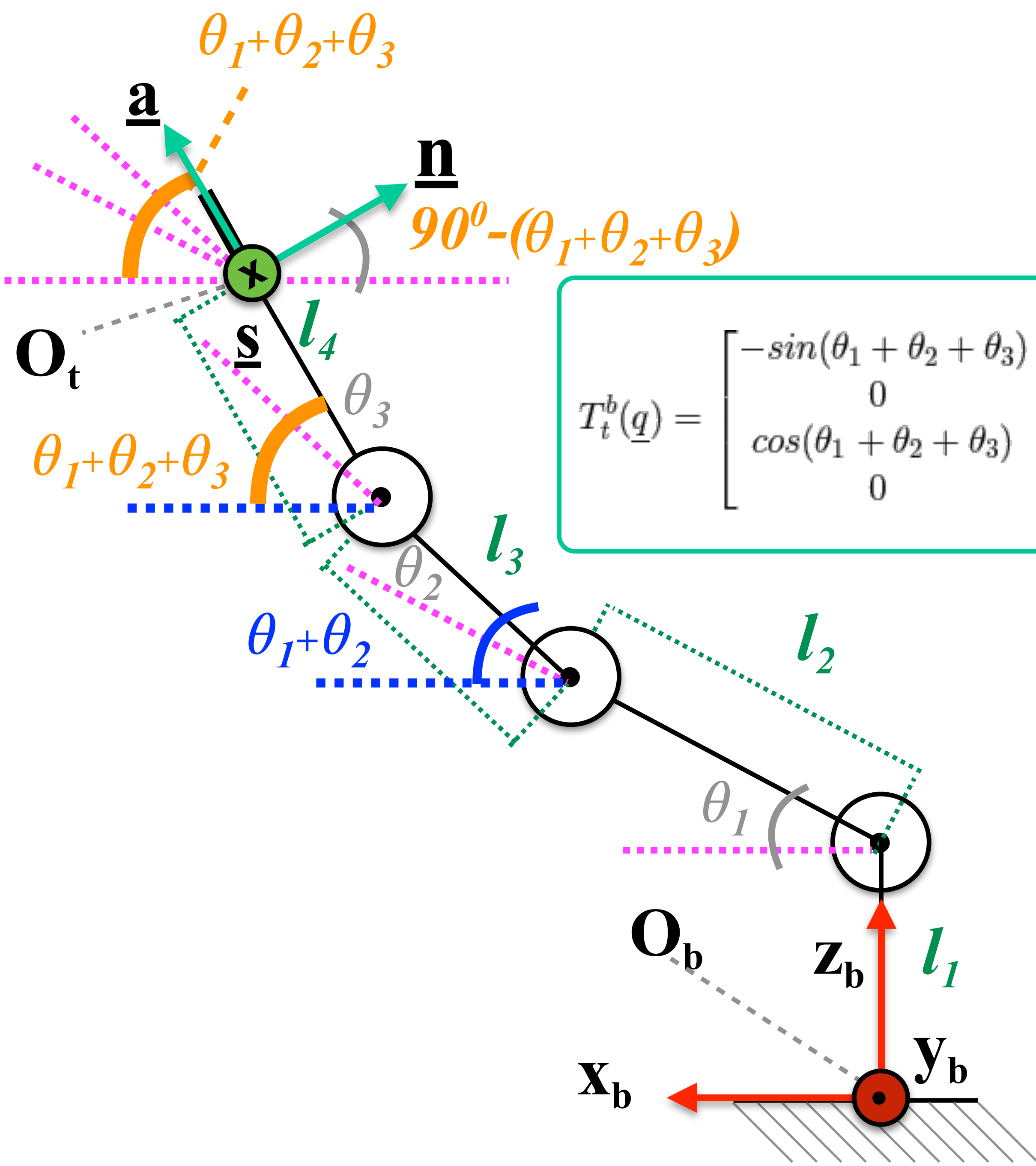


Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \begin{bmatrix} -cos[90^0-(\theta_1+\theta_2+\theta_3)] & 0 & cos(\theta_1+\theta_2+\theta_3) & l_2cos(\theta_1)+l_3cos(\theta_1+\theta_2)+l_4cos(\theta_1+\theta_2+\theta_3) \\ 0 & -1 & 0 & 0 \\ sin[90^0-(\theta_1+\theta_2+\theta_3)] & 0 & sin(\theta_1+\theta_2+\theta_3) & l_1+l_2sin(\theta_1)+l_3sin(\theta_1+\theta_2)+l_4sin(\theta_1+\theta_2+\theta_3) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2

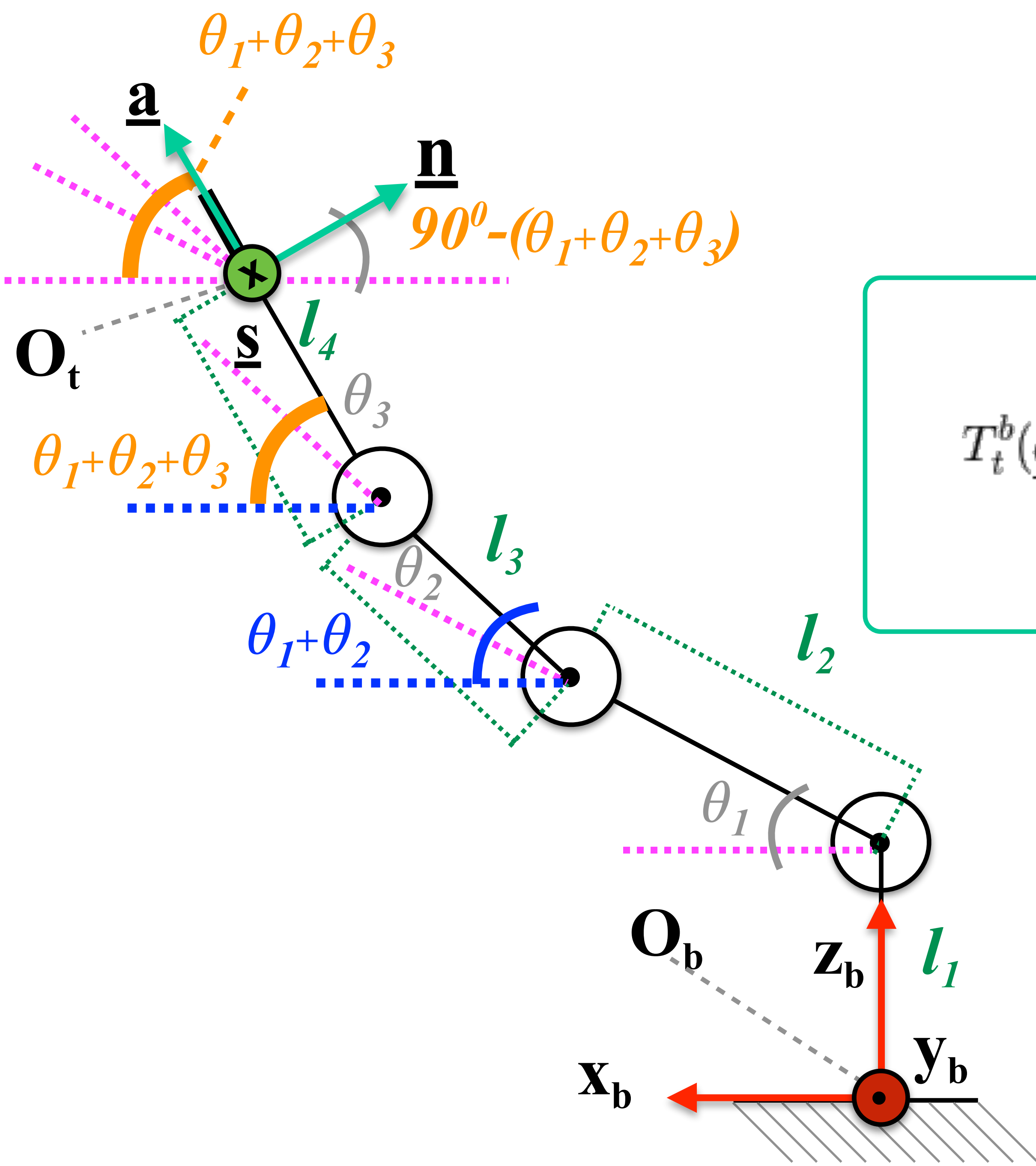


Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = \begin{bmatrix} -sin(\theta_1+\theta_2+\theta_3) & 0 & cos(\theta_1+\theta_2+\theta_3) & l_2cos(\theta_1)+l_3cos(\theta_1+\theta_2)+l_4cos(\theta_1+\theta_2+\theta_3) \\ 0 & -1 & 0 & 0 \\ cos(\theta_1+\theta_2+\theta_3) & 0 & sin(\theta_1+\theta_2+\theta_3) & l_1+l_2sin(\theta_1)+l_3sin(\theta_1+\theta_2)+l_4sin(\theta_1+\theta_2+\theta_3) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

Description of the Direct Kinematics of Anthropomorphic Robot Manipulators: Case 2



Determine the homogeneous transformation matrix

$$T_t^b(q) = \begin{bmatrix} -s_{123} & 0 & c_{123} & l_2c_1 + l_3c_{12} + l_4c_{123} \\ 0 & -1 & 0 & 0 \\ c_{123} & 0 & s_{123} & l_1 + l_2s_1 + l_3s_{12} + l_4s_{123} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_t^b(\underline{q}) = \left[\begin{array}{ccc:c} \underline{n}^b(\underline{q}) & \underline{s}^b(\underline{q}) & \underline{a}^b(\underline{q}) & \underline{o}_t^b(\underline{q}) \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$